

Grounds for Industrial Growth*

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Abstract

This paper examines the effects of land market liberalization on manufacturing firms through the staggered repeal of India’s Urban Land Ceiling and Regulation Act (UL-CRA), which severely restricted urban land transactions. The Repeal enabled more productive firms to expand land use and scale up production, thereby reducing land misallocation. This shift in land use also fueled business dynamism: treated firms experienced significantly higher product turnover, and market selection improved through the exit of unprofitable firms and the entry of more dynamic startups. To quantify the aggregate implications of the Repeal, we develop a general equilibrium model of firm dynamics with land market frictions disciplined by our reduced-form evidence. Through the lens of this model, the Repeal increased long-run manufacturing productivity growth by 0.32 percentage points and led to a 4.43% improvement in consumption-equivalent welfare.

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“Often the most valuable asset of an unprofitable firm was land. Since the [Urban Land Ceiling and Regulation] Act prohibited the firm from selling it at the market price, the firm chose not to exit and held onto the precious piece of land.”

– Arvind Panagariya, *India: The Emerging Giant*, 2010

1 Introduction

Restrictions on land property rights are common in developing countries, often characterized by insecure tenure, limits on land ownership, and restrictions on the ability to freely exchange land (World Bank, 2003, 2007).¹ While the development economics literature has long emphasized the importance of land property rights in *agricultural* production (de Janvry, 1981; Lin, 1992; Banerjee et al., 2002; Goldstein and Udry, 2008; Adamopoulos and Restuccia, 2020), there are few attempts to examine their consequences on *manufacturing* firms.²

In this paper, we address this gap by studying the micro- and macroeconomic impacts of India’s urban land market liberalization on manufacturing firms, both empirically and theoretically. Prior to the reform, urban land markets were tightly controlled by the 1976 Urban Land Ceiling and Regulation Act (ULCRA), which imposed stringent restrictions on land transactions. These restrictions limited firms’ ability to expand, adjust their landholdings, and exit the market—an issue eloquently captured in the epigraph from Panagariya’s book on India’s economic reforms. Considered a policy failure, the ULCRA was repealed by the central government in 1999.

To identify the causal effects of the liberalization, we assemble hand-collected data on the staggered repeal of the ULCRA across states and on variation in its enforcement within states. Our empirical strategy is a Difference-in-Differences design that compares firms in districts containing one of 64 Urban Agglomerations (UAs)³—where these land-market restrictions applied—to firms in districts without one, before and after the state-level repeal. Using non-UA firms as a comparison group allows us to isolate the effects of the liberalization from

¹Drawing on surveys from 140 countries, Prindex (2020) reports that roughly 20% of respondents believe they are at risk of losing the right to use their property against their will within five years. Appendix Figure B.6 shows the wide variation in tenure insecurity across countries and its strong negative correlation with GDP per capita.

²Three studies examine how land market frictions affect Indian manufacturing firms (Kimura, 2011; Duranton et al., 2017; Sood, 2022). We discuss these papers in detail below.

³An Urban Agglomeration (UA) comprises a core town and its surrounding suburban areas and may span multiple statutory towns and other urban entities such as railway colonies or university campuses. Conceptually, it is similar to a metropolitan statistical area in the United States.

other state-level shocks and contemporaneous policies. We show that UA and non-UA firms exhibit similar pre-trends prior to the Repeal, and that estimated effects on non-UA firms are small, which mitigates concerns about spillovers.

Our results show that, following the Repeal, treated plants expanded their landholdings by about 22% on average, thickening the right tail of the plant-size distribution. This reflected both within-plant expansion and a decline in the number of operating plants. The reform also facilitated land transactions: treated plants increased both land purchases and land sales, while land revaluation did not rise, consistent with lower frictions in buying and selling land rather than with a pure price effect. In addition, the Repeal improved the allocation of land: initially more productive plants increased their landholdings by 16 to 19%, while low-productivity plants did not. Consequently, the liberalization of the land market unleashed industrial growth. Treated plants expanded production by acquiring 16% more fixed assets (excluding land) and hiring 7% more labor, leading to a 17% increase in sales and a 13% increase in physical productivity.

The land market liberalization also significantly reshaped business dynamism and market selection. Following the Repeal, the rates at which plants introduced new products and discontinued old ones increased by roughly 14 and 15 percent, respectively, consistent with intensified creative destruction.⁴ These margins are directly observed from plants' product portfolios, an unusual advantage of the ASI data. To our knowledge, this provides some of the first causal evidence that factor-market distortions affect product innovation (Aghion et al., 2023; Bertrand et al., 2021). Furthermore, we find that the Repeal improved the selection of plants: unprofitable or dormant plants became more likely to exit, while the composition of new entrants shifted toward more innovative plants—entrants more likely to introduce new products over their life cycle.

To quantify the macroeconomic consequences of the Repeal on the Indian manufacturing sector, we develop a general equilibrium model of endogenous growth disciplined by our reduced-form causal estimates. Our model builds on the framework of Klette and Kortum (2004), which is particularly well suited to our setting as we directly observe plants' product portfolios and thus their product innovation decisions. The economy features two types of firms: *dynamic* firms that can expand into multi-product firms, and *stagnant* firms that operate a single product line throughout their lifecycle.⁵ We model the ULCRA as a

⁴The detailed product information also enables us to explore the broader consequences of the Repeal. The liberalization likely reduced marginal costs for treated plants, potentially allowing them to pass these gains on to consumers through lower prices. We find that treated plants reduced their product prices by roughly 10% without raising their markups, consistent with the gains being largely passed through to consumers.

⁵This distinction is motivated in part by the ASI evidence that more than half of plants never introduce

size-dependent wedge on the effective rental cost of land. As discussed in Section 2, this formulation captures the way the Act raised the cost of adjusting landholdings, especially for larger firms facing greater regulatory scrutiny.

We calibrate the model to match cross-sectional moments of the Indian manufacturing sector together with our DiD estimates of the Repeal’s causal effects on the marginal revenue product of land (MRPLand), product turnover, and entry and exit. The model closely replicates the main treatment effects from our empirical analysis. These moments discipline two distinct aspects of the land market distortions. The decline in the MRPLand indicates that the Repeal lowered the effective cost of acquiring and using land. In the model, this disciplines the average magnitude of the land wedge. The joint responses of product turnover, exit, and entry composition then determine whether the reform mattered disproportionately for more scalable plants. In the model, that pattern helps identify how sharply the land wedge rises with plant size. Beyond these targeted moments, the model also reproduces several untargeted features of the data, including the skewed product distribution and the flat life-cycle growth profile of Indian plants.

In the calibrated model, the counterfactual experiment eliminates the land market wedge. In the new steady state, the rate of creative destruction rises by 1.5 percentage points, as the higher return to expansion induces incumbent firms to introduce more new products and shifts entry toward dynamic firms. The model therefore predicts a compositional upgrading within the manufacturing sector: the dynamic share of entrants rises from 19.1% to 21.7%, and dynamic firms’ share of products rises from 40.1% to 44.5%. Together, these changes lead to a 0.32 percentage point acceleration in long-run aggregate productivity growth.

Accounting for transition dynamics—and, in our conservative baseline fiscal closure, rebating land-wedge revenue lump-sum to the household—the Repeal is equivalent to a 4.43% permanent increase in consumption per capita. Growth acceleration is the largest component, contributing 5.96 log points, while the change in resources allocated to product expansion and entry contributes -1.63 log points. The static gain from removing the cross-sectional dispersion of land distortions is negligible. Under the alternative assumption that wedge revenue finances wasteful government purchases, the consumption-equivalent gain rises to 7.30%.

a new product over their observed life cycle, while a smaller set of plants accounts for most product creation.

Literature review

Our study contributes to several strands of literature. First, it adds to the well-established literature on land market frictions and land reforms in developing countries (de Janvry, 1981; Lin, 1992; Besley and Burgess, 2000).⁶ In the agricultural sector, existing studies have highlighted the important role of property rights, particularly tenancy rights, in enhancing agricultural investment (Banerjee et al., 2002; Goldstein and Udry, 2008), as well as the importance of well-functioning land markets and clear land titles in resource and human capital allocation (de Janvry et al., 2015; Adamopoulos and Restuccia, 2020; Chari et al., 2021; Acampora et al., 2022; Adamopoulos et al., 2022; Liu et al., 2023; Adamopoulos et al., 2024). We extend this body of research by identifying the causal effects of land market frictions on *manufacturing* firms, leveraging novel landholding and land adjustment data in the Indian manufacturing census and the gradual repeal of the ULCRA as a natural experiment.

Second, we contribute to the expanding body of literature on resource misallocation and productivity growth among manufacturing firms in developing countries (Hsieh and Klenow, 2009; Hsieh and Olken, 2014; Hopenhayn, 2014; Restuccia and Rogerson, 2017; Verhoogen, 2023). In the Indian context, manufacturing firms tend to be inefficiently small and remain so throughout their lifecycle due to muted creative destruction (Akcigit et al., 2021; Peters and Zilibotti, 2021). Several recent studies have examined various forces that may hinder manufacturing growth, such as the role of contract enforcement (Boehm and Oberfield, 2018), delegation (Akcigit et al., 2021), industrial licensing (Chari, 2011; Alfaro and Chari, 2015), reservation laws (Martin et al., 2017), FDI restrictions (Bau and Matray, 2023), priority-sector lending (Banerjee and Duflo, 2014), and labor regulations (Amirapu and Gechter, 2019; Bertrand et al., 2021). We provide new evidence on the sources of misallocation and weak business dynamism in developing countries by exploring the role of land market frictions.

Three papers on land market frictions and Indian manufacturing are especially relevant. Kimura (2011) uses the ULCRA repeal to study land collateralization and credit reallocation across urban firms, showing that land deregulation affected land transactions and borrowing capacity. Duranton et al. (2017) quantify the misallocation of land and other factors in Indian manufacturing and relate it to land and property-market distortions. Sood (2022)

⁶India enacted the most extensive body of postcolonial land reforms compared to any other country, but empirical assessments are challenging due to state-specific variations in policies. Besley and Burgess (2000) categorize roughly 80 reforms into four broad categories and find mixed impacts of land reforms on economic growth. Using the same data, Ghatak and Roy (2007) find significant negative effects on agricultural productivity but emphasize the heterogeneity of effects across states and over time, cautioning against simplistic interpretations.

combines ASI land data with a structural model of land acquisition to study land aggregation costs, fragmentation, and firm growth. While these studies offer valuable insights, they do not consider that land market frictions may affect productivity growth by shaping firms’ innovation decisions. Our contribution is to provide causal evidence on how land market liberalization affects manufacturing plants’ land use, production, product turnover, and selection, and to discipline a model of endogenous growth—in which firms invest in innovation to raise their profits over time—with those estimates. To our knowledge, ours is the first study to document and quantify the *dynamic* effects of land market frictions.

A growing literature studies how distortionary regulations and market frictions shape innovation incentives and productivity dynamics (Mukoyama and Osotimehin, 2019; König et al., 2022; Aghion et al., 2023; Akcigit et al., 2023). Bento and Restuccia (2017) quantify how correlated distortions affect establishment-level productivity investment and aggregate output in India and the United States. In their stationary framework, the output loss associated with lower life-cycle productivity growth is fully offset by increased entry and reduced productivity dispersion; the remaining output loss operates through lower entry productivity and factor misallocation.⁷ Similarly, Bertrand et al. (2021) develop a model of product innovation to study Indian labor market frictions and find that cheaper contract labor raises innovation by high-type incumbents but reduces innovation by entrants, leaving aggregate innovation and long-run growth virtually unchanged. Our results stand in contrast to this pattern: growth acceleration is the largest component of the welfare improvement from the repeal of the ULCRA.

The remainder of the paper is organized as follows: Section 2 provides background information on the policy context and land market institutions in India. Section 3 describes our datasets and outlines the identification strategy. Section 4 presents our main empirical findings, highlighting the static and dynamic consequences of the Repeal. Section 5 outlines our model and discusses some of its implications. Section 6 details the estimation of the model’s parameters and quantifies both the equilibrium effects and welfare consequences of the Repeal.

⁷Atkeson and Burstein (2010) find a related offset in a model of international trade: lower trade costs raise process innovation among exporters but reduce product innovation through firm entry, so the two innovation responses largely offset one another in aggregate.

2 Institutional context

In this section, we describe the institutional origins of the ULCRA, the constraints it placed on urban land transactions, and the staggered process through which states repealed it.

The Urban Land (Ceiling and Regulation) Act (ULCRA). After gaining independence, India struggled with a severe urban housing shortage, with a deficit of 3 million housing units in 1971 (Tiwari, 2016).⁸ This shortage became more pressing due to rapid urbanization: between 1951 and 1971, the urban population expanded from 62 million to 109 million. At the same time, policymakers expressed growing concern over speculative landholding, as large landowners retained extensive tracts of vacant urban land, constraining supply and deepening the perceived housing “crisis.” Despite the enactment of numerous rural land legislation, urban land markets remained largely unregulated, with no statutory limits on land ownership or possession. This confluence of accelerating urban growth, acute housing shortages, and weak urban land regulation culminated in the enactment of the Urban Land (Ceiling and Regulation) Act in 1976.

From its inception, the ULCRA was fundamentally aimed at socializing urban land, an objective that became politically feasible during the National Emergency period (1975–1977), when democratic institutions were effectively suspended. The Act sought to address what policymakers viewed as an “excessive” concentration of urban landholdings, to curb speculative behavior, and to promote a more equitable distribution of land. Its provisions applied to designated UAs, defined as core cities and their contiguous suburban areas with populations exceeding 200,000 according to the 1971 Census. A total of 73 UAs were notified under the ULCRA to guide state-level implementation. Ultimately, the Act was adopted by 17 states and applied to all union territories, covering 64 UAs in practice. Appendix E.1 documents statutory coverage and the states and territories that fell outside it.

Ceilings on vacant landholdings. The ULCRA imposed stringent ceilings on *vacant* urban landholdings to limit the concentration of land within urban agglomerations. These ceilings were not aimed at productive land used by operating factories: land occupied by buildings and appurtenant land was generally excluded from the statutory definition of “vacant” land. These ceilings applied broadly to individuals, families, firms, companies, and associations, irrespective of their legal or incorporation status. Any land held in excess of

⁸The shortage of housing is measured as the number of households living in non-serviceable katcha (non-durable) houses, obsolescent structures, overcrowded dwellings, or without shelter.

the prescribed limits was subject to compulsory acquisition by the state, with compensation set at concessional rates. These ceiling limits were extremely restrictive, ranging from 500 to 2,000 square meters depending on the size of the urban agglomeration. Importantly, the Act defined landholdings broadly, encompassing not only owned land but also land held under lease or mortgage, thereby capturing virtually all forms of land possession within its scope. Appendix E.1 reports the ceiling schedule and compensation rules.

The definition of “vacant” land, however, remained ambiguous and was frequently interpreted to include any land not physically covered by structures (Ridhi, 2023). This ambiguity expanded bureaucratic discretion and often pushed larger parcels into litigation, exemption requests, or informal circumvention. In practice, the Act therefore increased the frictions associated with assembling, buying, selling, and developing urban land, particularly for larger firms and developers.

Transfer of property. The ULCRA also restricted how urban land could change hands. Vacant land within the ceiling could not be sold, mortgaged, gifted or leased without prior notice to the authorities, who could then pre-empt a sale at statutory rather than market prices. Land held above the ceiling could not be transferred at all until the excess had been declared and notified—a process slow enough to amount to an open-ended freeze—and a purchase that carried a firm above the ceiling exposed the excess to acquisition. Both restrictions reached only *vacant* land, but for a factory, that exclusion was narrow: it covered the land under a building already standing on the appointed day, together with at most five hundred square metres of appurtenant land. A plant’s yard, open storage and any land held for expansion therefore counted against ceilings of 500 to 2,000 square metres, so the Act bore on operating establishments and not only on idle holdings. Appendix E.2 documents these provisions, the relevant exemptions, and the judicial treatment of transfers involving buildings. These provisions exposed sellers to delay and to expropriation at below-market compensation, sharply reducing land market activity. Activity fell sharply: in some cities land transactions declined by as much as 75 percent, while land prices increased by 30 to 50 percent in the post-ULCRA period (Srinivas, 1991).

That these restrictions bore on operating plants, and not only on idle holdings, is visible in the proceedings themselves. In one adjudicated case an operating manufacturer of concrete pipes had 62 percent of its premises declared surplus vacant land; in another, a Thane factory was left with 38 percent of its site declared surplus even after the authority allowed appurtenant land at twice the built-up area, far beyond the five hundred square metres the statute prescribes. Nor were industrial holdings a marginal part of what the Act reached: in

Madras, of roughly 1,700 hectares of excess land identified, about 1,000 hectares were held by industry rather than by individuals (Acharya, 1989). Appendix E sets out these cases and their sources.

Repeal and aftermath. By the late 1990s, the shortcomings of the ULCRA had become increasingly apparent. Although the Act was intended to curb the concentration of urban land and promote equitable access, it largely failed to achieve these objectives. According to the Ministry of Urban Affairs, by 1999 only 20,654 hectares (approximately 80 square miles) of urban land had been acquired under the Act, and nearly half of this land remained undeveloped. A growing body of empirical evidence further documented the ULCRA’s failure to stimulate affordable housing construction, contributing to the central government’s decision to repeal the Act in 1999 (Srinivas, 1991; Joshi and Little, 1996; Sridhar, 2010; Siddiqi, 2013; Dutta and Gandhi, 2023).

The central government’s 1999 repeal was abrupt: the Repeal Act came into force on 11 January 1999. In its 1998–1999 working report, the Ministry of Urban Affairs acknowledged the absence of any systematic assessment of the repeal’s potential impact on land prices, noting that “[T]he Government have not made any analysis with regard to impact of the repeal Act on the prices of land as also the trend of land prices after the Act was implemented in 1976.” Whether the repeal was nonetheless anticipated by market participants is an empirical question the documentary record does not settle; the event studies below speak to it directly, showing flat pre-Repeal paths with no differential movement between UA and non-UA districts (Sections 3 and 4). The central repeal did not, however, itself lift the restrictions in the states. Under India’s constitutional framework, legislative authority over urban land rests with the states: the Repeal Act applied in the first instance only in the union territories and in Haryana and Punjab, the two states at whose request Parliament had passed it; in every other adopting state the ULCRA remained in force until that state adopted the Repeal Act by a resolution of its own legislative assembly, the procedure required under Article 252 of the Constitution for a law that Parliament had enacted at the states’ request.⁹

A majority of states and union territories adopted the Repeal Act in 1999, as summarized in Table B.14, while others followed in a staggered manner over the subsequent decade. Between 2000 and 2011, the Act was repealed in Chhattisgarh and Madhya Pradesh (2000),

⁹Article 252 reserved amendment or repeal of the parliamentary law to Parliament, while the timing of each state’s repeal resolution remained a state political choice. Appendix E.3 documents this mechanism and its legal basis.

Odisha (2002), Assam (2003), Bihar (2006), Maharashtra (2007), Andhra Pradesh and Telangana (2008),¹⁰ and Jharkhand (2011). West Bengal adopted the ULCRA but has still not adopted the Repeal Act, while Tamil Nadu never adopted the ULCRA in the first place and continues to operate its own land ceiling law; urban land ceilings therefore remain in effect in both states.¹¹

While the precise reasons for the delayed and uneven repeal across states are not fully understood, existing case studies suggest that political factors predominated over economic ones. The ULCRA generated substantial opportunities for rent-seeking and corruption, creating incentives for state-level elites to resist the Repeal (Siddiqi, 2013). In addition, partisan fragmentation and coalition politics often hindered the formation of legislative consensus necessary to dismantle the Act (The Economic Times, 2007).

3 Data and empirical strategy

This section describes the data and empirical strategy used in our analysis. We combine plant-level manufacturing data with a newly constructed dataset on the geographic coverage of the ULCRA and the staggered timing of its repeal across states. Our identification strategy exploits the interaction between repeal timing and local ULCRA exposure.

3.1 Data sources

ASI data. Our primary data source is the Annual Survey of Industries (ASI), which provides extensive information on manufacturing plants from 1993 to 2013 with district and plant identifiers.¹² The ASI provides a representative sample of registered manufacturing plants in India, with large plants covered yearly and smaller ones surveyed on a sampling basis. Because the large majority of registered Indian manufacturers operate a single plant, we use *plant* and *firm* interchangeably throughout the empirical analysis; the distinction matters only in Section 5, where a firm is a collection of product lines. Like other manufacturing surveys, the ASI provides detailed plant-level information, including fixed assets, wage bills,

¹⁰We date Andhra Pradesh and Maharashtra to their public statewide 2006 JNNURM repeal commitment in the baseline; using their formal repeal dates leaves the principal estimates essentially unchanged (Appendix Table B.30). Telangana is recorded as Andhra Pradesh throughout our sample. Appendix E.3 documents the chronology.

¹¹We exclude West Bengal and Tamil Nadu as institutional outliers; Appendix E.3 provides the historical and legal details.

¹²Survey years refer to the calendar year in which the April–March fiscal year begins. Appendix C.1 describes the panel construction and district harmonization.

number of employees, consumption of raw materials, and sales revenue.¹³

A distinctive feature of the ASI is its detailed disaggregation of plants’ capital stock into land, buildings, machinery, and other fixed assets, which allows us to separately identify land inputs. We use the average of the book value at the start and end of the year to measure land use, which reflects the acquisition cost of land rather than its current market value. This measure includes both owned and leased land that is recorded in the plant’s fixed-asset block and used in production, but excludes unused land.¹⁴ A limitation of the ASI is that it does not report land prices or physical land quantities, an issue we address in greater detail in the empirical analysis.

Another unique feature of the ASI dataset is that it provides information on both total product sales and total quantity sold at the product level, as mandated by the 1956 Companies Act. Products are defined based on 5-digit product codes, encompassing over 12,000 distinct products. Leveraging the plant-level panel structure of the data, we directly observe product innovation through changes in a plant’s product portfolio, including the addition and deletion of products. These product-based measures of innovation are crucial for us to examine the effects of land restrictions on innovation, especially since firms in developing countries rarely file patents or report their R&D expenditure. Table B.15 reports summary statistics for the main ASI variables used in our analysis.

ULCRA data. We manually collected and compiled a novel ULCRA dataset that records the timing of repeal across states and the geographic coverage of the 1976 Act.¹⁵¹⁶ The 1976 Act delineates numerous UAs and their associated towns and villages. We systematically gathered coordinates for over six hundred listed locales through extensive research. To ensure their accuracy, we used the Indian Village Directory to verify the existence and correct

¹³All financial amounts are expressed in 2004 Rupees following the procedures outlined in Allcott et al. (2016). Revenue is adjusted for inflation using 3-digit industry-level commodity price deflators. The deflators for materials reflect the average price changes in outputs from a given industry’s suppliers, as determined by the 1993-1994 input-output table. For capital, we use an implied national deflator based on capital formation provided by the Reserve Bank of India.

¹⁴Land is reported for about 58% of plant-years and is absent from the 1997 round. Appendix C.1 discusses revaluation, rental land, and missingness.

¹⁵We assign each formal repeal date to the ASI fiscal year containing it and code subsequent ASI years as post-Repeal. Maharashtra and Andhra Pradesh are dated to their public statewide 2006 JNNURM commitment; the principal estimates are robust to dating both states at their formal repeal years instead (Appendix Table B.30). Appendix C.1 details the coding, and Appendix E.3 documents the legal chronology.

¹⁶Collecting data on the timing of the repeal is challenging due to the lack of a centralized policy information platform in India, leading to discrepancies in previous studies regarding the timing of the repeal at the state level (Sridhar, 2010; Kimura, 2011; Duranton et al., 2017). We address this by consulting multiple legal databases and extracting repeal dates from public judicial records that explicitly mention these dates.

spellings of these places, especially when they were unavailable online. Historical name changes and alternate spellings presented additional challenges, which were addressed through targeted searches. We then mapped all UAs to district boundaries defined in the 1991 district map.

We use districts as our unit of treatment: a plant is classified as a UA plant if it is located in a district containing any UA locale, and is treated only once its state has also repealed the Act.¹⁷ This choice is guided by several considerations. First, the ASI reports plant locations only at the district level, preventing us from precisely identifying whether plants are located strictly *within* UA boundaries. Second, defining treatment at the district level mitigates potential bias from plants relocating between cities and surrounding suburbs.¹⁸ Finally, the effective reach of the ULCRA likely extended beyond formal UA boundaries. For example, following the Act’s introduction, developers reportedly hoarded land in adjacent areas, driving land prices up by 30–50% (Srinivas, 1991). In practice, using districts as the treatment unit effectively approximates the actual coverage of the ULCRA, as population census data show that over 60% of non-agricultural employment within districts is concentrated in UAs.¹⁹

3.2 Empirical strategy

To estimate the causal effects of the ULCRA repeal, we implement a difference-in-differences (DiD) strategy that exploits both the staggered timing of repeal across states and the fact that the ULCRA only applied to UAs. The key comparison is thus between plants located in UA districts, where these land-market restrictions applied, and plants in non-UA districts, before and after repeal within the same state. Our baseline empirical specification is given by the following equation:

$$Y_{idst} = \alpha + \beta_1 \cdot PostRepeal_{st} + \beta_2 \cdot PostRepeal_{st} \times UA_d + \gamma \cdot X_{idst} + \delta_t + \delta_d + \epsilon_{idst} \quad (1)$$

where Y_{idst} represents the various outcome variables for plant i in district d of state s in year t . $PostRepeal_{st}$ is an indicator variable that equals 1 if state s has adopted the repeal act

¹⁷Almost all UAs can be uniquely matched to a single district, meaning that all locales within a UA are situated in the same district. However, there are a few exceptions, such as the Calcutta UA and Madras UA, which span multiple districts—both of which fall in the two states we exclude from the sample.

¹⁸Cross-district relocation is rare in our data as less than 3% of plants have ever changed districts. This low rate may even be inflated by entry errors in district information.

¹⁹Figure B.7 plots village and town-level non-agricultural employment against their distance from the UAs’ centroids. Notably, employment decreases sharply to nearly zero outside a 5-kilometer radius from these centroids, suggesting that most non-agricultural activity occurs within or very close to UAs.

by year t , dated by the ASI year that contains the repeal.²⁰ UA_d is an indicator variable set to 1 if the plant resides in a district d that contains an UA; its main effect is absorbed by the district fixed effects. X_{idst} denotes the remaining controls, which we absorb as fixed effects: rural location, 2-digit industry, and the district’s pre-Repeal average plant land-size quartile interacted with year.²¹ δ_t and δ_d are year and district fixed effects, which account for any aggregate fluctuations and time-invariant cross-district heterogeneity. In specifications that focus on within-plant dynamics, we additionally include plant fixed effects δ_i . Plant-level regressions are weighted by the ASI sampling multiplier so that the estimates are representative of the population of registered plants.²² Standard errors are clustered at the state level—27 states and union territories—as the reform was adopted gradually by individual states and union territories.

Our primary coefficient of interest is β_2 (reported as ‘Repeal $\times$ UA’ in the tables, with β_1 reported as ‘Repeal’), which captures the differential response of plants in UA districts—those more directly exposed to ULCRA-related land frictions—relative to plants in non-UA districts, before and after repeal. This coefficient isolates the effect of relaxing those frictions in locations directly exposed to the policy. Because the ASI reports plant locations only at the district level, β_2 should be read as an intent-to-treat-style effect for plants in districts containing a UA, not as the effect on plants verified to lie within statutory UA boundaries; to the extent that some plants in UA districts fell outside the Act’s effective reach, their inclusion dilutes the estimate relative to the effect on directly covered plants. By contrast, β_1 measures the average post-repeal change for plants in non-UA districts. Because the ULCRA did not apply outside UAs, a nonzero estimate of β_1 may reflect contemporaneous state-level shocks, spillovers from treated UA districts to nearby non-UA areas, or biases associated with staggered adoption. For transparency, we report both coefficients in our baseline specification. We also show below that the estimates for β_2 are robust to including state-by-year fixed effects, which absorb β_1 and restrict identification to within-state-year differences between UA and non-UA districts: the production estimates are essentially unchanged, and the effect

²⁰The exceptions are Maharashtra and Andhra Pradesh, whose treatment onset we date to their 2006 statewide JNNURM repeal commitment rather than to their formal repeals in late 2007 and early 2008 (footnote 15; Section 2).

²¹Our industry classification is a time-consistent crosswalk between the 1987 and 1998 revisions of the National Industrial Classification, which yields 52 groups, roughly at the two-digit level. The land-size control is the quartile of the district’s average plant landholding in the pre-Repeal years, interacted with year fixed effects. Specifications that depart from this baseline set say so in their table or figure notes.

²²The exceptions are the heterogeneity splits by pre-Repeal productivity or MRPLand and the pre-trend tests, which are estimated unweighted: each conditions on plants for which a pre-Repeal measure exists and which are observed on both sides of the reform, so the estimation sample is a selected subsample rather than a representative cross-section and the multipliers do not recover a population quantity.

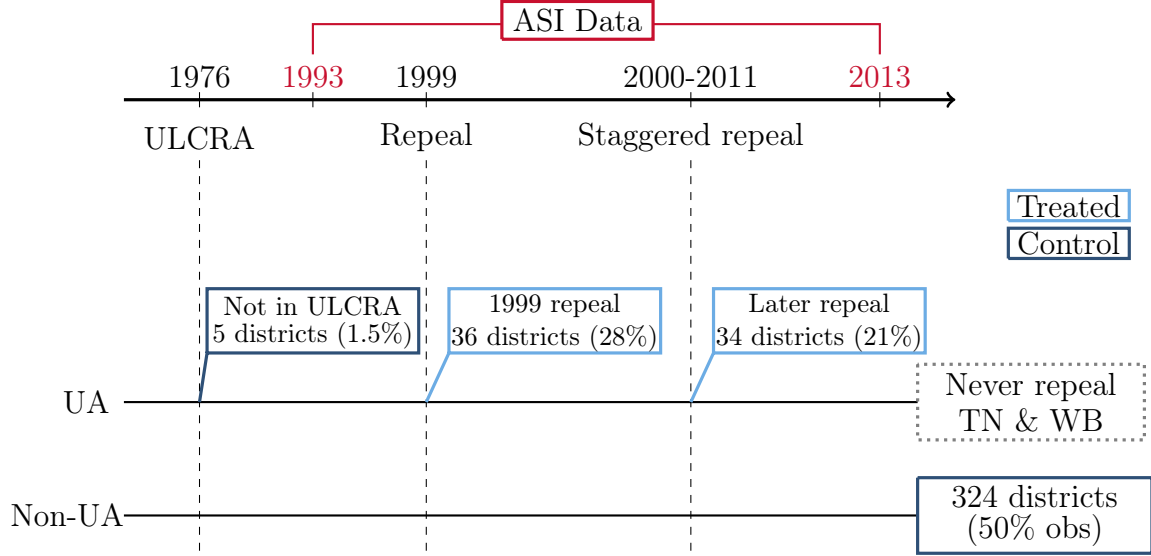


Figure 1: Timeline of the ULCRA and its Repeal

on the marginal revenue product of land, if anything, strengthens.

The identification of β_2 relies on a parallel-trends assumption: absent the Repeal, the difference in outcomes between plants in UA and non-UA districts would have evolved similarly across states that had already repealed and states that had not yet repealed the ULCRA. Equivalently, the UA–non-UA differential in treated states would have followed the same path as the corresponding differential in not-yet-treated states in the absence of the policy. This assumption rules out confounding shocks that differentially affect UA and non-UA districts and evolve systematically with repeal timing.

To illustrate our empirical design, we present the timeline of the ULCRA and its Repeal in Figure 1. The control group, non-UAs, consists of 324 districts, which constitutes approximately 50% of our ASI data. Jammu & Kashmir and Kerala, while having qualified UAs, never participated in the ULCRA, so the five UA districts in these two states have $PostRepeal_{st} = 0$ throughout and serve as controls as well. However, they constitute merely 1.5% of the data, and our results are robust to dropping them. A majority of the states repealed the ULCRA in 1999, as listed in Table B.14, encompassing 36 UA districts, or 28% of the observations. From 2000 to 2011, those lagging states implemented the Repeal gradually, including 34 UA districts, or 21% of the data. Tamil Nadu and West Bengal never adopted the Repeal Act and were excluded from the sample as discussed earlier. Therefore, our design essentially compares UAs vs. non-UAs, before and after these two episodes of repeal.

3.3 Threats to identification

In this section, we further elaborate on our identification strategy and address concerns about several potential sources of bias that could contaminate our results.

Non-random assignment of the Repeal. A potential concern is that firms in states that repealed the ULCRA earlier may differ systematically from those that repealed later, and that firms in UA districts may differ from firms in non-UA districts. However, district fixed effects (δ_d) account for any pre-existing, time-invariant differences (observed or unobserved) between firms across UAs, non-UAs, early-repeal states, and late-repeal states. Importantly, our identification strategy does not require the timing of repeal to be random or unrelated to underlying state-level trends. Instead, the estimate of β_2 remains unbiased as long as the factors driving earlier repeal do not generate differential changes in outcomes between UA and non-UA districts. A violation would arise, for example, if states with limited urban land supply and rising land prices in UAs were more likely to repeal earlier. Although we do not observe land prices directly, pre-existing differences in average plant landholdings across districts, and flexible trends by those differences, are absorbed by the district pre-Repeal land-size quartile-by-year fixed effects.

Validation of identification assumptions. Although the identifying assumptions cannot be tested directly, we provide several pieces of evidence consistent with their validity. First, our event study plots for β_1 show no visible pre-existing trends between non-UA firms in states that had repealed and those in states that had not, and the estimates for β_1 remain close to zero after repeal. Second, event study estimates for β_2 show no evidence of differential pre-trends between firms in UA and non-UA districts prior to repeal across states. Finally, we perform a range of robustness checks to demonstrate that our estimates remain stable when we include additional controls for differential trends at the district, industry, and state levels.

We further assess the validity of our identification strategy by testing for differential pre-trends between UA and non-UA districts prior to the Repeal. Using firm-level data from the pre-1998 period, we examine a range of outcomes, including output, landholdings, fixed assets, employment, and MRPLand.²³ For each outcome, we regress the variable on a UA indicator, a linear time trend, and their interaction, controlling for state and industry

²³MRPLand is measured using the average revenue product of land. Under a Cobb–Douglas production function, a higher average revenue product implies a higher marginal revenue product within industries.

fixed effects. As reported in Table B.16, the interaction terms are consistently small and statistically insignificant, providing no evidence of differential trends between UA and non-UA districts prior to the Repeal.

We also conduct additional checks to address concerns about the potential endogeneity of repeal timing, particularly whether early repeal could be driven by land shortages or unequal land access. To this end, we examine whether the timing of the repeal correlates with the pre-Repeal distribution of land and the degree of land productivity dispersion across firms. Specifically, we regress an indicator for repeal adoption on lagged (up to three years) measures of the standard deviation of landholdings, as well as the average and standard deviation of MRPLand, constructed at the state–industry level using ASI data. The results, presented in Table B.20, show no systematic relationship between these pre-Repeal characteristics and the timing of repeal. We further show, in the discussion of staggered adoption below, that our results are robust to using only the first episode of the repeal.

Staggered DiD. Recent work has shown that two-way fixed-effects (TWFE) difference-in-differences estimators can be biased when treatment adoption is staggered across units and treatment effects are heterogeneous over time or across groups (Athey and Imbens, 2018; Borusyak et al., 2023; Callaway and Sant’Anna, 2020; Goodman-Bacon, 2021; Sun and Abraham, 2021; de Chaisemartin and D’Haultfoeulle, 2020). In such settings, the TWFE estimand is a weighted average of underlying treatment effects, and those weights can be problematic when already-treated units serve as controls. We therefore treat the baseline TWFE results as our main specification and use alternative staggered-adoption estimators as robustness checks. Specifically, we show that our estimates remain similar when using the stacked DiD estimator of Cengiz et al. (2019), the imputation estimator of Borusyak et al. (2023), and the group-time estimator of Callaway and Sant’Anna (2020). We also show that the results remain similar when restricting the sample to the first repeal episode in 1999. Because that episode involves a single adoption cohort triggered by the central government’s own repeal, it removes comparisons between early- and late-treated states and further limits the scope for cross-state differences in repeal timing to drive our estimates.

4 The microeconomic impacts of the Repeal

In this section, we present empirical evidence on the microeconomic effects of repealing the ULCRA. We show that the Repeal liberalized urban land markets, improved the allocative efficiency of land, and triggered firm expansion, productivity growth, and greater business

dynamism.

4.1 Did the Repeal relax urban land constraints?

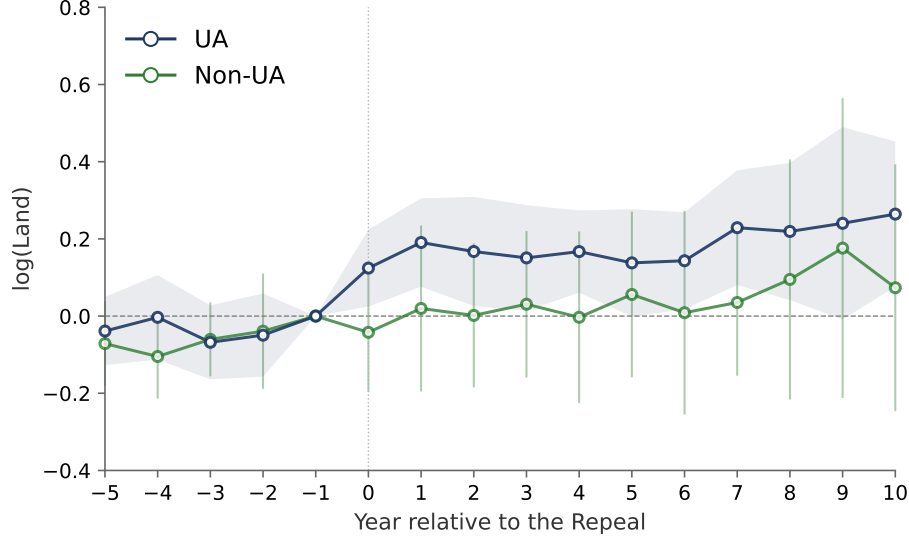
We begin by analyzing the impact of the liberalization on plants’ *average* land use. Using the event-study analogue of equation (1), which replaces the single $PostRepeal_{st}$ indicator with a full set of event-time indicators and their interactions with UA_d , we estimate dynamic effects on plants’ land use, measured by the stock of land employed in production. Figure 2 plots the estimated coefficients for plants in non-UA districts (β_1 , green) and the differential coefficients for plants in UA districts (β_2 , navy).

The event-study estimates show no visible differential effects prior to the Repeal for either group. The flat pre-Repeal path of β_2 indicates that, before liberalization, UA firms were not on obviously different land-use trajectories than non-UA firms; a joint Wald test confirms the visual impression, with the pre-Repeal β_2 leads jointly indistinguishable from zero ($\chi^2(4) = 7.14$, $p = 0.13$; Appendix B.2 reports this test for every displayed event study).²⁴ In addition, the pre-Repeal behavior of β_1 suggests no obvious differences in landholding trends between non-UA firms in earlier- and later-repealing states. Together, these patterns are consistent with the identifying assumption, as is the evidence in Appendix Table B.16.

Consistent with the ULCRA applying only to urban agglomerations, the reform has no discernible effect on plants’ land use in non-UA districts. Pooling the post-Repeal years into the difference-in-differences specification of equation (1)—reported in Column (1) of Table 1—the non-UA effect is $\beta_1 = 0.005$ (standard error 0.057), statistically indistinguishable from zero and with a confidence interval that excludes an effect of the magnitude we estimate for UA plants. Together with the β_1 profile in the event study, which is statistically indistinguishable from zero throughout, this lessens concerns that our estimates are being driven by common state-level shocks that affect treated and untreated districts similarly, or by large spillovers from UA to non-UA districts.²⁵ In contrast, plants in UA districts experience a sizable post-Repeal expansion in land use: the differential effect is $\beta_2 = 0.202$ (standard error 0.056), so land used in production by UA plants rose by roughly 22 percent relative to non-UA plants after the Repeal, corresponding to an increase of about 1.25 million Indian rupees

²⁴The first post-Repeal-year effect (0.124, 95% confidence interval [0.024, 0.224]) also remains significantly positive under the parallel-trends relaxations of Rambachan and Roth (2023): allowing the slope of the differential trend to change by up to 0.01 log points per year, or allowing post-Repeal violations up to 0.3 times the largest—and itself statistically insignificant—consecutive pre-Repeal deviation. See Appendix B.2.

²⁵Large spillovers are unlikely given the size of Indian districts—averaging roughly 5,000 square kilometers—and the substantial costs associated with cross-district firm relocation and migration (Kone et al., 2018).



Note: This figure reports the event study graph for the treatment effect of the Repeal Act on plants' landholdings. The dependent variable is the logarithm of the book value of land. Coefficients are normalized 1 year before the reform. The green line with error bars indicates the effects on plants in non-UA districts (β_1) and the navy line with the shaded band represents the differential effects on plants in UA districts (β_2). The regression includes ownership dummies, an urban dummy, industry fixed effects, district-level land endowment-by-year fixed effects, district fixed effects, and year fixed effects. Standard errors are clustered at the state level. The confidence interval is at the 95% level.

Figure 2: The effect of the Repeal on plant-level land use

for the average UA plant.²⁶ The event-study profile in Figure 2 shows that this average effect is not driven by any single post-Repeal year: the response emerges immediately after the reform and remains stable thereafter.

Why did average land use increase? At first glance, the post-Repeal increase in plants' average land use may seem puzzling, given that land supply is often thought to be inelastic in the short run. Two features of our empirical design help clarify this result. First, the event-study estimates in Figure 2 do not include plant fixed effects, so the observed increase reflects both within-plant expansion and changes in the composition of plants operating in UA districts. Second, Appendix Table B.17 shows that we cannot reject that district-level to-

²⁶Obtained by applying the estimated 22% effect to the mean pre-Repeal landholding of UA plants in repealing states (≈ 5.6 million deflated rupees, unweighted, over 61,043 plant-years). The landholding distribution is heavily right-skewed—the median is about 0.22 million—so this figure describes the mean plant rather than the typical one.

tal land used in production is unchanged after the Repeal, while the number of plants declines by roughly 10 percent, an effect significant at the 10 percent level. Taken together, these patterns suggest that the reform induced the exit of smaller plants and a thickening of the right tail of the land-use distribution among survivors. Appendix C.3 further shows that the Repeal increased the likelihood and scale of land purchases, and reports suggestive—though imprecisely estimated—evidence that treated plants became more responsive in adjusting landholdings after profitability shocks. Appendix Table B.29 makes the composition reading direct: with plant fixed effects on a consistent sample, the average incumbent’s landholding is flat while capital and wage bills rise significantly, so the average land response reflects reallocation—concentrated, per Table 2, in high-productivity incumbents—rather than uniform within-plant accumulation.

A natural concern is that the measured increase in land use could be driven by rising land prices rather than by a relaxation of the constraint itself. The marginal revenue product of land bears directly on this concern—subject to a measurement caveat we take up below—and it is the estimate our model will target. A cost-minimizing plant equates the marginal revenue product of land to the effective cost of land—the market price multiplied by any wedge the plant faces. If land had simply become more expensive, plants would economize on it until MRPLand *rose* to meet the higher price. Column (2) of Table 1 shows the opposite: relative to non-UA plants, MRPLand among UA plants *fell* by about 10% after the Repeal. Our land measure is deflated book value at acquisition cost rather than a current market valuation. For continuing holdings the denominator is therefore fixed at acquisition-vintage values, and a pure movement in land prices does not enter it; newly purchased parcels, however, enter book value at transaction prices, and the Repeal raised purchases. The scope of that channel is bounded by the flow of additions relative to the stock of land—see the measurement discussion in Appendix A.5—and the pattern of land transactions, district-level land totals, residential rents, and real input growth all point the same way (Appendix C.3; Appendix Tables B.17 and D.39). Under these measurement caveats, the decline in MRPLand is consistent with a fall in the wedge, and it is the moment we use to discipline the land distortion in Section 5.²⁷

Robustness to staggered adoption. As a robustness check, Appendix Figure B.8 reports event-study estimates using several DiD estimators designed for staggered treatment timing. Panel (a) compares the stacked DiD approach of Cengiz et al. (2019), the imputation estimator of Borusyak et al. (2023) and the group-time estimator of Callaway and Sant’Anna (2020),

²⁷Appendix A.5 explains the industry-by-year residualization and its model counterpart. Appendix Table B.18 reports robustness to progressively finer fixed effects.

	(1)	(2)	(3)	(4)	(5)	(6)
	Land	MRPLand	Capital	Employment	Wage Bills	Sales
Repeal	0.005 (0.057)	0.017 (0.044)	0.081 (0.104)	−0.029 (0.038)	−0.005 (0.051)	−0.021 (0.049)
Repeal × UA	0.202*** (0.056)	−0.102** (0.048)	0.145** (0.058)	0.067** (0.026)	0.083*** (0.025)	0.160*** (0.046)
Observations	338,686	298,863	530,008	533,802	531,119	464,755
R^2	0.183	0.053	0.254	0.148	0.214	0.244
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Size × Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Note: All dependent variables are in logarithms and deflated, and all columns estimate the same specification (year, district, rural, 2-digit industry and landholding-quartile-by-year fixed effects). Column (1) is land used in production. Column (2) is the marginal revenue product of land, measured as log sales over land and residualized on 2-digit-industry-by-year means so that the comparison is within industry and year; Appendix Table B.18 reports it under progressively finer fixed effects. Columns (3)–(6) are production outcomes; capital excludes land value, so the land and capital responses are measured on disjoint assets. ‘Land Size×Year’ are district average pre-treatment land-size bins interacted with year. Sample sizes differ across columns because the outcomes are reported by different sets of plants, so the coefficients are not algebraically comparable across columns. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table 1: Treatment effect of the Repeal on land use, MRPLand, and plant production

alongside two-way fixed effects as a benchmark and a single-cohort design that eliminates early–late comparisons by construction. Pre-Repeal coefficients are flat for every estimator, and the post-Repeal paths agree in timing and are similar in magnitude. Panels (b) and (c) replicate Figure 2 by jointly estimating β_1 and β_2 using the stacked DiD and imputation approaches, respectively. Taken together, these exercises show that the baseline findings are not sensitive to the particular estimator used to handle staggered adoption.

Land market allocative efficiency. We next examine whether the Repeal improved the allocation of land across firms. Following Chari et al. (2021), we study the differential impacts of the Repeal on plants based on their pre-Repeal total factor productivity (TFP), while including plant fixed effects to isolate within-plant adjustments in land use. The inclusion of plant fixed effects absorbs all time-invariant plant characteristics and restricts attention to firms that survive from the pre- to the post-Repeal period, allowing us to focus on reallocation among incumbents. We defer a discussion of changes in plant composition—including entry

and exit responses—to Section 4.3.²⁸

We measure productivity using both revenue-based TFP (TFPR) and quantity-based TFP (TFPQ) to account for the possibility that high TFPR reflects market power or product appeal rather than underlying technical efficiency. Both measures are production-function residuals, estimated separately within each two-digit industry using the control-function approach of Akerberg et al. (2015) as implemented by De Ridder et al. (2022), whose procedure—developed there for markup estimation—supplies the measurement-error correction and the price and market-share controls we use to recover quantity-based productivity.²⁹ We then estimate the effect of the Repeal on within-plant changes in land use, splitting the sample based on whether a plant’s average pre-Repeal TFPR(Q) lies above or below the district median. Table 2 reports the results: Columns (1)–(2) show effects for initially high-productivity plants, while Columns (3)–(4) correspond to less productive plants. We find that, following the Repeal, UA plants with higher initial TFPR(Q) expand their land use by 16–19 percent, whereas the effects for lower-productivity plants are small and statistically insignificant. The gradient is tested directly rather than inferred from the subsample significance pattern: a pooled specification that interacts every regressor and fixed effect with the high-productivity indicator—so that it nests the two subsample regressions exactly—rejects equality of the high- and low-group responses, with a High–Low difference of 0.105 (standard error 0.033, $p = 0.004$) for TFPR and 0.157 (standard error 0.040, $p = 0.001$) for TFPQ.

The precisely estimated null effect for non-UA plants further alleviates concerns that our findings reflect slow-moving land accumulation rather than the Repeal itself. If high-productivity plants were expanding landholdings even in the absence of liberalization, we would expect to observe similar increases among high-productivity firms in non-UA districts. Instead, we find no such effect. This placebo also disciplines a mechanical alternative: because the differential β_2 nets out any pattern common to high-productivity plants in UA and non-UA districts alike, mean reversion induced by sorting plants on estimated pre-Repeal productivity cancels by construction. Appendix Table B.22 shows that these results are robust across alternative measures of productivity. Event-study versions of these splits, joint pre-trend tests for every displayed event study, and formal sensitivity bounds are reported in Appendix B.2. Taken together, these findings suggest that the Repeal improved the allocative efficiency of land markets by enabling more productive firms to expand.

At first glance, it may seem surprising that less productive plants did not experience a

²⁸We do not split plants by pre-Repeal MRPLand because the corresponding event studies exhibit significant mean-reverting pre-trends; Appendix C.2 provides details.

²⁹Appendix C.1 provides the estimation and multi-product price-aggregation details.

	(1)	(2)	(3)	(4)
	High TFPR	High TFPQ	Low TFPR	Low TFPQ
Repeal	0.028 (0.042)	0.039 (0.036)	0.011 (0.040)	0.006 (0.052)
Repeal $\times$ UA	0.151*** (0.051)	0.177*** (0.047)	0.047 (0.047)	0.020 (0.056)
Observations	60,084	57,794	55,659	56,199
R^2	0.888	0.892	0.894	0.891
District FE	Yes	Yes	Yes	Yes
Size $\times$ Year FE	Yes	Yes	Yes	Yes
3-digit Ind. $\times$ Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes

Note: The dependent variable is log-landholdings. TFPR and TFPQ are production-function residuals estimated for each 2-digit industry by adapting the procedure of De Ridder, Grassi, and Morzenti (2022). The sample is split by whether a plant’s pre-Repeal average TFP lies above or below the district median. All regressions include plant fixed effects. A pooled specification that interacts every regressor and fixed effect with the High indicator—so that it nests the two subsample regressions exactly—directly tests equality of the high- and low-group Repeal $\times$ UA coefficients: the High–Low difference is 0.105 (SE 0.033, $p = 0.004$) for TFPR and 0.157 (SE 0.040, $p = 0.001$) for TFPQ. These regressions are unweighted. Because the sample is restricted to plants with a measurable pre-Repeal productivity level and observed on both sides of the reform, it is a selected subsample rather than a representative cross-section of registered plants, so the ASI sampling multipliers used elsewhere in the paper do not recover a population quantity here. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table 2: Within-plant land use growth by pre-Repeal TFP

decrease in their land use after the Repeal. This raises the question: where did these new plots of land come from? It is important to clarify that our analysis here, by including plant fixed effects, focuses only on the group of plants that have survived from the pre-Repeal to the post-Repeal periods—typically larger plants included in ASI’s census scheme.³⁰ Therefore, these *less* productive plants were not the *least* productive ones. In Section 4.3, we explore how the Repeal released resources from dormant plants (defined there as plants that still held onto their land and buildings but produced nothing) by driving them out of the market.

³⁰The ‘census scheme’ in the ASI refers to plants with more than 100 employees that are surveyed every year, as opposed to a ‘sample scheme’ which may include smaller plants surveyed less frequently.

4.2 Did the Repeal unleash industrial growth?

Manufacturing production. We now examine whether this land-market liberalization translated into increased manufacturing output. Because firms may have acquired land for speculative purposes rather than production, we test whether the Repeal relaxed economically meaningful land frictions for urban manufacturers. Using equation (1), we estimate the Repeal’s effect on fixed assets excluding land, employment, wage bills, and sales, alongside the land-use and MRPLand estimates of the preceding subsection. This allows us to distinguish between productive expansion and mere asset accumulation.

Table 1 shows that the Repeal had no detectable effect on plants in non-UA districts, consistent with our previous findings. Columns (1)–(2) repeat the land and MRPLand responses discussed above; Columns (3)–(6) report the production outcomes, estimated on the same specification. In contrast to non-UAs, the average plant in a UA experienced a 16% increase in its stock of buildings and machinery following the reform—an expansion in capital that is distinct from the land response, since our capital measure excludes land value. This capital expansion did not occur at the expense of labor; rather, employee counts and wage bills grew by roughly 7 and 9 percent, respectively, and, as a result, UA plants saw a 17% rise in total revenues. Collectively, these findings provide strong evidence that ULCRA-related land frictions materially constrained urban manufacturing prior to the reform.

These effects are not uniform across the plant size distribution: Appendix C.4 shows that they peak among upper-middle-sized plants, which also face the highest returns to land and therefore had the most to gain from cheaper land adjustment.

Appendix Figure B.9 presents the event study graphs for the treatment effects on firm production. Consistent with our DiD estimates, the Repeal did not affect plants in non-UA districts, while plants in UA districts ramped up production after the Repeal. Importantly, the UA differential β_2 —the parameter that identifies the effect—shows no discernible differential pre-Repeal trend, supporting the causal interpretation of our findings.³¹ Additionally, Appendix Table B.23 and Appendix Table B.24 show that our results are robust to adding state-by-year or sector-by-year fixed effects to flexibly control for any state-level or 2-digit-sector-level policy changes or shocks. Appendix Table B.27 and Appendix Table B.28 further report the imputation and Stacked DiD estimators, which address potential biases in staggered DiD: both deliver point estimates close to the baseline, though with less precision under

³¹Individual leads are occasionally distinguishable from zero—two in the wage-bill panel and one in the employment panel—but they are small and negative, and so run against the post-Repeal effects rather than generating them. The non-UA series β_1 for employment drifts modestly downward before the reform, which does not enter β_2 .

state-clustered inference: the land and capital effects remain significant in both designs, sales in the stacked design, while the employment and wage-bill effects are marginal ($p \approx 0.05$) there and insignificant under imputation.

Manufacturing productivity. Having shown that the Repeal Act eased land-market frictions and enabled plants to expand production, we now turn to its implications for plant-level physical productivity (TFPQ). We measure TFPQ using several approaches, with our baseline residual constructed by adapting the production-function procedure of [De Ridder et al. \(2022\)](#), as described above. Column (2) of Table 3, which uses the same district, year, and size-by-year fixed effects as our baseline, documents a 13% increase in TFPQ for UA plants following the Repeal, suggesting a substantial improvement in physical productivity; the effect is 11% under the most saturated specification, which adds industry-by-year fixed effects (Column 3). This finding is robust across three alternative TFPQ measures (Appendix Table B.25) and a range of model specifications. In addition, the event-study estimates displayed in Appendix Figure B.12 reveal no evidence of differential pre-trends, reinforcing a causal interpretation of the productivity gains.

One may question why the estimated effects on physical productivity are so large, given that sales and input usage rose by similar magnitudes in Table 1. This finding is particularly striking in light of [Bollard et al. \(2013\)](#), who found little evidence of reform-led productivity growth in Indian manufacturing. A plausible explanation is that efficiency gains translate primarily into lower prices rather than higher revenues, dampening observed changes in revenue-based productivity (TFPR). As emphasized by [Garcia-Marin and Voigtländer \(2019\)](#), more efficient producers may optimally charge lower prices, which compresses TFPR even as physical productivity improves. If the Repeal reduced marginal costs by lowering effective land use costs—and these reductions were fully passed through to prices—TFPR would respond only weakly despite substantial gains in TFPQ.

Consistent with this mechanism, Appendix Table B.26 shows that TFPR increases only modestly following the Repeal, with economically small point estimates that are significant in only two of the three specifications (Columns 1–3), while average output prices—measured relative to comparable products—decline by roughly 10%, though this last estimate is significant only at the 10 percent level. One caveat governs how much this decomposition can attribute. Because TFPQ is constructed by netting the firm-level price index out of revenue-based productivity, the price decline and the TFPQ increase are two views of the same underlying variation, not independent confirmation that marginal costs fell; and relative unit values can decline with lower marginal costs, but also with changes in product mix or in

	(1)	(2)	(3)
	TFPQ	TFPQ	TFPQ
Repeal	−0.004 (0.077)	0.002 (0.073)	−0.020 (0.052)
Repeal × UA	0.118* (0.063)	0.121** (0.055)	0.104** (0.044)
Observations	365,284	365,225	365,211
R^2	0.488	0.489	0.532
District FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Size × Year FE	No	Yes	Yes
Ind. × Year FE	No	No	Yes

Note: The dependent variable is TFPQ, a production-function residual estimated within 2-digit industries by adapting the procedure of De Ridder et al. (2022). ‘Land Size×Year’ are district average pre-treatment land-size bins interacted with year; ‘Industry×Year’ are 2-digit-industry-by-year fixed effects. TFPQ is quantity-based, so it nets out the firm price index whose response is reported in Table B.26; the increase here and the price decline there are two views of the same variation, not independent evidence. Additional controls: urban dummy, ownership fixed effects, and 2-digit industry fixed effects. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table 3: Treatment effect of the Repeal on productivity

quality within product codes. What the estimates establish is the rise in our quantity-based productivity measure itself, together with a price–TFPR split that is consistent with cost reductions being passed through to prices. We also find no statistically significant change in markups, which is likewise consistent with pass-through to consumers rather than with the gains being retained by firms as higher rents.

4.3 Selection and innovation after the Repeal

Weeding out “dormant” plants. As documented by Akcigit et al. (2021), India’s manufacturing sector exhibits weak selection, allowing stagnant plants with limited growth potential to survive for many years. In line with this evidence, Chatterjee et al. (2023) show that exit rates among Indian manufacturing plants are markedly lower than even those observed in other developing economies. By restricting land transfers, the ULCRA likely amplified the

costs of divestment and exit, further entrenching productive resources in unprofitable—or even dormant—plants (Panagariya, 2010). In this section, we put Panagariya’s conjecture to a formal test by examining how the Repeal affected plant exit behavior.

A valuable feature of the ASI data is that, since 1998, it distinguishes between plants that are no longer producing but still exist—which we classify as *dormant*—and plants that have ceased to exist and are removed from the survey—which we classify as *exits*. Appendix C.2 provides the exact status-code definitions. This measurement window carries an important limitation for what follows. Because the status codes that identify exit begin only in 1998, and because a plant cannot be observed exiting in the first year of the panel, no informative pre-Repeal exit variation exists for the seventeen states and union territories that repealed the Act in 1999: exit is zero by construction in the single pre-Repeal year available for that cohort. A parallel-trends check on exit is therefore unavailable for those states. Our estimates below rely on the design we use throughout, which compares UA with non-UA plants as repeal timing varies across states. Within that design, the identifying variation for exit comes disproportionately from the later-repealing cohorts, which are the ones with observable pre-Repeal years.

Table 4 reports these estimates, using our baseline empirical equation. Column (1) indicates that, after the Repeal, exit rates among UA plants rose by about 1.1 percentage points relative to non-UA plants (β_2)—roughly 23% of the sample mean exit rate of 4.7%. The response is much stronger among plants that had ever been in dormancy (Column (3)), which exhibit a 3.5 percentage point increase in exit rates, or roughly 53% of their mean exit rate of 6.6%. By comparison, exit responses among non-dormant plants (Column (2), about 1.1 percentage points) are similar to those in the overall sample.³² That the response is three times larger among plants with a history of dormancy is consistent with Panagariya’s conjecture: the additional exit fell disproportionately on plants that had at some point stopped producing while continuing to hold land and buildings.³³ Because the ever-dormant indicator classifies plants by their entire observed history—including post-Repeal years—the split in Columns (2)–(3) is a descriptive decomposition rather than a comparison across predetermined groups.³⁴ Column (4) limits how much reclassification the Repeal itself could have

³²Appendix Table B.31 shows that this non-dormant estimate is stable as the specification is saturated, moving from 1.1 to 0.8 percentage points and remaining significant at the 5 percent level once size-by-year and industry-by-year fixed effects are added.

³³Disproportionately, but not exclusively: plants with a dormancy history are only 13% of the sample, so in absolute terms the majority of the additional exits still occur among plants that were never recorded as dormant.

³⁴A check classifying dormancy strictly before repeal among late-treated states preserves the ordering, though inference is limited by the small number of treated states; see Appendix C.2.

induced, by using dormancy as the dependent variable. The estimate is economically small and statistically indistinguishable from zero (-0.002 , standard error 0.005 , against a base dormancy rate of 4.4%), so we find no evidence that the Repeal changed the probability that a plant *becomes* dormant; what it changed is the rate at which plants already in that state leave the market.³⁵ The reform therefore operated on the exit margin rather than on the incidence of dormancy, which is what the resource-release interpretation requires.³⁶

Appendix Figure B.13 provides descriptive corroboration from a different contrast built on the same repeal timing—comparing exit rates in states that repealed in 1999 with those in Andhra Pradesh and Maharashtra, the last large states to repeal. Exit rates run persistently higher in the early-repealing states through the 2000s and the differential closes after 2009, once the late-repealing states have themselves liberalized.³⁷

These exit responses also vary systematically with pre-Repeal profitability. Appendix Figure B.15 shows that plants with lower profitability prior to the liberalization experienced significantly larger increases in exit rates after the Repeal. Taken together, these results provide strong evidence that the Repeal intensified the process of creative destruction by easing the exit of unprofitable and dormant firms, in line with Panagariya’s observation.

Creative destruction through product turnover. Beyond plant exit, the Repeal also fueled business dynamism among surviving plants through product innovation. We measure the latter using the ASI’s detailed product-level data.³⁸ In particular, we construct two plant-year measures of product innovation: (i) the number of new products added to a plant’s portfolio and (ii) the number of discontinued products.³⁹ In our data, more than 50%

³⁵Column (4) excludes the 2010 and 2011 rounds, in which the ASI did not record the dormancy codes and the indicator is identically zero for every plant; retaining them would impose a mechanical zero on part of the post-Repeal window.

³⁶The estimation sample is restricted to plants with the landholding and employment information required by our controls, which excludes many non-reporting plants; its exit rate (4.7%) is accordingly lower than the rate in the unrestricted plant panel (about 9%). The estimates should therefore be read as effects among plants observed in production, a conservative sample for detecting exit.

³⁷Appendix Figure B.14 shows the corresponding series for dormancy rates. We treat both as suggestive context rather than as tests: they pool UA and non-UA plants, they use only two control states—the same two that supply much of the regression’s identifying pre-Repeal variation—and, for the reason given above, the exit figure cannot display a pre-Repeal year for the early cohort. Dormancy, by contrast, is observed contemporaneously from 1998, so the dormancy figure does normalize on a pre-Repeal year.

³⁸The ASI dataset provides product-level sales and quantities, as mandated by the 1956 Companies Act. Products are categorized by 5-digit product codes, with over 12,000 distinct products in our sample, differentiating items like men’s shirts (28232) from women’s shirts (28234).

³⁹Our measure of product innovation has several advantages over more conventional measures of innovation, such as patent counts and R&D spending, which are rarely pursued by firms in developing countries. For example, the ASI only began including questions about R&D activities in 2015, and merely 3% of plants reported having spent anything on R&D.

	Exit			Dormant
	(1)	(2)	(3)	(4)
	Overall	Non-dormant	Dormant	All plants
Repeal	−0.005 (0.008)	−0.007 (0.008)	−0.033 (0.019)	0.009 (0.005)
Repeal × UA	0.011** (0.004)	0.011** (0.004)	0.035** (0.016)	−0.002 (0.005)
Observations	287,316	249,183	38,128	253,368
R^2	0.060	0.055	0.154	0.037
Year FE	Yes	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes
Size × Year FE	No	No	No	No
Ind. × Year FE	No	No	No	No

Note: In columns (1)–(3) the dependent variable is a plant exit dummy (ASI ‘status of units’ codes 4, 6, 11, or 12): column (1) uses all plants, column (2) restricts to plants never flagged as dormant, and column (3) to plants ever flagged as dormant (status codes 2 or 3). Column (4) instead uses dormancy itself as the dependent variable, and asks whether the Repeal changed the probability of *being* dormant rather than of exiting; it excludes the 2010 and 2011 rounds, in which the ASI did not record the dormancy codes and the indicator is identically zero for every plant. Controls include a rural dummy, 2-digit industry fixed effects, firm age fixed effects, and landholding quartile-by-year fixed effects. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table 4: Treatment effect of the Repeal on plant exit and dormancy

of plants never introduce a new product over their entire life cycle. This is consistent with estimates in [Akcigit et al. \(2021\)](#) and [Peters and Zilibotti \(2021\)](#), which suggest that the majority of Indian plants are subsistence plants with low growth potential.

Table 5 reports the impact of the Repeal on product innovation in UA and non-UA districts, estimated by PPML on the 1995–2009 product panel using our baseline controls. Additions and deletions are constructed from the five-digit ASICC product schedules: a product counts as added in its first year of appearance and as deleted in its last, excluding a plant’s own first and last years in the product data, so a plant’s entry into or exit from the panel never mechanically registers its entire portfolio as turnover. Product shedding in the years before a plant exits remains recorded, as a real part of the exit process. The Repeal led to a 14% increase in the addition of new products (Column 1) and a 15% increase in the deletion of old products (Column 2) among plants in UA districts, consistent with intensified

	(1)	(2)
	#New Products	#Del. Products
Repeal	0.004 (0.049)	-0.035 (0.085)
Repeal $\times$ UA	0.132*** (0.051)	0.140** (0.058)
Observations	264,437	272,666
Year FE	Yes	Yes
District FE	Yes	Yes
Size $\times$ Year FE	Yes	Yes

Note: #New Products and #Del. Products are counts of 5-digit products added to / removed from a plant’s portfolio. Both columns are PPML estimates, based on data from 1995–2009. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table 5: Treatment effect of the Repeal on creative destruction

creative destruction.⁴⁰ Appendix Table B.32 shows that these effects remain positive under a more conservative definition of products based on 4-digit codes,⁴¹ suggesting that the effects reflect meaningful product turnover rather than reporting noise.

These aggregate responses combine within-plant portfolio changes with shifts in the composition of operating plants. Appendix Table B.33 separates the two by re-estimating both margins with plant fixed effects, restricting identification to within-plant variation among plants that ever innovate. The addition response is robust within plants (+0.173, $p = 0.006$): continuing plants genuinely refreshed their portfolios, rather than the aggregate reflecting composition alone. The deletion response turns negative (-0.142, $p = 0.015$) for two Repeal-driven reasons. First, deletions are recorded only while a plant remains in the sample, and the plants hit hardest by post-Repeal destruction exit quickly—so their largest deletion years are missing, while their pre-Repeal histories are complete. Second, continuing plants delete less by choice: once the land wedge no longer taxes scale, marginal product lines are worth keeping. Appendix C.2 details the reconciliation.

In contrast, Appendix Table B.34 shows no effect of the Repeal on the total number of products produced by a firm, suggesting that the liberalization reshaped product portfolios

⁴⁰Appendix C.2 details the construction and reconciles the baseline and within-plant estimates.

⁴¹Using 4-digit codes mitigates concerns that measurement error or minor misclassification—such as a change from men’s shirts (28232) to women’s shirts (28234)—may be spuriously recorded as innovation. Products defined at the 4-digit level typically correspond to distinct subclasses produced with different techniques (e.g., knitted versus non-knitted apparel), and thus capture more substantive product changes.

through higher turnover rather than through an expansion in product scope. Event-study estimates in Appendix Figure B.16 are consistent with these estimates, though noisier; Appendix B.2 reports the corresponding joint pre-trend tests. Overall, the evidence suggests that the land market liberalization increased business dynamism by inducing firms to refresh their product portfolios more frequently, adding new products while shedding old ones.

Entry and the selection of dynamic plants. The effect of the Repeal on plant entry is theoretically ambiguous. On the one hand, by easing land market restrictions, the Repeal reduced barriers to starting and scaling production, potentially attracting entrants with higher growth potential. At the same time, the intensification of creative destruction increased competitive pressures and the associated risk of failure, which may have deterred entry by less productive plants.

To study the impact of the Repeal on entry, we aggregate the ASI to the district-year level using sampling weights and estimate our DiD specification using Poisson Pseudo-Maximum Likelihood (PPML), which is well suited to count data with a high prevalence of zeros; district-year observations with fewer than 30 plants are excluded.⁴² Consistent with the presence of offsetting forces, the Repeal has no statistically significant effect on the *total* number of new entrant plants (Table 6, Column 1). Beneath this aggregate null, however, the Repeal reshaped the *composition* of entrants toward more dynamic plants.

We classify entrants by their *realized* dynamism—as *dynamic* if they introduced any new product over their life cycle, and as *stagnant* otherwise. Table 6 shows that the number of dynamic entrants rises significantly in UA districts ($\text{Repeal} \times \text{UA} = 0.243$, Column 3), while total and stagnant entry are unchanged (Columns 1–2); the corresponding event study appears in Appendix Figure B.17. The Repeal thus shifted the composition of new entrants toward greater dynamism, drawing in more dynamic plants without changing the total inflow.⁴³

Because realized dynamism is measured over a plant’s life cycle and could in principle

⁴²This size filter uses contemporaneous plant counts, which the Repeal could in principle move across the threshold. Fixing each district in or out of the sample by its pre-repeal (1993–97) mean plant count instead moves 665 of 7,608 district-year cells and leaves the results essentially unchanged: the dynamic-entry estimate below becomes 0.228 (standard error 0.115, $p = 0.048$, $N = 5,141$) against 0.243 ($p = 0.035$, $N = 5,198$), and total and stagnant entry remain null.

⁴³The null on *total* entry and the positive effect on *dynamic* entry are mutually consistent within the ASI sample: dynamic entrants account for about 26% of entry, so the 0.243 log-point increase in that component implies a rise of roughly 7% in total entry—comfortably inside the confidence interval of the Column (1) estimate (−0.020, standard error 0.125). The ASI covers the organized manufacturing sector only, so entry by informal plants and household enterprises is outside the scope of the analysis.

	(1)	(2)	(3)
	All	Stagnant	Dynamic
Repeal	0.054 (0.118)	−0.047 (0.133)	0.056 (0.114)
Repeal × UA	−0.020 (0.125)	−0.028 (0.163)	0.243** (0.115)
Observations	5,410	5,379	5,198
Year FE	Yes	Yes	Yes
District FE	Yes	Yes	Yes
Dist. size × Year FE	Yes	Yes	Yes
Private share × Year FE	Yes	Yes	Yes

Note: District-year Poisson Pseudo-Maximum Likelihood (PPML) estimates. The district-year entry counts are population-projected sums constructed with the ASI sampling multipliers; the PPML step itself is unweighted. District-years with fewer than 30 plants are excluded. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table 6: Treatment effect of the Repeal on plant entry

respond to the Repeal itself, we corroborate this result with a predicted measure whose classification rule is fixed before the reform. We train a LASSO on pre-Repeal data to predict dynamism from plant characteristics observed at entry—ownership and organizational form, importing activity, computer use, and the number of initial products, together with their interactions and quadratic terms—and this predicted measure closely tracks realized dynamism both before and after the Repeal (Appendix Figure B.18).⁴⁴ Sorting entrants into quartiles of predicted dynamism yields the same qualitative pattern: entry declines among the least dynamic quartile, changes little in the middle quartiles, and rises among the most dynamic entrants (Appendix Figure B.19), and a binary LASSO classification confirms it (Appendix Table B.35). Both the realized and predicted classifications thus deliver the same conclusion—that the Repeal draws in more dynamic entrants—mitigating concerns that the result is an artifact of an endogenous ex-post classification.

⁴⁴Training on pre-Repeal data fixes the prediction rule, though the predictors are measured at entry and can themselves respond to the reform for post-Repeal entrants; the exercise therefore rules out classification based on realized post-entry outcomes, not treatment-induced changes in entrant characteristics.

4.4 Further remarks on the impacts of the Repeal

While our empirical analysis in this paper focuses on the impacts of the Repeal Act on manufacturing firms, the ramifications of this legislative change extended well beyond the manufacturing sector. The ULCRA’s regulation of urban land markets might influence aggregate productivity through both agglomeration and congestion forces, which have been extensively discussed in the urban literature. Furthermore, the expansion of urban manufacturing firms could potentially act as a catalyst for rural-to-urban migration and structural changes, thereby reshaping the spatial distribution of labor. Apart from these economic implications, the redevelopment of urban land holds significant relevance for the living conditions of urban inhabitants. For instance, [Gechter and Tsivanidis \(2023\)](#) provides a case study of the redevelopment of 60 textile mills in Mumbai. Their study highlights that while such urban redevelopment projects can benefit the city’s residents by increasing housing and floor space supply, they can concurrently impose substantial costs on slum dwellers who are forcibly relocated. These dimensions present interesting avenues for future research, yet fall outside the scope of this study.

Socioeconomic impacts of the Repeal. For interested readers, Appendix Section [D](#) documents the broader socioeconomic effects of the Repeal using village-level employment and amenity data from three rounds of the Population Census, household wealth data from the 2011 Socio-Economic and Caste Census, and household expenditure data from the National Sample Surveys. The evidence shows that the Repeal accelerated the reallocation of agricultural labor into non-agricultural sectors. Notably, villages even 25 kilometers away from UAs experienced significant structural changes after the Repeal.

At the same time, we find no evidence of short- to medium-term improvements in local amenities—such as roads, electricity, or schools—around Urban Agglomerations. The null effects on amenities also mitigate concerns that our firm results are driven by infrastructure investments made after the Repeal. Additionally, we find no post-Repeal increase in residential rent levels, and at most weak evidence of a rise in the housing-services expenditure share ([Appendix D.6](#)). In the single round of the Socio-Economic and Caste Census, which permits only a cross-sectional comparison of municipalities closer to and farther from UAs rather than our difference-in-differences design, we likewise find no deterioration in housing conditions, and we find higher household incomes and lower poverty rates closer to UAs.

Jawaharlal Nehru National Urban Renewal Mission (JNNURM). One potential confounding policy is the Jawaharlal Nehru National Urban Renewal Mission (JNNURM), a nationwide urban development initiative launched by the federal government from December 2005 to March 2011. The JNNURM had two missions: Urban Infrastructure and Governance (UIG), which aimed to improve water supply and sanitation, solid waste management, road networks, urban transport, and the redevelopment of old city areas, and Basic Services to the Urban Poor (BSUP), which focused on slum development. One of the conditions for states to receive JNNURM funding was the repeal of the ULCRA if it had not already been repealed. The JNNURM targeted 65 cities in total, which included cities with populations exceeding one million in 2001, as well as cities of religious, historic, and touristic significance. The list of JNNURM cities had a partial overlap with the UAs affected by the ULCRA.

This overlap raises the concern that our estimates may reflect the effects of JNNURM-funded infrastructure investments rather than land market liberalization. However, JNNURM implementation progressed slowly over our study period, which ends in 2013. According to the government’s audit report, by 2011, only 17.8% of approved UIG projects and 1.5% of BSUP projects had been completed ([Comptroller and Auditor General of India, 2012](#)). Using the mission’s detailed implementation data, we assess the sensitivity of our estimates to JNNURM exposure—excluding the JNNURM-covered urban agglomerations and controlling for measured implementation intensity. These exercises check the stability of the Repeal coefficient rather than estimate the JNNURM’s own effects; as discussed in [Appendix F](#), they indicate that JNNURM-funded investment is unlikely to bias our production results.

5 Theoretical framework

In this section, we present a general equilibrium model of firm growth through product innovation with land market frictions. The model builds on the framework of [Klette and Kortum \(2004\)](#) and introduces frictions to land transactions that distort firms’ expansion strategies.

5.1 The economic environment

Preferences. Consider a continuous-time economy in which time is indexed by $t \in \mathbb{R}_+$. The economy is populated by an infinitely-lived representative household of unit measure

with logarithmic preferences over aggregate consumption C_t :

$$U_0 = \int_0^\infty e^{-\rho t} \ln(C_t) dt \quad (2)$$

where $\rho > 0$ is the household's rate of time preference. This household inelastically supplies $H > 0$ unit of labor and $L > 0$ units of land per period.

Production technology. The economy is composed of a final sector producing a manufacturing final good according to a Cobb–Douglas technology that combines differentiated products indexed by $i \in [0, 1]$:

$$\ln(Y_t) = \int_0^1 \ln(y_{it}) di \quad (3)$$

where y_{it} is the quantity of product i used in production. Each product is manufactured by a single firm using labor and land according to the following Cobb–Douglas technology:

$$y_{it} = z_{it} h_{it}^\alpha \ell_{it}^{1-\alpha} \quad (4)$$

where z_{it} is the process efficiency at which product i is manufactured, h_{it} and ℓ_{it} are the quantity of labor and land used in production, respectively, and $\alpha \in (0, 1)$.

Expansion technology. A firm $f \in \mathcal{F}_t$ consists of a portfolio of products collected in the set $\mathcal{P}_{ft}$ with cardinality $n_{ft} = |\mathcal{P}_{ft}| \in \mathbb{N}$ and corresponding process efficiencies $\{z_{it}\}_{i \in \mathcal{P}_{ft}}$. Firms can grow over time by adding more products to their portfolio. Specifically, a firm holding n_{ft} products must allocate y_{ft}^X units of final goods to expand into new product lines at rate X_{ft} . The expansion technology is given by:

$$y_{ft}^X = \frac{X_{ft}^2 Y_t}{2\theta_X n_{ft}} \quad (5)$$

where $\theta_X > 0$ measures the efficiency of the expansion technology of incumbent firms. In this specification, the quadratic cost of expansion is scaled by the firm's current number of active product lines. This scaling reflects the idea that larger firms may leverage their organizational infrastructure to expand more efficiently into new markets. Upon successful expansion, the newly operational product line displaces a randomly selected incumbent product through creative destruction. That new product improves on the displaced incumbent's process efficiency by a factor of $\lambda > 1$.

Entry, exit, and firm type. At every instant, a unit mass of potential entrants direct Y_t^E units of final goods toward entry to achieve an entry rate of e_t :

$$Y_t^E = \frac{e_t^2 Y_t}{2\theta_E} \quad (6)$$

where $\theta_E > 0$ measures the efficiency of the entry technology. Upon successful entry, each entrant endogenously selects a type that determines its growth prospects. There are two types of firms: *dynamic* firms have access to the expansion technology in equation (5) and can grow into multi-product firms, while *stagnant* firms operate a single product line for the remainder of their lifecycle. To achieve a probability d_t of drawing the dynamic type, an entrant must pay a type-selection cost of Y_t^T units of final goods, given by:

$$Y_t^T = -\frac{[\ln(1 - d_t) + d_t]Y_t}{\theta_T} \quad (7)$$

where $\theta_T > 0$ determines the efficiency of the type-selection technology. The cost of type selection is zero when $d_t = 0$ and diverges as $d_t \rightarrow 1$. When an entrant successfully introduces a new product, it displaces a randomly selected incumbent product through creative destruction and improves on its process efficiency by the same factor of $\lambda > 1$. Finally, firms exit when they lose all their products to creative destruction.

Resource constraints. Land is in fixed (inelastic) supply:

$$\int_0^1 \ell_{it} di \leq L. \quad (8)$$

The resource constraint for labor is given by:

$$\int_0^1 h_{it} di \leq H. \quad (9)$$

The final-good resource constraint is:

$$C_t + Y_t^X + Y_t^E + e_t \cdot Y_t^T \leq Y_t \quad \text{where} \quad Y_t^X \equiv \int_{f \in \mathcal{F}_t} y_{ft}^X df. \quad (10)$$

5.2 The market equilibrium allocation

We now characterize the agents’ decision problems, which jointly determine equilibrium prices and quantities on the product, labor, and land markets. While the labor and land markets are perfectly competitive, product markets feature monopolistic competition.

Representing the ULCRA. A literal representation of the ULCRA as a binding ceiling on landholdings would be misleading—though not because the ceiling left manufacturing untouched. It applied to *vacant* land, which excluded a plant’s buildings but not its yard, its open storage, or the land it held for expansion, and the thresholds were low enough that a working site of any size exceeded them.⁴⁵ What made the ceiling something other than a hard cap was enforcement: by 1999 only 20,654 hectares had been acquired nationally, and firms routinely circumvented the ceiling through bribes and political connections. The constraint bound in law and was evaded at a price, which is why we represent it as a wedge on the cost of land rather than as a limit on the quantity a firm may hold.

The more economically significant friction came from the Act’s *transfer restrictions*. Vacant land could not be sold without first notifying the authorities, who could then pre-empt the sale at statutory rather than market prices, and land held above the ceiling could not be transferred at all until the holder’s excess had been declared and notified—a process that was slow and heavily litigated. Acquiring land carried the mirror constraint, since a purchase that took a firm above the ceiling exposed the excess to acquisition.⁴⁶ These provisions exposed sellers to delay and to expropriation at below-market compensation. The ceiling compounded them by giving officials bureaucratic leverage: the low statutory thresholds (500 to 2,000 square meters) and the ambiguous definition of “vacant” land provided a legal pretext to scrutinize firms, initiate acquisition proceedings, and extract rents.

We capture these frictions as a size-dependent wedge $\tau(n) \geq 0$ on the effective rental cost of land for a firm operating n product lines. A firm with n products pays a factor $1 + \tau(n)$ above the market rental rate per unit of land. This wedge captures approval delays, mandatory disclosures, expropriation risk, and the corruption costs that pervaded land transactions under the Act. Both dynamic and stagnant firms face this wedge, but since stagnant firms always hold a single product, they face $\tau(1)$. The government collects the land tax $\tau(n)q_t\ell_{it}$

⁴⁵See Section 2. Private industrial firms had no exemption as of right, and sought relief instead through the discretionary and revocable exemptions of Section 20.

⁴⁶Section 2 gives the statutory detail. The provision often described as the Act’s central transfer restriction, the prior-permission requirement of Section 27, was inoperative during our sample: the Supreme Court struck it down in 1980 for land within the ceiling area, and it was in any case time-limited by its own terms.

and rebates the proceeds lump-sum to the household.⁴⁷

The wedge is increasing in n , reflecting the fact that the ULCRA's regulatory burden fell disproportionately on larger firms. The size gradient comes from the ceiling rather than from the notification requirement: notifying a transfer costs the same whatever the holding, whereas the ceiling machinery bites only once a firm's vacant landholdings approach and exceed the statutory threshold, at which point new acquisitions become liable to acquisition and existing holdings cannot be freely transferred. That burden therefore scales with a firm's land footprint—and so with its product scope, since each product line requires land as a factor of production. Because enforcement was weak and evasion common, the constraint operated as a rising cost of holding and transacting land rather than as a hard cap, which is what a smooth wedge is meant to capture. We parameterize this wedge as a saturating function $\tau(n) = \bar{\tau}(1 - e^{-\kappa n})$, where $\bar{\tau} > 0$ is the asymptotic wedge level and $\kappa > 0$ governs the rate of saturation. This functional form nests a constant wedge ($\kappa \rightarrow \infty$) and ensures that the wedge is bounded. We restrict the parameter space so that $(1 + \bar{\tau})^{1-\alpha} < \lambda$. Because $\tau(n) < \bar{\tau}$ for every finite n , this sufficient global condition ensures that the leader's technological advantage exceeds the largest possible land-cost penalty.

The household's problem. Taking prices as given, the household's problem is to choose a time path of consumption to maximize lifetime utility:

$$\max_{\{C_t\}_{t \geq 0}} \int_0^\infty e^{-\rho t} \ln(C_t) dt \quad \text{s.t.} \quad \dot{A}_t + P_t C_t \leq r_t A_t + w_t H + q_t L + B_t^G$$

where P_t denotes the price of the final good, w_t the wage, q_t the land rental rate, A_t is the value of corporate assets, and r_t is the rate of return on those assets:

$$A_t = \int_{f \in \mathcal{F}_t} V_{ft} df \quad \text{where} \quad \lim_{t \rightarrow \infty} e^{-\int_0^t r_{t'} dt'} A_t = 0.$$

Here, V_{ft} denotes the value of firm f . This problem delivers the usual Euler equation:

$$\dot{C}_t = (r_t - \rho)C_t.$$

The final sector's problem. The final good is produced competitively according to the technology in equation (3). Taking prices as given, the final sector chooses its demand for

⁴⁷We also report an alternative fiscal closure in which the tax revenue finances wasteful government expenditures.

each product to maximize profits:

$$\max_{\{y_{it}\}_{i \in [0,1]}} P_t Y_t - \int_0^1 p_{it} y_{it} di \quad \text{s.t.} \quad \ln(Y_t) = \int_0^1 \ln(y_{it}) di.$$

This problem delivers the following demand schedule for each product:

$$y_{it} = \frac{P_t Y_t}{p_{it}}, \quad \forall i \in [0, 1]$$

where the price of the final good is given by:

$$\ln(P_t) \equiv \int_0^1 \ln(p_{it}) di.$$

The final good is chosen as the numéraire such that $P_t = 1$ for all t .

The firm's problem. Each product line features a competitive fringe that uses the same production technology (4) as the leader but at its previous process efficiency level (i.e., the fringe's process efficiency for product i is z_{ift}/λ). The fringe is a reduced-form outside option used only to determine the leader's limit price: it is not an active firm type in the model's firm distribution, is distinct from stagnant firms, and is not directly mapped to NSS establishments. We maintain the assumption that this outside option faces no ULCRA wedge, capturing small producers that operate on existing plots and need not acquire additional land through the regulated transfer process. Stagnant firms, by contrast, operate active product lines and face $\tau(1)$. The fringe prices competitively, so the leader captures the entire market by pricing just below the fringe's marginal cost:

$$p_{ift} = \frac{\lambda \varsigma_t}{z_{ift}} \quad \text{where} \quad \varsigma_t \equiv \left(\frac{w_t}{\alpha}\right)^\alpha \left(\frac{q_t}{1-\alpha}\right)^{1-\alpha}.$$

The parameter restriction $(1 + \bar{\tau})^{1-\alpha} < \lambda$ implies that the leader's unit cost lies strictly below this limit price for every firm size, validating this allocation and ensuring positive operating profits. The final sector's Cobb–Douglas aggregator implies that every product commands the same revenue. Since the leader's own unit cost is $\varsigma_t[1 + \tau(n_{ft})]^{1-\alpha}/z_{ift}$, where n_{ft} is the number of products held by the leading firm, per-product flow profit is:

$$\pi_t(n_{ft}) = Y_t \left[1 - \frac{T(n_{ft})}{\lambda}\right] \quad \text{where} \quad T(n_{ft}) \equiv [1 + \tau(n_{ft})]^{1-\alpha}.$$

Since $T(n_{ft}) \geq 1$ for all n_{ft} while the fringe is undistorted, the wedge strictly reduces per-product profits for every firm, including stagnant firms, which face $\tau(1) > 0$. Recall that the wedge $\tau(n)$ is parameterized as a saturating function:

$$\tau(n) = \bar{\tau}(1 - e^{-\kappa n}).$$

That is, the wedge rises steeply for small firms and saturates near $\bar{\tau}$ for large ones. Consequently, the profit penalty $T(n)/\lambda$ is increasing in n —each additional product line lowers the per-product margin—so the ULCRA acts as a progressive tax on product scope.

Let δ_t denote the aggregate rate of creative destruction. The Hamilton–Jacobi–Bellman (HJB) equation for a dynamic firm with $n \geq 1$ products is:

$$\begin{aligned} r_t V_t^D(n) &= n\pi_t(n) + \dot{V}_t^D(n) \\ &\quad + \max_{X_t \geq 0} \{X_t[V_t^D(n+1) - V_t^D(n)] - X_t^2 Y_t / (2\theta_X n)\} \\ &\quad - n\delta_t[V_t^D(n) - V_t^D(n-1)] \end{aligned}$$

with boundary condition $V_t^D(0) = 0$. The three terms on the right-hand side are (1) flow profits plus capital gains, (2) the option value of expansion net of its cost in final goods, and (3) the expected loss in value from creative destruction. The first-order condition for the per-product expansion rate $x_t(n) \equiv X_t/n$ yields:

$$x_t(n) = \frac{\theta_X [V_t^D(n+1) - V_t^D(n)]}{Y_t}. \quad (11)$$

By contrast, a stagnant firm holds a single product ($n = 1$), has no access to the expansion technology, and has value V_t^S satisfying:

$$(r_t + \delta_t)V_t^S = \pi_t(1) + \dot{V}_t^S.$$

The entrant’s problem. Potential entrants direct final goods toward entry to maximize expected profits net of entry costs:

$$\max_{e_t \geq 0, d_t \in [0,1)} \{e_t[d_t V_t^D(1) + (1 - d_t)V_t^S - Y_t^T] - Y_t^E\}.$$

The first-order condition for the dynamic share d_t balances the value premium of being dynamic against the type-selection cost:

$$d_t = \frac{\theta_T[V_t^D(1) - V_t^S]}{\theta_T[V_t^D(1) - V_t^S] + Y_t}.$$

Since the ULCRA depresses the value of dynamic firms more than that of stagnant firms (dynamic firms expect to grow into higher- n states where the wedge binds more severely), the premium $V_t^D(1) - V_t^S$ is compressed under the Act, reducing d_t and thereby the expected value of entry. The first-order condition for the entry rate e_t equates the marginal entrant's expected value to its marginal cost:

$$e_t = \theta_E \cdot \left\{ \frac{d_t V_t^D(1) + (1 - d_t) V_t^S}{Y_t} + \frac{\ln(1 - d_t) + d_t}{\theta_T} \right\}.$$

Equilibrium definition. Now that all decision problems have been described, we can define the concept of a market equilibrium allocation.

Definition 5.1. Given initial conditions for the process efficiency of each product $\{z_{i0}\}_{i \in [0,1]}$, the distribution of product across dynamic firms $\{m_0^D(n)\}_{n \geq 1}$, and the measure of stagnant firms M_0^S , a market equilibrium allocation consists of time paths for allocations, prices, and the distributions $\{m_t^D(n)\}_{n \geq 1}$, M_t^S , and $\phi_t(s)$ such that the following conditions hold:

1. $\{C_t\}_{t \geq 0}$ solve the household's problem.
2. $\{\{y_{it}\}_{i \in [0,1]}\}_{t \geq 0}$ solve the final sector's problem.
3. $\{\{\{h_{ift}, \ell_{ift}, p_{ift}\}_{i \in \mathcal{P}_{ft}}, X_{ft}\}_{f \in \mathcal{F}_t}\}_{t \geq 0}$ solve the firm's problem.
4. $\{e_t, d_t\}_{t \geq 0}$ solve the entrant's problem.
5. The dynamic-firm distribution $\{m_t^D(n)\}_{n \geq 1}$, the measure of stagnant firms M_t^S , and the log-productivity density $\phi_t(s)$ evolve according to the Kolmogorov–Forward equations (A.31)–(A.33) and (A.38).
6. $\{\{p_{it}\}_{i \in [0,1]}\}_{t \geq 0}$ clear the product markets.
7. $\{w_t\}_{t \geq 0}$ clear the labor market.
8. $\{q_t\}_{t \geq 0}$ clear the land market.
9. $\{r_t\}_{t \geq 0}$ clear the asset market.

10. The government's budget is balanced each instant.

Aggregate output and misallocation. Despite the rich heterogeneity in firm size and land distortion exposure, the market equilibrium allocation of our model admits a clean aggregation. As derived in appendix A.2, aggregate output at every instant satisfies:

$$Y_t = \frac{Z_t H^\alpha L^{1-\alpha}}{\mathcal{T}_t} \quad \text{where} \quad \mathcal{T}_t \equiv \mathcal{H}_t^\alpha \cdot \mathcal{L}_t^{1-\alpha} \quad (12)$$

where Z_t is the geometric mean of process efficiencies across the product continuum, which we refer to as the economy's total factor productivity (TFP):

$$\ln(Z_t) \equiv \int_0^1 \ln(z_{it}) di.$$

The second term, $\mathcal{T}_t$, is an aggregate misallocation wedge that satisfies $\mathcal{T}_t \geq 1$ with equality if and only if τ is identical across all product lines with positive product weight (see appendix A.2). This wedge is composed of two product-weighted averages:

$$\begin{aligned} \mathcal{H}_t &\equiv \sum_{n \geq 1} [1 + \tau(n)]^{1-\alpha} n m_t^D(n) + [1 + \tau(1)]^{1-\alpha} M_t^S \\ \mathcal{L}_t &\equiv \sum_{n \geq 1} [1 + \tau(n)]^{-\alpha} n m_t^D(n) + [1 + \tau(1)]^{-\alpha} M_t^S. \end{aligned}$$

In the absence of the ULCRA (when $\tau(n) = 0$ for all n), the aggregate wedge equals one, so aggregate output reaches the productivity frontier $Z_t H^\alpha L^{1-\alpha}$. The structure of the wedge reveals the source of the factor-misallocation output loss: if τ were identical across all firms, then $\mathcal{T}_t = 1$ regardless of its level, because the exponents on $1 + \tau$ cancel exactly. This cross-sectional distortion arises entirely from the *size-dependence* of $\tau(n)$: larger firms face a higher effective land cost and use too little land per product, while smaller firms use too much. Since $\mathcal{T}_t$ depends on the current firm-size distribution, this aggregate distortion evolves endogenously as the distribution changes. A uniform positive wedge therefore creates no static factor-misallocation loss, although it still reduces leaders' profit margins relative to the undistorted fringe and can affect innovation and entry.

TFP growth and creative destruction. Having identified Z_t as TFP, we now characterize its growth rate. At every instant, the rate of creative destruction (the rate at which

existing products are displaced by innovations) is the sum of incumbent expansion and entry:

$$\delta_t = \sum_{n=1}^{\infty} x_t(n) n m_t^D(n) + e_t = \bar{x}_t^D N_t^D + e_t \quad (13)$$

where $\bar{x}_t^D$ is the product-weighted mean expansion rate of dynamic firms, N_t^D is the share of products held by dynamic firms, and e_t is the entry rate. As shown in appendix A.2, each displacement raises process efficiency by a factor of λ , so growth rate of TFP satisfies:

$$g_t = \delta_t \cdot \ln(\lambda) \quad (14)$$

at all times. This decomposition highlights the two margins through which the ULCRA may suppress growth. First, the land distortion reduces per-product profits:

$$\pi_t(n) = Y_t \left[1 - \frac{T(n)}{\lambda} \right]$$

lowering the value of expanding into new product lines and thereby depressing the expansion rate $\bar{x}_t^D$. Second, lower firm values reduce the entry rate e_t and distort the composition of entrants toward the stagnant type.

Disciplining the land wedge with MRPLand and dynamic responses. The aggregate wedge $\mathcal{T}_t$ in equation (12) summarizes cross-sectional factor misallocation. Their common structural parameters $\bar{\tau}$ and κ are not directly observable. The model links these parameters to two distinct sets of empirical moments. The first is the marginal revenue product of land (MRPLand) difference-in-differences (DiD) coefficient estimated in Section 4. In the model, a product manufactured by a firm with n products uses:

$$\ell_t(n) = \frac{(1 - \alpha) Y_t}{\lambda q_t [1 + \tau(n)]^\alpha}$$

units of land (see appendix A.1), such that its log marginal revenue product of land is:

$$\ln(\text{MRPLand}) = \underbrace{\ln(\lambda q_t)}_{\text{absorbed by FEs}} + \alpha \ln[1 + \tau(n)].$$

In the DiD estimation exercise, the constant is absorbed by fixed effects and the treatment consists of eliminating the wedge. The regression's dependent variable is the within-industry-by-year residual of log MRPLand; because the model has no industry dimension, that cell

mean is a function of time alone and is absorbed by the same fixed effects, so the mapping below is unchanged (appendix A.5). Therefore, the model counterpart of the MRPLand DiD coefficient is:

$$\beta_{DiD} = -\alpha \cdot \mathbb{E}_{\text{firm}}[\ln(1 + \tau(n))] \quad (15)$$

where the expectation is firm-weighted among dynamic firms on the pre-repeal balanced growth path. Since the land share $1 - \alpha$ is externally calibrated, the MRPLand DiD estimate directly disciplines the average log-wedge $\mathbb{E}_{\text{firm}}[\ln(1 + \tau(n))]$, which in turn pins down its overall magnitude $\bar{\tau}$. MRPLand alone, however, does not determine how the wedge varies with firm size. That information comes from the joint response of product creation, product destruction, exit, and dynamic entry documented in Section 4. These moments help distinguish a roughly uniform decline in effective land costs from one that disproportionately relaxes constraints on more scalable firms. Within the model, they therefore help pin down how sharply the wedge rises with product scope, as governed by κ .

Welfare decomposition. The ULCRA reduces welfare through cross-sectional factor misallocation and through dynamic effects on R&D, entry, and TFP growth. Define $\mathcal{C}_t \equiv C_t/Y_t$ as the consumption share of output. Let γ denote the consumption-equivalent welfare gain, defined as the factor by which pre-repeal consumption must be permanently scaled to make the household indifferent between the pre-repeal stationary allocation and the post-repeal path. As a useful steady-state benchmark, comparing the pre- and post-repeal balanced growth paths yields (see appendix A.4):

$$\ln(\gamma) = \underbrace{\ln(\mathcal{T})}_{\text{Misallocation}} + \underbrace{\frac{g^{\text{post}} - g^{\text{pre}}}{\rho}}_{\text{Growth}} + \underbrace{\ln\left(\frac{\mathcal{C}^{\text{post}}}{\mathcal{C}^{\text{pre}}}\right)}_{\text{Consumption share}}. \quad (16)$$

The first term captures the static gain from eliminating cross-sectional factor misallocation, the second is the dynamic gain from permanently faster TFP growth, and the third reflects the change in final goods allocated to expansion and entry.⁴⁸

5.3 Calibration

The model features six structural parameters $(\lambda, \theta_X, \theta_E, \theta_T, \bar{\tau}, \kappa)$ that we jointly calibrate by matching ten empirical moments. Two additional parameters are set externally: the discount

⁴⁸If the tax proceeds instead financed wasteful government expenditures G_t , define their output share as $\mathcal{G}_t \equiv G_t/Y_t$. The welfare gain would then include the additional term $\ln[\mathcal{C}^{\text{pre}}/(\mathcal{C}^{\text{pre}} - \mathcal{G}^{\text{pre}})]$.

rate $\rho = 0.05$ and the labor share $\alpha = 0.81$, following [Sood \(2022\)](#) for Indian manufacturing. The model’s dynamic and stagnant firms are latent structural types, distinguished by whether they have access to the product-expansion technology. Our calibration adopts a dual-sample empirical mapping: we use organized-sector plants covered by the Annual Survey of Industries (ASI) as the empirical analogue of dynamic firms and combine ASI and National Sample Survey (NSS) data for economy-wide moments. This mapping is a sampling approximation rather than a one-to-one classification. In particular, a structurally dynamic firm need not realize a product addition during its observed life, and an ASI plant that does not add a product is not thereby classified as structurally stagnant. We estimate the six parameters by minimizing the weighted sum of squared percentage deviations between model and data moments. Appendix [A.5](#) derives the model counterpart of each targeted moment.

Targeted moments. The ten moments fall into two distinct groups. The first five are cross-sectional features of the pre-repeal economy. The economy-wide TFP growth rate of 3.5% is taken from [Bollard et al. \(2013\)](#) for 1980–1992. The average number of products per firm is computed twice: once across all firms (ASI and NSS combined, capturing both organized and unorganized manufacturing) and once among dynamic (ASI) firms only. The entry rate is the share of organized-sector ASI plants observed in their first year of operation, so it captures entry within registered manufacturing rather than economy-wide firm entry. The old-to-young ratio is the number of dynamic firms aged 40–45 relative to those aged 1–5, capturing the shape of the age distribution.

The remaining five moments are difference-in-differences coefficients from Section 4, measuring the causal effect of the Repeal on treated firms. The product creation and destruction DiDs capture the change in new products introduced and existing products discontinued, estimated by Poisson pseudo-maximum likelihood. The entry DiD measures the change in dynamic entry. The exit DiD captures the change in firm exit rates from a linear probability model. The MRPLand DiD measures the change in the log marginal revenue product of land.

Identification. Although all parameters are jointly calibrated, some moments are especially informative about particular parameters. The process efficiency step size $\lambda > 1$ scales TFP growth through $g = \delta \cdot \ln(\lambda)$. The expansion efficiency θ_X governs how many products dynamic firms accumulate, primarily disciplined by $\mathbb{E}[n \mid \text{dynamic}]$. The entry efficiency θ_E is disciplined by the entry rate, while the type-selection efficiency parameter θ_T shapes the age distribution through its effect on the dynamic share of entrants, as reflected in the

Parameter	Description	Value	Target / Source
<i>Panel A. Externally calibrated</i>			
ρ	Discount rate	0.05	Standard
$1 - \alpha$	Land share	0.19	Sood (2022)
<i>Panel B. Internally calibrated</i>			
λ	Innovation step size	1.231	TFP growth rate
θ_X	Expansion efficiency	0.073	Avg. products (dynamic)
θ_E	Entry efficiency	0.100	Entry rate
θ_T	Type-selection efficiency	0.887	Old-to-young ratio
$\bar{\tau}$	Asymptotic land wedge	0.231	MRPLand DiD
κ	Wedge saturation rate	1.374	Product turnover and entry-exit DiDs

Note: Panel A parameters are set externally. Panel B parameters are jointly calibrated by minimizing the weighted sum of squared percentage deviations between model and data moments. The “Target / Source” column indicates the primary identifying moment for each parameter.

Table 7: Summary of parameters

old-to-young ratio. The ULCRA parameters $\bar{\tau}$ and κ are disciplined by the causal moments from the Repeal. The MRPLand DiD is informative about the average magnitude of the effective land-cost distortion through equation (15). The product creation, destruction, exit, and dynamic-entry DiDs then help pin down how strongly that distortion rose with firm size, because they reveal whether the Repeal primarily relaxed constraints on incumbent expansion, market selection, or both.

Calibrated parameters and model fit. Table 7 reports the externally set and calibrated parameter values. The calibrated step size $\lambda = 1.231$ implies that each innovation raises process efficiency by roughly 23.1%. The expansion and entry efficiencies $\theta_X = 0.073$ and $\theta_E = 0.100$ generate modest rates of product creation, consistent with the slow firm growth documented in Indian manufacturing. The type-selection parameter $\theta_T = 0.887$ governs the cost of drawing a dynamic type upon entry. The ULCRA parameters $\bar{\tau} = 0.231$ and $\kappa = 1.374$ imply a saturating wedge schedule: a single-product firm faces a wedge of 0.173, rising to 0.231 for a five-product firm and 0.231 for a ten-product firm.

Table 8 compares the targeted moments in the data and the model. The model reproduces the innovation DiD within about 5% and the destruction DiD within about 10% of their empirical counterparts. The entry DiD is about 20% below the data, while the exit DiD is close in absolute terms. The model overstates the magnitude of the MRPLand decline by about 39%. Among the cross-sectional moments, the average product counts are well

matched (within 1–6%), and the old-to-young ratio is within about 29%. The TFP growth rate and entry rate deviate by about 24% and 43%, respectively. These discrepancies reflect a tension between matching the pre-repeal level of firm dynamics and the DiD treatment effects with only six free parameters.

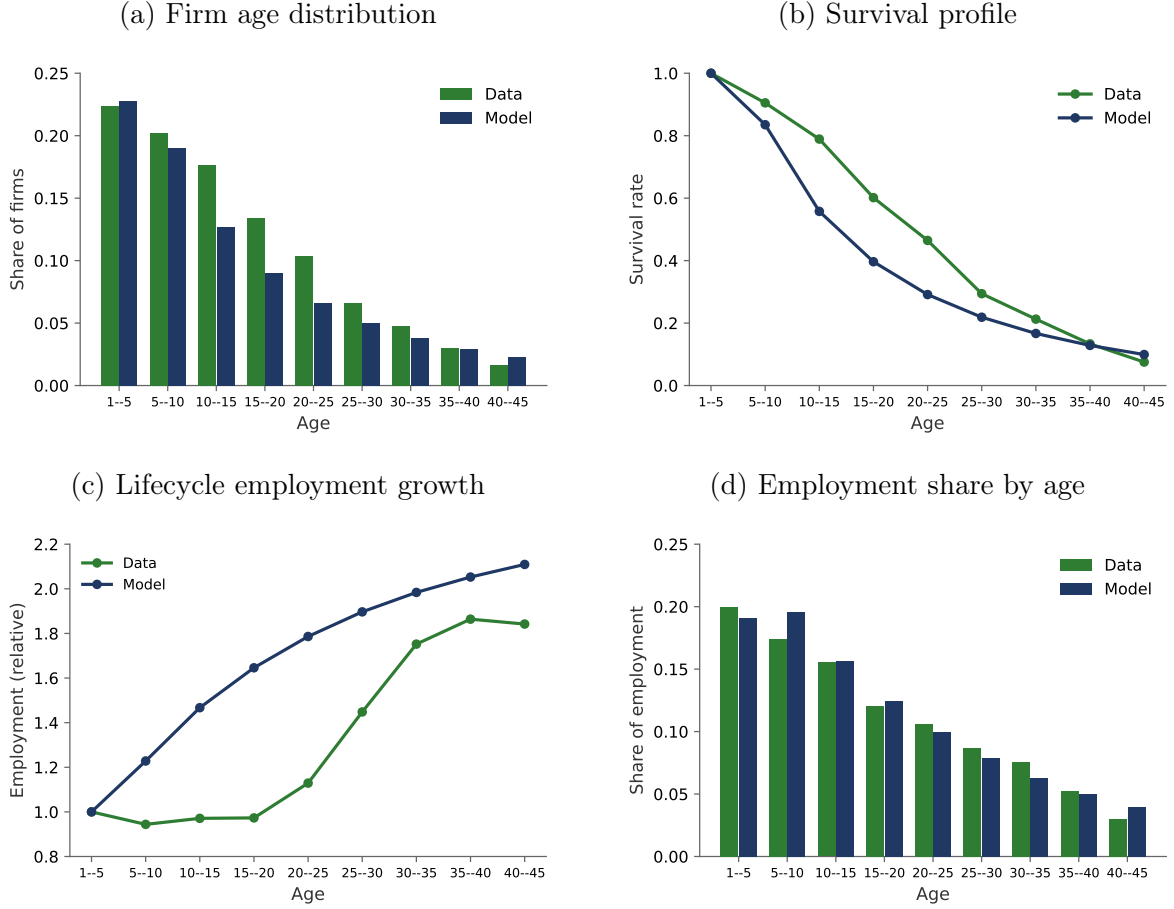
Moment	Data	Model	Sample
<i>Panel A. Cross-sectional moments</i>			
TFP growth rate	0.035	0.027	Economy-wide
$\mathbb{E}[n \mid \text{all}]$	1.143	1.210	Economy-wide
$\mathbb{E}[n \mid \text{dynamic}]$	1.748	1.760	Dynamic firms
Entry rate	0.052	0.075	Dynamic firms
Old-to-young ratio	0.077	0.099	Dynamic firms
<i>Panel B. Difference-in-differences estimates</i>			
Product creation DiD	0.132	0.138	Dynamic firms
Product destruction DiD	0.140	0.126	Dynamic firms
Entry DiD	0.243	0.194	Dynamic firms
Exit DiD	0.011	0.009	Dynamic firms
MRPLand DiD	−0.102	−0.142	Dynamic firms

Note: “Economy-wide” moments use ASI and NSS combined data; “Dynamic firms” moments use ASI (organized sector) only. Panel A moments are pre-repeal cross-sectional features. Panel B moments are DiD coefficients from Section 4. The TFP growth rate is from [Bollard et al. \(2013\)](#).

Table 8: Targeted moments: data vs. model

Untargeted moments. The model also matches several patterns not directly targeted in the calibration. A dynamic firm with n products charges a markup over its actual marginal cost of $\mu(n) = \lambda/T(n)$. Averaging this markup over the pre-Repeal dynamic-firm distribution gives 1.191; the corresponding product-weighted mean is 1.188. Estimating markups directly from the ASI data using the production function approach of [De Ridder et al. \(2022\)](#) yields a pre-Repeal average of 1.34, about 11% above the model’s firm-weighted value. The product distribution among dynamic firms is heavily right-skewed, with 62.0% of dynamic firms holding a single product (compared to 62.5% in the data). The model generates a flat lifecycle employment profile consistent with the well-documented pattern of slow firm growth in India ([Hsieh and Klenow, 2014](#)). Firm survival rates decline steeply with age, driven by the high rate of creative destruction faced by single-product firms. The age distribution of firms is also well-matched. Figure 3 plots the four age and lifecycle profiles against their

empirical counterparts; Appendix [A.6](#) reports the product distribution.



Note: Data: ASI pre-1999 urban-area sample (organized sector only). Model: dynamic firms only. In all four panels, the ASI-to-dynamic mapping is maintained: no NSS (unorganized sector) data enters. Panel (a): share of firms in each age bin. Panel (b): cross-sectional age distribution normalized by the youngest bin, which equals the survival function in steady state. Panel (c): average employment per firm by age, normalized by the youngest cohort; the model counterpart is average products n per firm. Panel (d): share of total employment by age; the model counterpart is the share of total products.

Figure 3: Untargeted moments: model vs. data

	Pre	Post	Change
<i>Panel A. Innovation and growth</i>			
Creative destruction rate (δ)	12.8%	14.3%	+1.5 (p.p.)
Product-weighted expansion rate ($\bar{x}^D$)	8.3%	9.4%	+1.1 (p.p.)
Entry rate (e)	9.5%	10.1%	+0.7 (p.p.)
TFP growth rate (g)	2.7%	3.0%	+0.32 (p.p.)
<i>Panel B. Selection</i>			
Dynamic product share (N^D)	40.1%	44.5%	+4.4 (p.p.)
Dynamic share of entrants (d)	19.1%	21.7%	+2.6 (p.p.)
$\mathbb{E}[n \mid \text{dynamic}]$	1.760	1.785	+1.4 (%)

Note: “Pre” and “Post” denote the balanced growth paths with and without the ULCRA wedge $\tau(n)$, respectively. All structural parameters are held fixed at their calibrated values.

Table 9: Equilibrium effects of the Repeal on firm dynamics

6 The macroeconomic impacts of the Repeal

We now use the calibrated model to quantify the aggregate effects of the Repeal. The counterfactual experiment sets the land market wedge to zero while holding all structural parameters at their calibrated values. This exercise should be interpreted as translating the causal microeconomic pattern documented in the data—lower effective land costs, greater product turnover, and stronger selection among scalable plants—into aggregate dynamics. We first examine how the Repeal reshapes firm dynamics in the new steady state, then compute the welfare gain along the transition path.

6.1 The long-run effects of the Repeal

Table 9 presents different moments of the pre- and post-repeal stationary market equilibrium allocations. Panel A focuses on product innovation and growth. The Repeal raises the growth rate of TFP by 0.32 percentage points, from 2.66% to 2.97%. Since $g = \delta \cdot \ln(\lambda)$ and λ is a structural parameter, the entire growth effect operates through an increase in the rate of creative destruction δ .

Why does creative destruction accelerate? Recall from equation (13) that $\delta = \bar{x}^D N^D + e$. Removing the wedge $\tau(n)$ raises per-product profits for all firms, increasing the marginal value of adding a product line. Dynamic firms respond by expanding faster, with the product-

weighted expansion rate $\bar{x}^D$ rising by 1.1 percentage points. The entry rate e also rises modestly (+0.7 p.p.), as higher expected firm values make entry more attractive.

At first glance, the simultaneous increase in both expansion and creative destruction may seem paradoxical: in the standard framework of [Klette and Kortum \(2004\)](#), higher δ shortens the expected lifetime of each product line, reducing the return to expansion. This business-stealing force is present and quantitatively important, but it does not reverse the sign. Because dynamic firms hold only about two-fifths of all products before the Repeal, faster expansion raises the creative destruction rate by less than one for one. Meanwhile, the higher per-product profit flow anchors the franchise value of each product line independently of δ , preventing the business-stealing feedback from spiraling.

Why does the growth effect survive the business-stealing offset, when prior work on Indian factor market distortions finds near-zero or ambiguous effects on productivity growth? A key mechanism is the presence of stagnant firms, which hold roughly three-fifths of all product lines before the Repeal but perform no product expansion. When dynamic firms further expand after the Repeal, they displace products held by both dynamic and stagnant firms. But stagnant firms do not respond to displacement by creating new products—they simply exit when they lose their only product. This asymmetry weakens the usual feedback loop: in a model where all firms expand, each displaced product triggers a competitive response that partially undoes the initial expansion impulse. Here, a substantial share of the products captured by dynamic firms comes from a non-reactive segment of the economy, allowing the growth acceleration to persist.

These product expansion effects are further reinforced by a compositional shift (Panel B). The dynamic product share N^D rises by 4.4 percentage points (from 40.1% to 44.5%), as dynamic firms expand into a larger fraction of the product space. This shift is driven in part by the dynamic share of entrants d rising from 19.1% to 21.7%: the Repeal increases the value premium of being dynamic, since the option to expand is more valuable when land costs do not penalize growth. The average product count among dynamic firms, $\mathbb{E}[n \mid \text{dynamic}]$, rises only modestly (from 1.760 to 1.785), as new single-product entrants dilute the firm-size distribution. This compositional upgrading is consistent with the empirical evidence in [Section 4](#): the Repeal increased the entry of dynamic firms while leaving subsistence entry roughly unchanged.

6.2 Welfare gains from the Repeal

To quantify the welfare improvement from the Repeal, we solve for the full transition path of the reform. On the date of the Repeal, the wedge $\tau(n)$ is immediately removed, but the pre-determined firm-size distribution adjusts only gradually. As the distribution converges to the new steady state, the creative destruction rate δ_t , the product expansion rates $x_t(n)$, and the consumption share $\mathcal{C}_t \equiv C_t/Y_t$ all adjust along the transition. We define the welfare gain γ as the permanent proportional increase in pre-repeal consumption that would make the household indifferent between the pre-repeal stationary allocation and the post-repeal transition path. In appendix A.4, this welfare variation is decomposed into three components:

$$\ln(\gamma) = \underbrace{\ln(\mathcal{T})}_{\text{Misallocation}} + \underbrace{\ln(\lambda) \int_0^\infty e^{-\rho t} (\delta_t^{\text{post}} - \delta^{\text{pre}}) dt}_{\text{Growth acceleration}} + \underbrace{\int_0^\infty \rho e^{-\rho t} \ln\left(\frac{\mathcal{C}_t^{\text{post}}}{\mathcal{C}^{\text{pre}}}\right) dt}_{\text{Consumption share of output}}. \quad (17)$$

Table 10 reports this decomposition where the component rows are additive contributions to $\ln(\gamma)$, whereas the total improvement in welfare is converted into a 4.43% permanent increase in consumption per capita. The growth acceleration contributes 5.96 log points, while the R&D and entry allocation effect contributes -1.63 log points: faster growth changes the share of final goods devoted to product expansion and entry along the transition. If the land tax revenue was instead wasted rather than rebated to the household, the consumption-equivalent welfare gain would rise to 7.30%. Indeed, we would see an additional contribution of 2.71 log points from eliminating wasteful government expenditures. Figure A.5 plots the transition paths of the TFP growth rate and the consumption share of output.

The contribution from removing cross-sectional factor misallocation is negligible (+0.00 log points). This component equals $\ln(\mathcal{T}^{\text{pre}})$. Since labor and land are inelastically supplied, a uniform wedge leaves $\mathcal{T} = 1$; only cross-sectional dispersion in $\tau(n)$ affects this component.⁴⁹ This does not imply that a uniform wedge is innocuous: it still reduces leaders' profit margins relative to the undistorted fringe and can affect innovation and entry.

The importance of the growth acceleration is the central quantitative finding of this exercise. Several recent studies have quantified the welfare effects of factor market distortions in India: Bertrand et al. (2021) estimate a 5.9% static gain from contract labor reform with zero dynamic gain; Akcigit et al. (2021) find that improving the managerial environment to U.S. levels would close approximately 11% of the India–U.S. income gap, with roughly equal

⁴⁹A second-order approximation to the misallocation component yields $\ln(\mathcal{T}) \approx \alpha(1-\alpha)\mathbb{V}(\tau)/[2(1+\mathbb{E}[\tau])^2]$, where both moments are computed under the product distribution.

Component	Lump-sum rebate	Wasteful use
Static misallocation removal	+0.00	+0.00
Government waste removal	0.00	+2.71
R&D and entry allocation	−1.63	−1.63
Growth acceleration	+5.96	+5.96
Total log-welfare gain	+4.33	+7.04
Consumption-equivalent gain	+4.43	+7.30

Note: Component rows and the total log-welfare row report contributions to $\ln(\gamma)$ multiplied by 100. The final row converts the total into a permanent consumption-equivalent percentage gain. In the baseline, wedge revenue is rebated lump-sum to the household. In the alternative column, it finances wasteful government purchases. See appendix A.4 for the derivation.

Table 10: Welfare under alternative uses of land-wedge revenue

static and dynamic components; [Bau and Matray \(2023\)](#) estimate 3–17% static gains from FDI liberalization. In contrast, the ULCRA repeal produces a welfare gain of comparable magnitude but primarily through a *dynamic* mechanism. This difference arises because the ULCRA distorts the returns to expansion, directly suppressing the product expansion engine that drives long-run growth.

7 Conclusion

This paper studies the micro- and macroeconomic consequences of land market liberalization on manufacturing firms, leveraging the staggered repeal of India’s Urban Land Ceiling and Regulation Act as a natural experiment. Using a Difference-in-Differences framework, we show that the Repeal liberalized urban land markets, improved allocative efficiency, and unleashed industrial growth: treated plants expanded their land use, capital, employment, sales, and productivity. Beyond these static effects, we provide, to our knowledge, some of the first causal evidence that land market frictions affect firms’ product innovation decisions. The ULCRA’s repeal intensified creative destruction—both the creation and destruction of product lines accelerated—and shifted the composition of entrants toward more dynamic plants, those more likely to introduce new products over their life cycle.

To quantify the aggregate implications of the Repeal, we developed and calibrated a general equilibrium model of endogenous growth in which the ULCRA acts as a size-dependent wedge on the effective cost of land. This formulation captures the land-market frictions created by the Act and their disproportionate burden on larger firms. The calibrated model

indicates that the Repeal raised the growth rate of Indian manufacturing by 0.32 percentage points and generated a 4.43% improvement in consumption-equivalent welfare. Growth acceleration is the largest component of this gain, while the static productivity gain from removing cross-sectional factor misallocation is negligible.

Several directions remain for future research. Our analysis focuses on the manufacturing sector, but the ramifications of land market policies extend to urbanization, structural transformation, and housing markets; we provide preliminary evidence on some of these broader consequences in the appendix. More fundamentally, the interaction between land market reforms and other economic frictions—such as labor regulations, credit constraints, and infrastructure provision—could amplify or attenuate the growth effects we document. Understanding these complementarities is essential for designing effective reforms in developing economies.

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A Theoretical appendix

This appendix provides complete derivations of all optimality conditions, aggregation results, the balanced growth path system, and the welfare decomposition stated in the main text. Each derivation proceeds from the optimization problem defined in Section 5 to the closed-form condition used in the calibration.

A.1 Derivation of optimality conditions

The household's problem. The representative household chooses a consumption path $\{C_t\}_{t \geq 0}$ to maximize lifetime utility subject to a flow budget constraint:

$$\int_0^\infty e^{-\rho t} \ln(C_t) dt \quad \text{s.t.} \quad \dot{A}_t = r_t A_t + w_t H + q_t L - C_t.$$

Form the current-value Hamiltonian with costate variable μ_t :

$$\mathcal{H}_t = \ln(C_t) + \mu_t(r_t A_t + w_t H + q_t L - C_t). \quad (\text{A.1})$$

The necessary conditions for an interior optimum are:

$$\text{FOC w.r.t. } C_t: \quad \frac{\partial \mathcal{H}_t}{\partial C_t} = \frac{1}{C_t} - \mu_t = 0, \quad (\text{A.2})$$

$$\text{Costate equation:} \quad \dot{\mu}_t = \rho \mu_t - \frac{\partial \mathcal{H}_t}{\partial A_t} = (\rho - r_t) \mu_t. \quad (\text{A.3})$$

Combining both conditions gives the Euler equation:

$$\frac{\dot{C}_t}{C_t} = r_t - \rho. \quad (\text{A.4})$$

The final sector's problem. The competitive final sector takes prices as given and chooses its demand for products to maximize profits subject to the Cobb–Douglas technology:

$$\max_{\{y_{it}\}_{i \in [0,1]}} P_t Y_t - \int_0^1 p_{it} y_{it} di \quad \text{s.t.} \quad \ln(Y_t) = \int_0^1 \ln(y_{it}) di$$

where P_t is the price of the final good. The first-order condition for each product i implies:

$$p_{it} y_{it} = P_t Y_t, \quad \forall i \in [0, 1].$$

To derive the price index, substitute $y_{it} = P_t Y_t / p_{it}$ into the technology $\ln(Y_t) = \int_0^1 \ln(y_{it}) di$:

$$\ln(Y_t) = \int_0^1 [\ln(P_t Y_t) - \ln(p_{it})] di = \ln(P_t Y_t) - \int_0^1 \ln(p_{it}) di.$$

Rearranging gives:

$$\ln(P_t) = \int_0^1 \ln(p_{it}) di. \quad (\text{A.5})$$

That is, the price of the final good is the geometric mean of product prices. Setting $P_t = 1$ (the numéraire normalization) imposes $\int_0^1 \ln(p_{it}) di = 0$. With this normalization:

$$p_{it} y_{it} = Y_t, \quad \forall i \in [0, 1]. \quad (\text{A.6})$$

The firm's problem. Consider a leading n -product firm with process efficiency z_{it} manufacturing product i . The leader minimizes total cost subject to meeting demand y_{it} :

$$\min_{h_{it}, \ell_{it} \geq 0} \{w_t h_{it} + [1 + \tau(n)] q_t \ell_{it}\} \quad \text{s.t.} \quad z_{it} h_{it}^\alpha \ell_{it}^{1-\alpha} \geq y_{it}.$$

Let μ_{it} denote the Lagrange multiplier on the production constraint. The first-order conditions are:

$$\text{FOC w.r.t. } h_{it}: \quad w_t = \mu_{it} \alpha z_{it} h_{it}^{\alpha-1} \ell_{it}^{1-\alpha} \quad (\text{A.7})$$

$$\text{FOC w.r.t. } \ell_{it}: \quad [1 + \tau(n)] q_t = \mu_{it} (1 - \alpha) z_{it} h_{it}^\alpha \ell_{it}^{-\alpha}. \quad (\text{A.8})$$

Taking the ratio of equations (A.7) and (A.8) eliminates μ_{it} :

$$\frac{h_{it}}{\ell_{it}} = \frac{\alpha}{1 - \alpha} \cdot \frac{[1 + \tau(n)] q_t}{w_t}. \quad (\text{A.9})$$

Substituting the optimal factor ratio back into the production function and solving for the cost-minimizing inputs gives:

$$w_t h_{it} = \alpha \cdot \frac{\varsigma_t(n)}{z_{it}} \cdot y_{it}, \quad (\text{A.10})$$

$$[1 + \tau(n)] q_t \ell_{it} = (1 - \alpha) \cdot \frac{\varsigma_t(n)}{z_{it}} \cdot y_{it} \quad (\text{A.11})$$

where the distorted unit cost index is:

$$\varsigma_t(n) \equiv \varsigma_t \cdot T(n). \quad (\text{A.12})$$

Here, we have defined:

$$\varsigma_t \equiv \left(\frac{w_t}{\alpha}\right)^\alpha \left(\frac{q_t}{1-\alpha}\right)^{1-\alpha} \quad \text{and} \quad T(n) \equiv [1 + \tau(n)]^{1-\alpha}.$$

The competitive fringe uses the same Cobb–Douglas technology as the leader but operates at process efficiency z_{it}/λ and faces no ULCRA wedge. Its marginal cost is therefore:

$$\text{MC}_{\text{fringe}} = \frac{\lambda \varsigma_t}{z_{it}}. \quad (\text{A.13})$$

The leader sets price equal to the fringe's marginal cost (limit pricing):

$$p_{it} = \frac{\lambda \varsigma_t}{z_{it}}. \quad (\text{A.14})$$

From equations (A.6), (A.11) and (A.14), solving for ℓ_{it} :

$$\ell_{it} = \frac{(1-\alpha)\varsigma_t(n)y_{it}}{[1+\tau(n)]q_t z_{it}} = \frac{(1-\alpha)Y_t}{\lambda q_t [1+\tau(n)]^\alpha}. \quad (\text{A.15})$$

Similarly, from equations (A.6), (A.10) and (A.14), solving for h_{it} :

$$h_{it} = \frac{\alpha \varsigma_t(n)y_{it}}{w_t z_{it}} = \frac{\alpha [1+\tau(n)]^{1-\alpha} Y_t}{\lambda w_t}. \quad (\text{A.16})$$

Revenue per product equals Y_t by equation (A.6). Production cost per product is the leader's marginal cost times quantity:

$$\frac{\varsigma_t \cdot T(n)}{z_{it}} \cdot y_{it} = \frac{\varsigma_t \cdot T(n)}{z_{it}} \cdot \frac{Y_t z_{it}}{\lambda \varsigma_t} = \frac{T(n)Y_t}{\lambda}.$$

We restrict the parameter space so that $(1+\bar{\tau})^{1-\alpha} < \lambda$. Since $T(n) = [1+\tau(n)]^{1-\alpha} < (1+\bar{\tau})^{1-\alpha}$ for every finite n , the leader's marginal cost is strictly below the fringe's limit price and operating profits are positive at every firm size. Per-product profit is revenue minus cost:

$$\pi_t(n) = Y_t \left[1 - \frac{T(n)}{\lambda} \right]. \quad (\text{A.17})$$

Because per-product profit depends on n but not on the individual z_{it} 's, the dynamic firm's value $V_t^D(n)$ depends only on the product count n . The corresponding HJB equation is:

$$\begin{aligned} r_t V_t^D(n) = & n\pi_t(n) + \dot{V}_t^D(n) \\ & + \max_{X_t \geq 0} \{X_t[V_t^D(n+1) - V_t^D(n)] - X_t^2 Y_t / (2\theta_X n)\} \\ & - n\delta_t[V_t^D(n) - V_t^D(n-1)] \end{aligned} \quad (\text{A.18})$$

with boundary condition $V_t^D(0) = 0$. The three groups of terms on the right-hand side are: (1) flow profits plus capital gains, (2) the option value of expansion net of its cost in final goods, and (3) the expected loss from creative destruction. The maximand in the second line is concave in X_t , so the first-order condition is necessary and sufficient:

$$\frac{\partial}{\partial X_t} \left\{ X_t[V_t^D(n+1) - V_t^D(n)] - \frac{X_t^2 Y_t}{2\theta_X n} \right\} = V_t^D(n+1) - V_t^D(n) - \frac{X_t Y_t}{\theta_X n} = 0.$$

Solving for X_t and defining the per-product expansion rate $x_t(n) \equiv X_t/n$:

$$x_t(n) = \frac{\theta_X[V_t^D(n+1) - V_t^D(n)]}{Y_t}. \quad (\text{A.19})$$

A stagnant firm holds instead a single product, has no access to the expansion technology, and its value function satisfies:

$$(r_t + \delta_t)V_t^S = \pi_t(1) + \dot{V}_t^S.$$

The entrant's problem. The unit mass of potential entrants maximizes expected profit net of entry costs:

$$\max_{e_t \geq 0, d_t \in [0,1]} \{e_t[d_t V_t^D(1) + (1-d_t)V_t^S - Y_t^T] - Y_t^E\} \quad (\text{A.20})$$

where $Y_t^E = e_t^2 Y_t / (2\theta_E)$ and $Y_t^T = -[\ln(1-d_t) + d_t]Y_t / \theta_T$ from equation (6). Taking the first-order condition with respect to d_t :

$$e_t[V_t^D(1) - V_t^S] - \frac{e_t d_t Y_t}{\theta_T(1-d_t)} = 0.$$

The factor e_t cancels (assuming $e_t > 0$). Defining $\Delta V_t \equiv V_t^D(1) - V_t^S$ and solving for d_t :

$$d_t = \frac{\theta_T \cdot \Delta V_t}{\theta_T \cdot \Delta V_t + Y_t}. \quad (\text{A.21})$$

An interior solution $d_t > 0$ requires $\Delta V_t > 0$: the dynamic type must be strictly more valuable than the stagnant type. Taking the first-order condition with respect to e_t :

$$d_t V_t^D(1) + (1 - d_t) V_t^S - Y_t \left[\frac{e_t}{\theta_E} - \frac{\ln(1 - d_t) + d_t}{\theta_T} \right] = 0.$$

Solving for e_t , we obtain:

$$e_t = \theta_E \cdot \left\{ \frac{V_t^S + d_t \cdot \Delta V_t}{Y_t} + \frac{\ln(1 - d_t) + d_t}{\theta_T} \right\}. \quad (\text{A.22})$$

A.2 Aggregation

Labor market clearing. The labor market clearing condition yields:

$$\int_0^1 h_{it} di = H. \quad (\text{A.23})$$

Substituting equation (A.16) for h_{it} , we obtain:

$$\int_0^1 h_{it} di = \frac{\alpha Y_t \mathcal{H}_t}{\lambda w_t} = H \quad (\text{A.24})$$

where the aggregate labor distortion $\mathcal{H}_t$ is defined as:

$$\mathcal{H}_t \equiv \left\{ \sum_{n=1}^{\infty} [1 + \tau(n)]^{1-\alpha} n m_t^D(n) + [1 + \tau(1)]^{1-\alpha} M_t^S \right\}. \quad (\text{A.25})$$

Here, $m_t^D(n)$ is the measure of dynamic firms with n products and M_t^S is the measure of stagnant firms. The aggregate labor distortion $\mathcal{H}_t$ is a product-weighted average of the ULCRA wedges across firms. We can solve for the wage rate in equation (A.24) as:

$$w_t = \frac{\alpha Y_t \mathcal{H}_t}{\lambda H}.$$

Land market clearing. The land market clearing condition yields:

$$\int_0^1 \ell_{it} di = L. \quad (\text{A.26})$$

Substituting equation (A.15) for ℓ_{it} , we obtain:

$$\int_0^1 \ell_{it} di = \frac{(1-\alpha)Y_t \mathcal{L}_t}{\lambda q_t} = L \quad (\text{A.27})$$

where the aggregate land distortion $\mathcal{L}_t$ is defined as:

$$\mathcal{L}_t \equiv \left\{ \sum_{n=1}^{\infty} [1 + \tau(n)]^{-\alpha} n m_t^D(n) + [1 + \tau(1)]^{-\alpha} M_t^S \right\}. \quad (\text{A.28})$$

We can solve for the land rent in equation (A.27) as:

$$q_t = \frac{(1-\alpha)Y_t \mathcal{L}_t}{\lambda L}.$$

Government revenue and the baseline rebate. The government collects the wedge payment $\tau(n)q_t \ell_t(n)$ from each product operated by an n -product firm. Under the baseline fiscal closure, it rebates this revenue lump-sum to the household through a transfer B_t^G . Its balanced budget is:

$$B_t^G = \sum_{n=1}^{\infty} \tau(n)q_t \ell_t(n) n m_t^D(n) + \tau(1)q_t \ell_t(1) M_t^S. \quad (\text{A.29})$$

Under the wasteful-use alternative below, this revenue instead finances government purchases $G_t = B_t^G$. Substituting land demand from (A.15), the associated government expenditure share is:

$$\begin{aligned} \mathcal{G}_t &\equiv \frac{G_t}{Y_t} = \frac{B_t^G}{Y_t} \\ &= \frac{1-\alpha}{\lambda} \left\{ \sum_{n=1}^{\infty} \tau(n) [1 + \tau(n)]^{-\alpha} n m_t^D(n) + \tau(1) [1 + \tau(1)]^{-\alpha} M_t^S \right\} \\ &= \frac{1-\alpha}{\lambda} (\mathcal{H}_t - \mathcal{L}_t). \end{aligned} \quad (\text{A.30})$$

The last equality uses $(1 + \tau)^{1-\alpha} - (1 + \tau)^{-\alpha} = \tau(1 + \tau)^{-\alpha}$. Under the baseline closure, $G_t = 0$ and B_t^G instead enters household income, so the revenue cancels from the consolidated household–government resource constraint.

Final goods market clearing. Aggregate output is split between consumption, expansion, entry, and type-selection spending. Total incumbent expansion spending is:

$$Y_t^X = \int_{f \in \mathcal{F}_t} \frac{X_{ft}^2 Y_t}{2\theta_X n_{ft}} df = \frac{Y_t}{2\theta_X} \cdot \sum_{n=1}^{\infty} x_t(n)^2 n m_t^D(n).$$

Total entry spending is:

$$Y_t^E = \frac{e_t^2 Y_t}{2\theta_E}.$$

Type-selection spending per entrant is:

$$Y_t^T = -\frac{[\ln(1 - d_t) + d_t] Y_t}{\theta_T}.$$

Therefore, the final goods market clearing condition is:

$$Y_t = C_t + \frac{Y_t}{2\theta_X} \cdot \sum_{n=1}^{\infty} x_t(n)^2 n m_t^D(n) + \frac{e_t^2 Y_t}{2\theta_E} - e_t \cdot \frac{[\ln(1 - d_t) + d_t] Y_t}{\theta_T}.$$

Creative destruction. The aggregate rate of creative destruction is the sum of two flows: incumbent expansion and entry. Each dynamic firm with n products expands into new product lines at total rate $x_t(n)n$, displacing a randomly selected incumbent product. Entry creates new products at rate e_t , and each entrant product displaces a randomly selected incumbent product. Therefore, the aggregate creative destruction rate is:

$$\delta_t = \sum_{n=1}^{\infty} x_t(n) n m_t^D(n) + e_t.$$

Kolmogorov-Forward (KF) equation. Let $m_t^D(n)$ denote the measure of dynamic firms with n products. The KF equation for $m_t^D(n)$ at $n \geq 2$ is:

$$\begin{aligned} \dot{m}_t^D(n) &= x_t(n-1)(n-1)m_t^D(n-1) \\ &\quad - [x_t(n)n + \delta_t n]m_t^D(n) \\ &\quad + \delta_t(n+1)m_t^D(n+1). \end{aligned} \tag{A.31}$$

The first term is the inflow from expansion of $(n-1)$ -product firms, the second is the outflow from firms at n that either expand or lose a product, and the third is the inflow from $(n+1)$ -product firms that lose a product. At $n = 1$, there are two sources of inflow: entry of dynamic firms at rate $e_t d_t$ and two-product firms losing a product at rate $2\delta_t$. The outflow consists of

expansion to $n = 2$ at rate $x_t(1)$ and exit upon losing the sole product at rate δ_t . Therefore, the KF equation at $n = 1$ is:

$$\dot{m}_t^D(1) = e_t d_t - [x_t(1) + \delta_t] m_t^D(1) + 2\delta_t m_t^D(2). \quad (\text{A.32})$$

The measure of stagnant firms simply satisfies:

$$\dot{M}_t^S = e_t(1 - d_t) - \delta_t M_t^S. \quad (\text{A.33})$$

Product shares. The total share of products held by dynamic and stagnant firms must exhaust the unit product continuum. Therefore, we have:

$$N_t^D + N_t^S = 1 \quad \text{where} \quad N_t^D \equiv \sum_{n=1}^{\infty} n m_t^D(n) \quad \text{and} \quad N_t^S \equiv M_t^S. \quad (\text{A.34})$$

Aggregate output. The numéraire normalization implies that the price of the final good is $P_t = 1$, which yields:

$$\int_0^1 \ln(p_{it}) di = 0.$$

Substituting the limit price from equation (A.14) gives:

$$\ln(\lambda \varsigma_t) = \int_0^1 \ln(z_{it}) di.$$

Define the geometric mean of process efficiencies as:

$$\ln(Z_t) \equiv \int_0^1 \ln(z_{it}) di \quad (\text{A.35})$$

and substitute the definition of ς_t from equation (A.12):

$$\lambda \left(\frac{w_t}{\alpha} \right)^\alpha \left(\frac{q_t}{1 - \alpha} \right)^{1 - \alpha} = Z_t.$$

Using the wage and land rent from equations (A.24) and (A.27) gives:

$$Y_t = \frac{Z_t H^\alpha L^{1 - \alpha}}{\mathcal{T}_t} \quad \text{where} \quad \mathcal{T}_t \equiv \mathcal{H}_t^\alpha \cdot \mathcal{L}_t^{1 - \alpha}. \quad (\text{A.36})$$

In the absence of ULCRA wedges, $\mathcal{H}_t = \mathcal{L}_t = 1$ and aggregate output is equal to:

$$Y_t = Z_t H^\alpha L^{1-\alpha}.$$

With ULCRA wedges, aggregate output is distorted by the factor $\mathcal{T}_t$, which we now show is weakly greater than one.

Bound on the aggregate wedge.

Proposition A.1. *For any $\alpha \in (0, 1)$, any product weights $\{w_i\}_{i \in I}$ with $w_i \geq 0$ and $\sum_{i \in I} w_i = 1$, and any wedges $\{\tau_i\}_{i \in I}$ with $\tau_i \geq 0$:*

$$\mathcal{T} \equiv \left[\sum_i w_i (1 + \tau_i)^{1-\alpha} \right]^\alpha \left[\sum_i w_i (1 + \tau_i)^{-\alpha} \right]^{1-\alpha} \geq 1$$

Defining the positive-weight support as $I_+ \equiv \{i \in I : w_i > 0\}$, equality holds if and only if τ_i is constant over I_+ .

Proof. Let $x_i \equiv 1 + \tau_i$, drawn with probability w_i , and define the log-moment function $b(t) \equiv \ln(\mathbb{E}[x^t])$. Since $x_i > 0$, b is well-defined for all $t \in \mathbb{R}$. The function b is convex: for any $t_1, t_2 \in \mathbb{R}$ and $\nu \in [0, 1]$, Hölder's inequality with exponents $1/\nu$ and $1/(1-\nu)$ gives:

$$\mathbb{E}[(x^{t_1})^\nu \cdot (x^{t_2})^{1-\nu}] \leq \mathbb{E}[x^{t_1}]^\nu \cdot \mathbb{E}[x^{t_2}]^{1-\nu}$$

which is $b(\nu t_1 + (1-\nu)t_2) \leq \nu b(t_1) + (1-\nu)b(t_2)$. Now observe that:

$$\ln(\mathcal{T}) = \alpha b(1-\alpha) + (1-\alpha)b(-\alpha) \geq b(\alpha(1-\alpha) + (1-\alpha)(-\alpha)) = b(0) = 0$$

where the inequality comes from the convexity of b and the last step uses $b(0) = \ln(\mathbb{E}[x^0]) = 0$. Hölder's inequality holds with equality if and only if x_i is constant almost surely under the probability weights w_i , i.e., if and only if the τ_i 's are equal over I_+ . $\square$

Second-order approximation of $\ln(\mathcal{T})$. Proposition A.1 shows that a uniform wedge $\tau_i = \mu$ for all i yields $\mathcal{T} = 1$ exactly, regardless of the common level μ . This result is a consequence of inelastic factor supply: since aggregate labor are fixed, a uniform wedge merely scales factor prices without distorting aggregate output. When the wedge varies across product lines, only its *cross-sectional dispersion* generates a loss. The following second-order approximation makes this precise.

Write $\tau_i = \mu + \varepsilon_i$ where $\mu \equiv \mathbb{E}[\tau_i]$ is the product-weighted mean and $\mathbb{E}[\varepsilon_i] = 0$, $\mathbb{E}[\varepsilon_i^2] = \sigma^2 \equiv \mathbb{V}(\tau)$. For a smooth function h , the second-order expansion of $\ln(\mathbb{E}[h(\tau)])$ around the degenerate distribution $\tau_i = \mu$ is:

$$\ln(\mathbb{E}[h(\tau)]) \approx \ln(h(\mu)) + \frac{\sigma^2}{2} \cdot \frac{h''(\mu)}{h(\mu)}.$$

Applying this to the two components of $\ln(\mathcal{T}) = \alpha \ln(\mathcal{H}) + (1 - \alpha) \ln(\mathcal{L})$:

$$\begin{aligned} h(\tau) = (1 + \tau)^{1-\alpha} &\implies \frac{h''(\mu)}{h(\mu)} = \frac{-\alpha(1-\alpha)}{(1+\mu)^2} \\ g(\tau) = (1 + \tau)^{-\alpha} &\implies \frac{g''(\mu)}{g(\mu)} = \frac{\alpha(1+\alpha)}{(1+\mu)^2}. \end{aligned}$$

The zeroth-order terms cancel: $\alpha \ln(h(\mu)) + (1-\alpha) \ln(g(\mu)) = [\alpha(1-\alpha) - \alpha(1-\alpha)] \ln(1+\mu) = 0$, confirming that $\mathcal{T} = 1$ when $\tau_i = \mu$. Collecting the second-order terms:

$$\ln(\mathcal{T}) \approx \frac{\sigma^2}{2(1+\mu)^2} \cdot [\alpha \cdot (-\alpha)(1-\alpha) + (1-\alpha) \cdot \alpha(1+\alpha)] = \frac{\alpha(1-\alpha)}{2(1+\mu)^2} \cdot \mathbb{V}(\tau) \quad (\text{A.37})$$

where $\mu = \mathbb{E}[\tau]$ and $\mathbb{V}(\tau)$ are both computed under the product distribution.

Growth of aggregate productivity. Let $\phi_t(s)$ denote the density of the logarithm of process efficiency $s \equiv \ln(z)$ across the product continuum. When creative destruction hits a product at rate δ_t , its log-productivity jumps by $\ln(\lambda)$. The KF equation for ϕ_t is:

$$\dot{\phi}_t(s) = \delta_t \cdot \phi_t(s - \ln(\lambda)) - \delta_t \cdot \phi_t(s). \quad (\text{A.38})$$

The first term is the inflow from products whose log-productivity was $s - \ln(\lambda)$ before being hit; the second is the outflow from products at s that are displaced. Taking the derivative of $\ln(Z_t)$ with respect to time:

$$\begin{aligned} \frac{\dot{Z}_t}{Z_t} &= \int s \cdot \dot{\phi}_t(s) ds \\ &= \int s \cdot [\delta_t \cdot \phi_t(s - \ln(\lambda)) - \delta_t \cdot \phi_t(s)] ds \\ &= \delta_t \int s \phi_t(s - \ln(\lambda)) ds - \delta_t \ln(Z_t). \end{aligned}$$

Substituting $u = s - \ln(\lambda)$ in the first integral gives:

$$\frac{\dot{Z}_t}{Z_t} = \delta_t \int [u + \ln(\lambda)] \phi_t(u) du - \delta_t \ln(Z_t) = \delta_t \cdot \ln(\lambda). \quad (\text{A.39})$$

This holds at all times: aggregate productivity grows at rate $\delta_t \ln(\lambda)$.

The differential-algebraic system. Equations (A.4), (A.17)–(A.19), (A.24)–(A.33), and (A.39), together with the final goods market clearing condition and the stagnant firm’s HJB, form a differential-algebraic system in the time-varying unknowns:

$$\{m_t^D(n), M_t^S, \delta_t, x_t(n), e_t, d_t, V_t^D(n), V_t^S, w_t, q_t, r_t, Y_t, C_t\}_{t \geq 0}.$$

From (A.39) and the price normalization, the output growth rate equals $\delta_t \ln(\lambda)$ whenever the firm-size distribution—and hence the product-weighted cost aggregates in equations (A.24) and (A.27)—is stationary.

A.3 Balanced growth path

Normalization. On the balanced growth path (BGP), Y_t , C_t , w_t , q_t , $V_t^D(n)$, and V_t^S all grow at the common constant rate $g \equiv \delta \ln(\lambda)$, so their ratios to C_t are time-invariant. Define the normalized quantities:

$$\mathcal{V}_t^D(n) \equiv \frac{V_t^D(n)}{C_t}, \quad \mathcal{V}_t^S \equiv \frac{V_t^S}{C_t}, \quad \mathcal{Y}_t \equiv \frac{Y_t}{C_t}, \quad \omega_t^H \equiv \frac{w_t}{C_t}, \quad \omega_t^L \equiv \frac{q_t}{C_t}.$$

On the BGP, these are all constant such that we suppress the time subscript when no ambiguity arises. The interest rate satisfies $r = \rho + g$ by the Euler equation. To derive the normalized HJB, note that $\dot{V}_t^D(n) = gV_t^D(n)$ on the BGP, so dividing equation (A.18) by C_t and using $r - g = \rho$ gives:

$$\rho \mathcal{V}^D(n) = n\mathcal{Y}[1 - T(n)/\lambda] + x(n)^2 n\mathcal{Y}/(2\theta_X) - n\delta[\mathcal{V}^D(n) - \mathcal{V}^D(n-1)]$$

where we have substituted the optimal per-product expansion rate from equation (A.19):

$$x(n) = \frac{\theta_X[\mathcal{V}^D(n+1) - \mathcal{V}^D(n)]}{\mathcal{Y}}.$$

A stagnant firm's normalized value function instead satisfies:

$$\mathcal{V}^S = \frac{\mathcal{Y}[1 - T(1)/\lambda]}{\rho + \delta}.$$

The BGP system. The stationary system of equations is:

1. *Dynamic firm's HJB equation:*

$$\rho \mathcal{V}^D(n) = n\mathcal{Y}[1 - T(n)/\lambda] + x(n)^2 n\mathcal{Y}/(2\theta_X) - n\delta[\mathcal{V}^D(n) - \mathcal{V}^D(n-1)]. \quad (\text{A.40})$$

2. *Optimal expansion rate:*

$$x(n) = \theta_X[\mathcal{V}^D(n+1) - \mathcal{V}^D(n)]/\mathcal{Y}. \quad (\text{A.41})$$

3. *Stagnant firm's value function:*

$$\mathcal{V}^S = \mathcal{Y}[1 - T(1)/\lambda]/(\rho + \delta). \quad (\text{A.42})$$

4. *Optimal type selection:*

$$d = \frac{\theta_T \Delta \mathcal{V}}{\theta_T \Delta \mathcal{V} + \mathcal{Y}} \quad \text{where} \quad \Delta \mathcal{V} \equiv \mathcal{V}^D(1) - \mathcal{V}^S. \quad (\text{A.43})$$

5. *Optimal entry rate:*

$$e = \frac{\theta_E}{\mathcal{Y}} \cdot \left\{ \mathcal{V}^S + d\Delta \mathcal{V} + \frac{\mathcal{Y}[\ln(1-d) + d]}{\theta_T} \right\}. \quad (\text{A.44})$$

6. *Stationary firm-product distribution:*

$$0 = \begin{cases} (ed)/M^D - [\delta + x(1)]f^D(1) + 2\delta f^D(2) & \text{if } n = 1, \\ x(n-1)(n-1)f^D(n-1) - [x(n)n + \delta n]f^D(n) & \\ \quad + \delta(n+1)f^D(n+1) & \text{if } n \geq 2 \end{cases} \quad (\text{A.45})$$

where $f^D(n)$ is the fraction of dynamic firms with n products; $\sum_{n=1}^{\infty} f^D(n) = 1$.

7. *Product shares:*

$$N^S = e(1-d)/\delta \quad \text{and} \quad N^D = M^D \sum_{n=1}^{\infty} n f^D(n) = 1 - N^S. \quad (\text{A.46})$$

8. *Creative destruction:*

$$\delta = \bar{x}^D N^D + e \quad \text{where} \quad \bar{x}^D \equiv \frac{\sum_{n=1}^{\infty} x(n) n f^D(n)}{\sum_{n=1}^{\infty} n f^D(n)}. \quad (\text{A.47})$$

9. *Growth rate:*

$$g = \delta \cdot \ln(\lambda). \quad (\text{A.48})$$

10. *Goods market clearing:*

$$\frac{1}{\mathcal{Y}} + M^D \sum_{n=1}^{\infty} \frac{x(n)^2 n f^D(n)}{2\theta_X} + \frac{e^2}{2\theta_E} - e \cdot \frac{[\ln(1-d) + d]}{\theta_T} = 1. \quad (\text{A.49})$$

11. *Labor and land market clearing:*

$$\omega^H = \frac{\alpha \mathcal{YH}}{\lambda H} \quad \text{and} \quad \omega^L = \frac{(1-\alpha) \mathcal{YL}}{\lambda L}. \quad (\text{A.50})$$

Aggregate scale of the economy. The BGP system pins down the ratios of all variables to consumption C_t , but not the aggregate scale of the economy. To determine the scale, we can rewrite equation (A.36) as:

$$C_t = \frac{Z_0 e^{gt} H^\alpha L^{1-\alpha}}{\mathcal{YT}}.$$

Therefore, the aggregate scale is determined by the initial level of aggregate productivity.

A.4 Welfare

Define the household's lifetime utility when their consumption is perpetually multiplied by a factor of $\gamma > 0$ as:

$$U_0(\gamma) = \int_0^{\infty} e^{-\rho t} \ln(\gamma \cdot C_t) dt.$$

The consumption-equivalent welfare gain from the Repeal of the ULCRA is the value of γ that makes the household indifferent between the pre-repeal stationary allocation and the

post-repeal transition path:

$$U_0^{\text{pre}}(\gamma) = U_0^{\text{post}}(1).$$

In other words, γ is the factor by which we must scale up consumption in the pre-repeal stationary allocation to make the household as well off as they would be under the post-repeal transition path. The pre-repeal lifetime utility is a simple exponential integral:

$$\begin{aligned} U_0^{\text{pre}}(\gamma) &= \int_0^\infty e^{-\rho t} \ln(\gamma C_t^{\text{pre}}) dt \\ &= \int_0^\infty e^{-\rho t} \ln \left(\frac{\gamma Z_0 e^{gt} H^\alpha L^{1-\alpha}}{\mathcal{T} \mathcal{Y}^{\text{pre}}} \right) dt \\ &= \frac{1}{\rho} \cdot \left[\ln \left(\frac{\gamma Z_0 H^\alpha L^{1-\alpha}}{\mathcal{T} \mathcal{Y}^{\text{pre}}} \right) + \frac{\delta^{\text{pre}} \ln(\lambda)}{\rho} \right]. \end{aligned}$$

Since $\tau(n) = 0$ for all n after the repeal, $\mathcal{T}_t^{\text{post}} = 1$ for all $t \geq 0$. The post-repeal lifetime utility is therefore:

$$\begin{aligned} U_0^{\text{post}}(1) &= \int_0^\infty e^{-\rho t} \ln(C_t^{\text{post}}) dt \\ &= \int_0^\infty e^{-\rho t} \ln \left(\frac{Z_t^{\text{post}} H^\alpha L^{1-\alpha}}{\mathcal{Y}_t^{\text{post}}} \right) dt \\ &= \int_0^\infty e^{-\rho t} \ln \left(\frac{Z_0 e^{\ln(\lambda) \int_0^t \delta_s^{\text{post}} ds} H^\alpha L^{1-\alpha}}{\mathcal{Y}_t^{\text{post}}} \right) dt \\ &= \frac{1}{\rho} \cdot \left[\ln(Z_0 H^\alpha L^{1-\alpha}) - \int_0^\infty \rho e^{-\rho t} \ln(\mathcal{Y}_t^{\text{post}}) dt + \ln(\lambda) \int_0^\infty \rho e^{-\rho t} \int_0^t \delta_s^{\text{post}} ds dt \right]. \end{aligned}$$

Combining the two expressions and solving for γ gives:

$$\ln(\gamma) = \ln(\mathcal{T}) + \int_0^\infty \rho e^{-\rho t} \ln(\mathcal{C}_t^{\text{post}} / \mathcal{C}^{\text{pre}}) dt + \ln(\lambda) \int_0^\infty e^{-\rho t} (\delta_t^{\text{post}} - \delta^{\text{pre}}) dt$$

where we have defined the consumption share of aggregate output as:

$$\mathcal{C}^{\text{pre}} \equiv \frac{1}{\mathcal{Y}^{\text{pre}}} \quad \text{and} \quad \mathcal{C}_t^{\text{post}} \equiv \frac{1}{\mathcal{Y}_t^{\text{post}}}.$$

Under the baseline rebate closure, the government transfer is not a resource use. Final-goods market clearing therefore implies, both before and after the Repeal:

$$\mathcal{C}_t = 1 - \frac{Y_t^X + Y_t^E + e_t Y_t^T}{Y_t}.$$

The consumption-path term therefore measures the reallocation of final goods to expansion and entry directly. The baseline welfare gain has three components: (1) removal of cross-sectional factor misallocation $\mathcal{T}$, (2) reallocation of final goods to expansion and entry along the transition, and (3) faster productivity growth.

Wasteful-use alternative. As an alternative fiscal closure, suppose that the government uses the wedge revenue in (A.29) to purchase $G_t = B_t^G$ units of final goods that provide no household utility or productive services, and removes the transfer from household income. Consolidated final goods market clearing then includes G_t as an additional use of output. This alternative does not alter the effective land price $(1 + \tau(n))q_t$ faced by a firm and therefore does not undo the land distortion. It also leaves the real innovation equilibrium unchanged: dividing each consumption-normalized firm value by $\mathcal{Y} = Y/C$ makes the HJB equations, expansion and entry conditions, and Kolmogorov–Forward equations independent of $\mathcal{Y}$, while goods market clearing determines only the consumption-output ratio. Because none of the calibration targets is a consumption-share moment, the calibrated parameters and model fit are unchanged across the two closures.

Under wasteful use, the pre-Repeal consumption share is $\mathcal{C}^{\text{pre}} - \mathcal{G}^{\text{pre}}$, rather than $\mathcal{C}^{\text{pre}}$ under the baseline rebate. The post-Repeal allocation is identical because immediate repeal implies $\mathcal{G}_t^{\text{post}} = 0$. Hence the wasteful-use welfare gain satisfies:

$$\ln(\gamma^{\text{waste}}) = \ln(\gamma) + \ln\left(\frac{\mathcal{C}^{\text{pre}}}{\mathcal{C}^{\text{pre}} - \mathcal{G}^{\text{pre}}}\right). \quad (\text{A.51})$$

Thus the wasteful-use alternative adds exactly the government-waste component to log welfare while preserving the static-misallocation, R&D-and-entry, and growth components.

A.5 Calibration

This section derives the model counterpart of each empirical moment targeted in the calibration. We adopt a *dual-sample* approach that reflects the structure of Indian manufacturing data. The model’s dynamic and stagnant firms are latent structural types, distinguished by access to the product-expansion technology. We use organized-sector plants in the Annual Survey of Industries (ASI) as the empirical analogue of dynamic firms and combine ASI and National Sample Survey (NSS) data for economy-wide moments. This is a sampling approximation, not a one-to-one classification by realized innovation: a structurally dynamic firm need not add a product during its observed life, and an ASI plant that does not do so is

not thereby assigned to the model's stagnant type. Some moments are computed over all firms (ASI+NSS), while cross-sectional and DiD moments condition on the ASI analogue of dynamic firms. Superscripts “pre” and “post” denote the pre-repeal and post-repeal BGPs, respectively.

TFP growth rate.

From equation (A.48), the economy-wide TFP growth rate is:

$$g = \delta \cdot \ln(\lambda). \quad (\text{A.52})$$

Average number of products per firm (all firms).

The total mass of products is equal to one and the total measure of firms is $M^D + M^S$, so the economy-wide average (ASI+NSS) is:

$$\mathbb{E}[n \mid \text{all firms}] = \frac{1}{M^D + M^S}. \quad (\text{A.53})$$

Average number of products per dynamic firm.

Among dynamic (organized-sector) firms, the average number of products is:

$$\mathbb{E}[n \mid \text{dynamic}] = \frac{N^D}{M^D} = \frac{\sum_{n=1}^{\infty} n m^D(n)}{\sum_{n=1}^{\infty} m^D(n)}. \quad (\text{A.54})$$

This moment, together with $\mathbb{E}[n \mid \text{all}]$, pins down the stagnant share $\omega \equiv M^S/(M^D + M^S) = (\mathbb{E}[n \mid \text{dynamic}] - \mathbb{E}[n \mid \text{all}])/(\mathbb{E}[n \mid \text{dynamic}] - 1)$.

Entry rate (dynamic firms).

The empirical entry rate is the share of ASI firms observed in their first year of operation. Since the ASI covers organized-sector (dynamic) firms, its model counterpart is the age share among dynamic firms:

$$s^D[0, 1] = \frac{\int_0^1 m^D(a) da}{M^D} \quad (\text{A.55})$$

where $m^D(a)$ is the age measure density of dynamic firms derived in the old-to-young section below.

Old-to-young firm ratio.

When the expansion rate $x(n)$ depends on n , a firm's current product count is correlated with its age. To compute the old-to-young ratio, we must therefore track the *joint* distribution of firm age and product count. The derivation proceeds in four steps: (1) define the age-product density, (2) write its KF equation, (3) sum the KF equation to obtain an equation for total survival, and (4) incorporate stagnant firms.

Let $m^D(n, a)$ denote the measure density of dynamic firms with exactly $n \in \{1, 2, 3, \dots\}$ products at age $a \in \mathbb{R}_+$, defined so that $m^D(n, a)da$ is the measure of such firms with age in $[a, a+da)$. (To distinguish from the product-count measure $m^D(n) \equiv \int_0^\infty m^D(n, a)da$ defined earlier, the age-product density always carries two arguments in this section.) On the BGP, all flow rates are time-invariant, and the entry flow of dynamic firms is $e \cdot d$ per unit time. Since every dynamic firm enters with exactly one product at age zero, the initial condition is:

$$m^D(n, 0) = \begin{cases} e \cdot d & \text{if } n = 1, \\ 0 & \text{if } n \geq 2. \end{cases} \quad (\text{A.56})$$

As a cohort of dynamic firms ages from a to $a + da$, its product count changes through two channels: *expansion* (gaining one product at total rate $n \cdot x(n)$ for a firm currently at state n) and *creative destruction* (losing one product at total rate $n \cdot \delta$). A firm at $n = 1$ that loses its sole product exits permanently. The KF equation governing the evolution of $m^D(n, a)$ in age a is:

$$\partial_a m^D(1, a) = 2\delta m^D(2, a) - [x(1) + \delta]m^D(1, a) \quad (\text{A.57})$$

for $n = 1$, and for $n \geq 2$:

$$\begin{aligned} \partial_a m^D(n, a) = & (n-1)x(n-1)m^D(n-1, a) \\ & + (n+1)\delta m^D(n+1, a) \\ & - n[x(n) + \delta]m^D(n, a). \end{aligned} \quad (\text{A.58})$$

Define the total measure density of dynamic firms at age a :

$$m^D(a) \equiv \sum_{n=1}^{\infty} m^D(n, a). \quad (\text{A.59})$$

We derive an equation for $\partial_a m^D(a)$ by summing the KF equation (A.57)–(A.58) over each

n . Summing the expansion inflow and outflow terms across all n :

$$\begin{aligned}\text{Expansion inflow: } & \sum_{n=2}^N (n-1)x(n-1)m^D(n-1, a) = \sum_{m=1}^{N-1} mx(m)m^D(m, a) \\ \text{Expansion outflow: } & \sum_{n=1}^N nx(n)m^D(n, a).\end{aligned}$$

The net contribution of expansion is:

$$\sum_{m=1}^{N-1} mx(m)m^D(m, a) - \sum_{n=1}^N nx(n)m^D(n, a) = -Nx(N)m^D(N, a). \quad (\text{A.60})$$

As $N \rightarrow \infty$, the density $m^D(N, a)$ decays to zero (all eigenvalues of the sub-generator have strictly negative real part), so the net expansion contribution vanishes:

$$\lim_{N \rightarrow \infty} Nx(N)m^D(N, a) = 0. \quad (\text{A.61})$$

This telescoping holds for any function $x(n)$, not only for constant x . The reason is purely algebraic: each expansion event moves exactly one unit of mass from state n to state $n+1$, so the in-flow at n from expansion is exactly the out-flow at $n-1$ from expansion. Summing over all n , the interior terms cancel pairwise, leaving only the boundary term at N , which vanishes in the limit. Summing the creative destruction inflow and outflow terms across all n :

$$\begin{aligned}\text{Destruction inflow: } & \sum_{n=1}^N (n+1)\delta m^D(n+1, a) = \delta \sum_{m=2}^{N+1} mm^D(m, a) \\ \text{Destruction outflow: } & \sum_{n=1}^N n\delta m^D(n, a) = \delta \sum_{n=1}^N nm^D(n, a).\end{aligned}$$

The net destruction contribution is:

$$\delta \left[\sum_{m=2}^{N+1} mm^D(m, a) - \sum_{n=1}^N nm^D(n, a) \right] = \delta [(N+1)m^D(N+1, a) - m^D(1, a)].$$

As $N \rightarrow \infty$, the first term vanishes, leaving:

$$\text{Net destruction} = -\delta \cdot m^D(1, a). \quad (\text{A.62})$$

The interpretation is clear: the *only* way a dynamic firm can leave the system is by losing its sole product when at $n = 1$, which occurs at rate $\delta \cdot m^D(1, a)$. All other destruction events merely shift firms from $n + 1$ to n without removing them.

Summing the KF equation (A.57)–(A.58) over n and using (A.61) and (A.62):

$$\partial_a m^D(a) = -\delta \cdot m^D(1, a) \quad \text{where} \quad m^D(0) = e \cdot d. \quad (\text{A.63})$$

This equation holds for *any* expansion rate schedule $x(n)$. It states that the rate at which the surviving dynamic firm density decreases with age equals the exit rate of single-product firms. However, equation (A.63) is *not* a closed equation for $m^D(a)$ alone, because $m^D(1, a)$ depends on the entire product-count distribution. To close it, one must solve the full infinite-dimensional system (A.57)–(A.58).

Stagnant firms enter at flow rate $e(1 - d)$ with a single product and exit at rate δ . Their age measure density is:

$$m^S(a) = e(1 - d)e^{-\delta a} \quad (\text{A.64})$$

with total mass $M^S = \int_0^\infty m^S(a) da = e(1 - d)/\delta$. Since the old-to-young ratio is computed from the ASI (organized sector), we condition on dynamic firms only. Define the dynamic-firm age share:

$$s^D[a_1, a_2] \equiv \frac{\int_{a_1}^{a_2} m^D(a) da}{M^D}. \quad (\text{A.65})$$

The old-to-young ratio targeted in the calibration is $s^D[40, 45]/s^D[1, 5]$. Because M^D cancels:

$$\frac{s^D[40, 45]}{s^D[1, 5]} = \frac{\int_{40}^{45} m^D(a) da}{\int_1^5 m^D(a) da}. \quad (\text{A.66})$$

When the expansion rate $x(n)$ depends on n , the dynamic measure density $m^D(a)$ has no closed-form expression: as noted above, equation (A.63) is not a closed ODE. Computing the ratio (A.66) therefore requires solving the KF system (A.57)–(A.58) numerically. We truncate the product-count state space at $\bar{n} = 20$, impose the terminal value boundary $V^D(\bar{n} + 1) = V^D(\bar{n})$, and hence set $x(\bar{n}) = 0$. The terminal KF equation has no inflow from $\bar{n} + 1$. We discretize the age axis on a non-uniform grid, solve the resulting finite-dimensional linear ODE system forward in age using the Crank–Nicolson scheme, and evaluate the integrals in (A.66) by quadrature. At the calibrated pre-Repeal equilibrium, only 0.0004% of dynamic-firm mass lies at $n = 20$. Increasing the truncation to $\bar{n} = 25$ changes the creative-destruction rate from 0.127771 to 0.127779 and the old-to-young ratio from 0.100492 to 0.100465, relative changes of 0.006% and 0.028%, respectively.

Innovation DiD (dynamic firms).

The expected number of new products per dynamic firm is:

$$\bar{A}^D \equiv \frac{\sum_{n=1}^{\infty} nx(n)m^D(n)}{M^D} = \sum_{n=1}^{\infty} nx(n)f^D(n). \quad (\text{A.67})$$

Since the innovation regression uses ASI (organized-sector) data, the DiD is conditioned on dynamic firms: $\beta_{\text{DiD}}^{\text{innov}} = \ln(\bar{A}^{D,\text{post}}) - \ln(\bar{A}^{D,\text{pre}})$.

Destruction DiD (dynamic firms).

Products held by dynamic firms are destroyed at rate δ per product. The destruction rate per dynamic firm is $\bar{D}^D \equiv \delta \cdot \mathbb{E}[n \mid \text{dynamic}] = \delta \cdot N^D / M^D$. The DiD is $\beta_{\text{DiD}}^{\text{destr}} = \ln(\bar{D}^{D,\text{post}}) - \ln(\bar{D}^{D,\text{pre}})$.

Dynamic entry DiD.

The flow of dynamic entrants per unit time is $e \cdot d$. This moment does not require sample conditioning since it identifies the entry of innovative (dynamic) firms directly. The entry DiD moment is $\beta_{\text{DiD}}^{\text{entry}} = \ln(e^{\text{post}}d^{\text{post}}) - \ln(e^{\text{pre}}d^{\text{pre}})$.

Exit DiD (dynamic firms).

Among dynamic firms, only those at $n = 1$ exit (by losing their sole product at rate δ). The exit rate among dynamic firms is:

$$\bar{E}^D \equiv \delta \cdot f^D(1) \quad (\text{A.68})$$

where $f^D(1) = m^D(1)/M^D$ is the fraction of dynamic firms with a single product. The exit DiD is a level difference (matching the LPM specification): $\beta_{\text{DiD}}^{\text{exit}} = \bar{E}^{D,\text{post}} - \bar{E}^{D,\text{pre}}$.

MRPLand DiD (dynamic firms).

The marginal revenue product of land for product i held by firm f is:

$$\text{MRPLand}_{ift} \equiv \frac{(1 - \alpha)p_{ift}y_{ift}}{\ell_{ift}} = \frac{(1 - \alpha)Y_t}{\ell_{ift}}.$$

From the cost-minimization first-order condition for land, the price-to-marginal-cost ratio times the effective land cost gives:

$$\text{MRPLand}(n) = \lambda q_t [1 + \tau(n)]^\alpha.$$

Taking logs: $\ln[\text{MRPLand}(n)] = \ln(\lambda) + \ln(q_t) + \alpha \ln[1 + \tau(n)]$. Under the parallel-trends assumption, time fixed effects absorb the common change in $\ln q_t$. Since the regression uses ASI data, the DiD coefficient is the dynamic-firm-weighted average:

$$\beta_{\text{DiD}}^{\text{MRPLand}} = -\alpha \cdot \sum_{n=1}^{\infty} \ln[1 + \tau(n)] f^D(n) = -\alpha \cdot \mathbb{E}_D[\ln(1 + \tau(n))]. \quad (\text{A.69})$$

The residualized outcome. The regression's dependent variable is not $\ln(\text{MRPLand})$ itself, but its sampling-weighted within-industry-by-year residual,

$$\ln(\widetilde{\text{MRPLand}}_{it}) = \ln(\text{MRPLand}_{it}) - \mathbb{E}^{\text{mult}}[\ln(\text{MRPLand}) \mid s(i), t],$$

where $s(i)$ is the plant's industry. This leaves equation (A.69) unchanged. The model has no industry dimension: every product shares the land elasticity $1 - \alpha$ and faces the same land rent q_t , so the industry-by-year cell mean of $\ln \text{MRPLand}$ is a function of t alone and is absorbed by the same time fixed effects that already absorb $\ln(q_t)$. Subtracting it removes nothing that varies with $\tau(n)$, so the model counterpart of the residualized-outcome DiD is the same $\beta_{\text{DiD}}^{\text{MRPLand}}$.

What the residualization buys is therefore external to the model. In the data, land elasticities and land-price dynamics differ across industries, so an industry-wide shock to the revenue-to-land ratio would otherwise enter the estimate; netting out the industry-by-year mean restricts the comparison to plants observed in the same industry and the same year. Because the model carries a single land share, this is a property of the empirical design rather than one the theory reproduces.⁵⁰

Measurement: book-value land. The empirical denominator is deflated book value at acquisition cost, not a physical land quantity. Writing $B_{it} = P_{it}^B L_{it}$, where P_{it}^B is the acquisition-vintage price index embedded in plant i 's book value, measured $\log \text{MRPLand}$

⁵⁰Only the dependent variable is residualized: the treatment indicators enter the additive fixed-effect regression directly, so the estimator is not numerically identical to placing industry-by-year fixed effects on the full regression. The two agree closely on our sample, -0.102 against -0.105 .

satisfies:

$$\ln(\widehat{\text{MRPLand}}_{it}) = \ln(\text{MRPLand}_{it}^{\text{phys}}) - \ln(P_{it}^B),$$

so the DiD coefficient on measured MRPLand equals its physical counterpart minus the differential change in $\ln(P_{it}^B)$. For continuing holdings P_{it}^B is fixed at historical acquisition values and contributes nothing to the differential. The index moves only when parcels enter the stock at current transaction prices, so the differential price component is proportional to the treatment-induced flow of additions relative to the stock of land, scaled by the gap between transaction and vintage prices. Since land sales rose alongside purchases (Table C.36) and departing parcels leave at low vintage values, the sales margin partially offsets the purchases margin. Mapping the measured coefficient directly to equation (A.69) therefore assumes the net differential movement in $\ln(P_{it}^B)$ is small relative to the estimated -0.102 ; the bound above is what that assumption requires, and the transaction, revaluation, district-total, and rent evidence cited in the main text speaks to it. A direct selection check complements the bound: regressing an indicator for contributing to the land sample (nonmissing, positive book value; 57.5% of plant-years) on the baseline specification over the full panel gives a $\text{Repeal} \times \text{UA}$ effect of 0.001 (standard error 0.013), so the Repeal did not differentially move plants into or out of land reporting. The estimate is similarly null when Andhra Pradesh and Maharashtra are dated to their formal repeals or excluded.

Loss function and moment weights.

The six parameters $(\lambda, \theta_X, \theta_E, \theta_T, \bar{\tau}, \kappa)$ are estimated by minimizing:

$$\mathcal{L}(\theta) = \sum_{j=1}^{10} w_j \left(\frac{m_j(\theta)}{d_j} - 1 \right)^2 \quad (\text{A.70})$$

where $m_j(\theta)$ is the model-implied moment and d_j is its empirical counterpart. The squared percentage deviation $(m_j/d_j - 1)^2$ provides a scale-free, sign-robust loss that treats all moments in comparable units.

The weight vector w reflects two principles. First, the total weight is split 2/3 to the cross-sectional moments and 1/3 to the DiD moments, proportional to the number of parameters each panel primarily identifies (four structural parameters and two policy parameters, respectively). Second, within each panel, weights reflect the information content of each moment.

Within the cross-sectional panel, the TFP growth rate, average products among dynamic

firms, and entry rate each receive weight $1/4$, as each primarily identifies a distinct parameter (λ , θ_X , and θ_E). The average products across all firms and the old-to-young ratio each receive weight $1/8$, as both primarily identify the type-selection parameter θ_T through their dependence on the stagnant-to-dynamic firm composition.

Within the DiD panel, moments are weighted by $(d_j/\hat{\sigma}_j)^2$, proportional to the squared signal-to-noise ratio of each DiD estimate. Here $\hat{\sigma}_j$ is the clustered standard error from the corresponding regression. Since the loss function uses percentage deviations, $\hat{\sigma}_j/|d_j|$ is the relevant standard error of the percentage miss (by the delta method), and weighting by its inverse variance $(d_j/\hat{\sigma}_j)^2$ gives more influence to more precisely estimated moments. Table A.11 reports the standard errors and resulting weights.

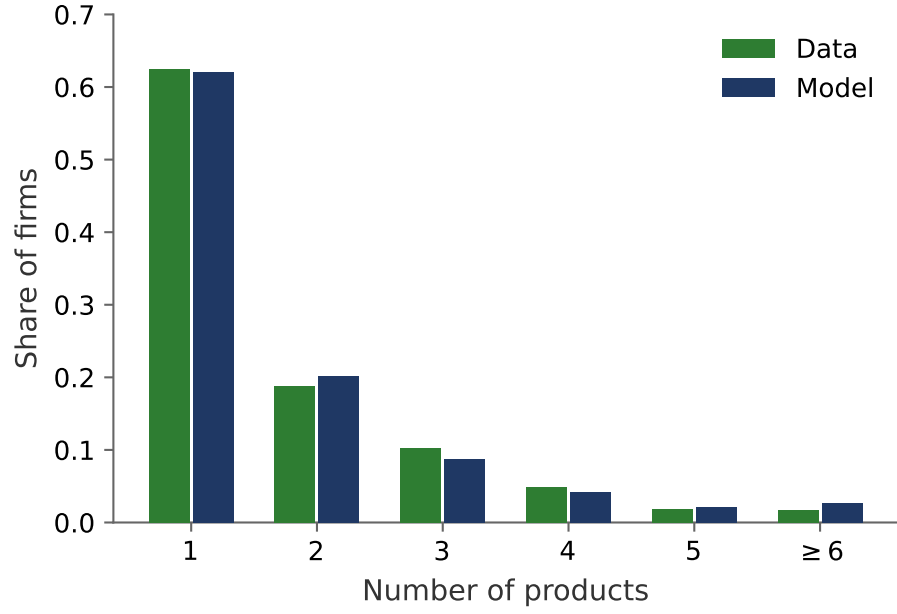
Moment	$\hat{\sigma}_j$	Within-panel	Overall w_j
<i>Panel A. Cross-sectional (2/3 total weight)</i>			
TFP growth rate	—	1/4	1/6
$\mathbb{E}[n \mid \text{all}]$	—	1/8	1/12
$\mathbb{E}[n \mid \text{dynamic}]$	—	1/4	1/6
Entry rate	—	1/4	1/6
Old-to-young ratio	—	1/8	1/12
<i>Panel B. Difference-in-differences (1/3 total weight)</i>			
Product creation DiD	0.051	0.230	0.077
Product destruction DiD	0.058	0.200	0.067
Entry DiD	0.115	0.154	0.051
Exit DiD	0.004	0.260	0.087
MRPLand DiD	0.048	0.155	0.052

Note: Panel A moments have no reported standard errors; within-panel weights reflect the parameter identification structure. Panel B within-panel weights are proportional to $(d_j/\hat{\sigma}_j)^2$, the squared signal-to-noise ratio, and normalized to sum to one. Overall weights sum to one.

Table A.11: Calibration weights

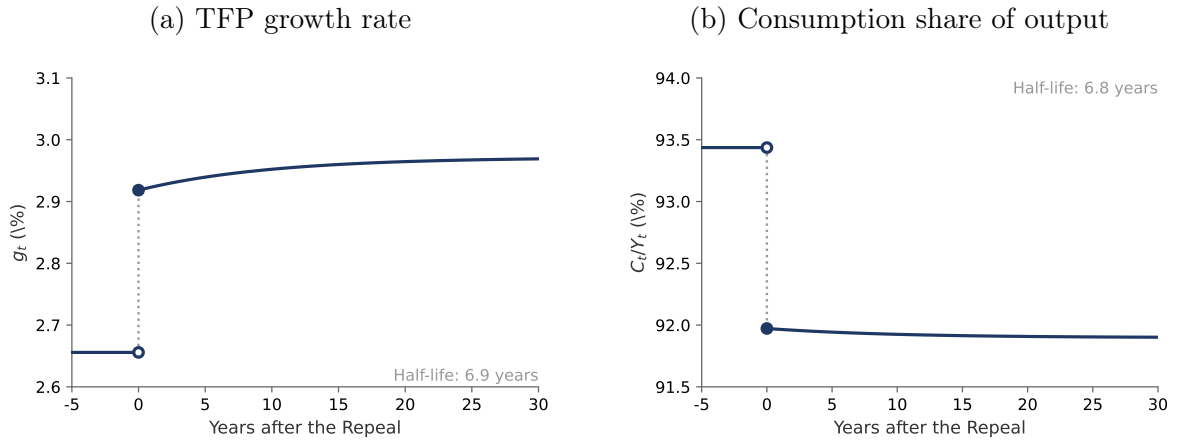
A.6 Model fit and dynamics

This appendix collects the model's fit to moments it was not calibrated to match, and the transition path between the pre- and post-Repeal balanced growth paths.



Note: Data: ASI pre-1999 urban-area sample (organized sector). Model: dynamic firm distribution $f^D(n)$ on the pre-repeal balanced growth path.

Figure A.4: Product distribution



Note: Transition paths after the unexpected permanent repeal of the ULCRA wedge $\tau(n)$ at $t = 0$. Panel (a): TFP growth rate $g_t = \delta_t \cdot \ln(\lambda)$. Panel (b): consumption share of output C_t/Y_t .

Figure A.5: Transition dynamics

B Additional figures and tables

B.1 Additional figures

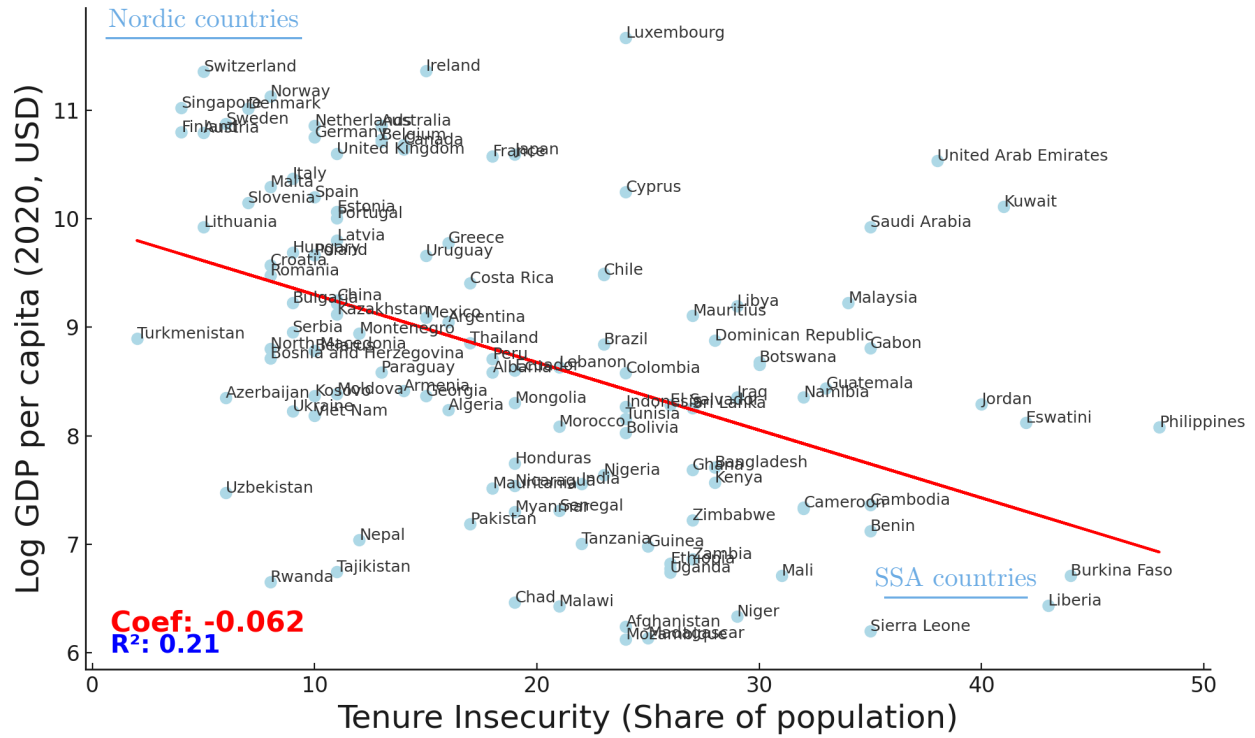


Figure B.6: Cross-country relationship between GDP per capita and tenure insecurity

Note: This figure illustrates the cross-country relationship between log GDP per capita and tenure insecurity, defined as the percentage of the population expecting to lose property rights against their will within 5 years. Tenure insecurity data is sourced from [Prindex](#) (2020).

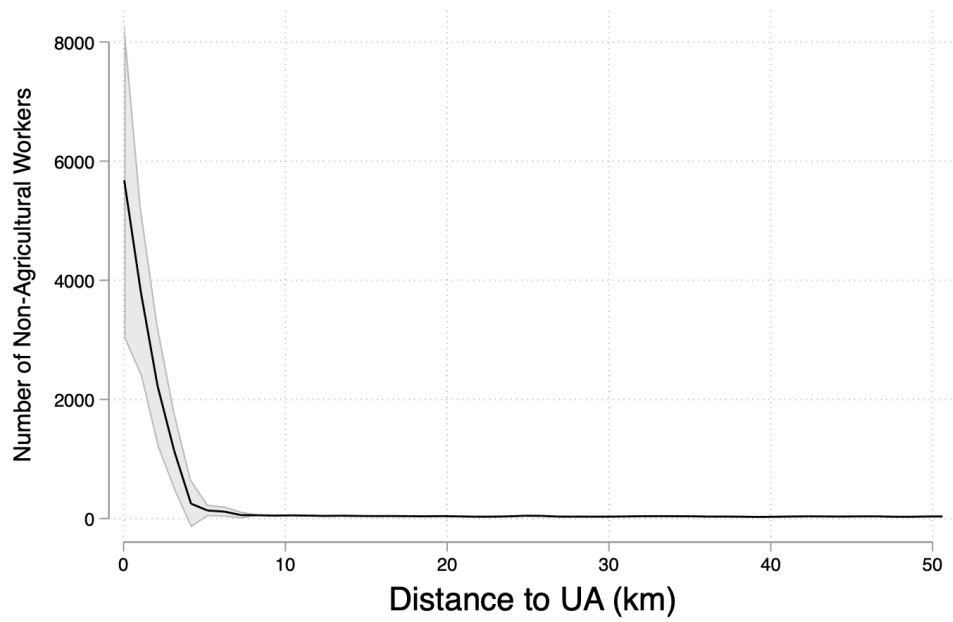


Figure B.7: Non-agricultural employment and distance to UAs

Note: This figure presents the distribution of non-agricultural employment at the village and town level against their distance from the centroids of Urban Agglomerations (UAs), using village-level population census data from 1991. The black line represents a local polynomial fitted line, and the shaded area indicates the 95% confidence interval.

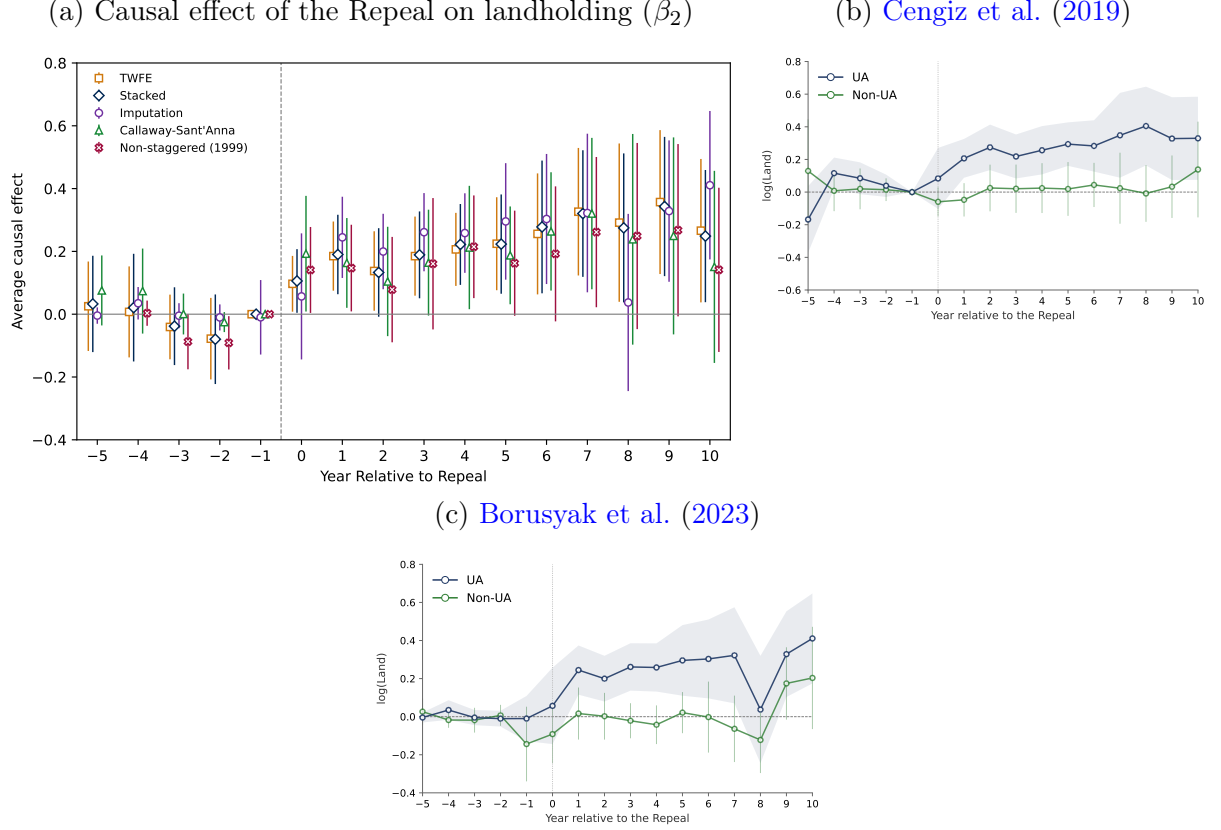


Figure B.8: Event study graphs for landholdings (staggered DiD)

Note: Panel (a) reports event-study estimates of the Repeal’s effect on UA landholdings from equation (1) using five staggered-DiD estimators: two-way fixed effects (orange square), the stacked DiD of [Cengiz et al. \(2019\)](#) (navy diamond), the imputation estimator of [Borusyak et al. \(2023\)](#) (purple circle), [Callaway and Sant’Anna \(2020\)](#) (green triangle), and a single-cohort triple-difference restricted to the 1999 repeal cohort (maroon cross). The two-way, stacked and single-cohort series identify the differential β_2 ; the imputation series reports the UA-specific effect, because that routine supplies no standard error for the difference; and the group-time series reports the effect on treated UA plants against a control pool of non-UA and never-treated plants, which nets out much of the non-UA response without being β_2 . The series share an event window, one set of nonmissing observations, and our baseline fixed effects—district, year, rural, industry, and land-size quartile interacted with year—so point estimates differ through the estimator rather than the specification. Three departures remain, each intrinsic to the estimator that carries it. `csdid` cannot absorb high-dimensional categorical controls at this sample size, so that series shares the sample but carries no covariates at all. The stacked estimator interacts every fixed effect with the sub-experiment but, like the other series, clusters at the state level, so every replicated copy of a state’s rows shares one cluster; it also omits the standalone year effect, which the land-size-quartile-by-year effect already spans, leaving the estimates unchanged. The single-cohort design is restricted by construction to the 1999 repeal cohort and the never-treated. All five series are unweighted, whereas the regression tables weight by the ASI sampling multiplier, so levels are not directly comparable with Table 1: the pooled specification on this sample gives 0.228 weighted against 0.300 unweighted. Panels (b) and (c) report both β_1 and β_2 for the stacked and imputation estimators respectively. All dependent variables are in logarithms; coefficients in panel (a) are normalized to zero one year before the reform; confidence intervals are at the 95% level.

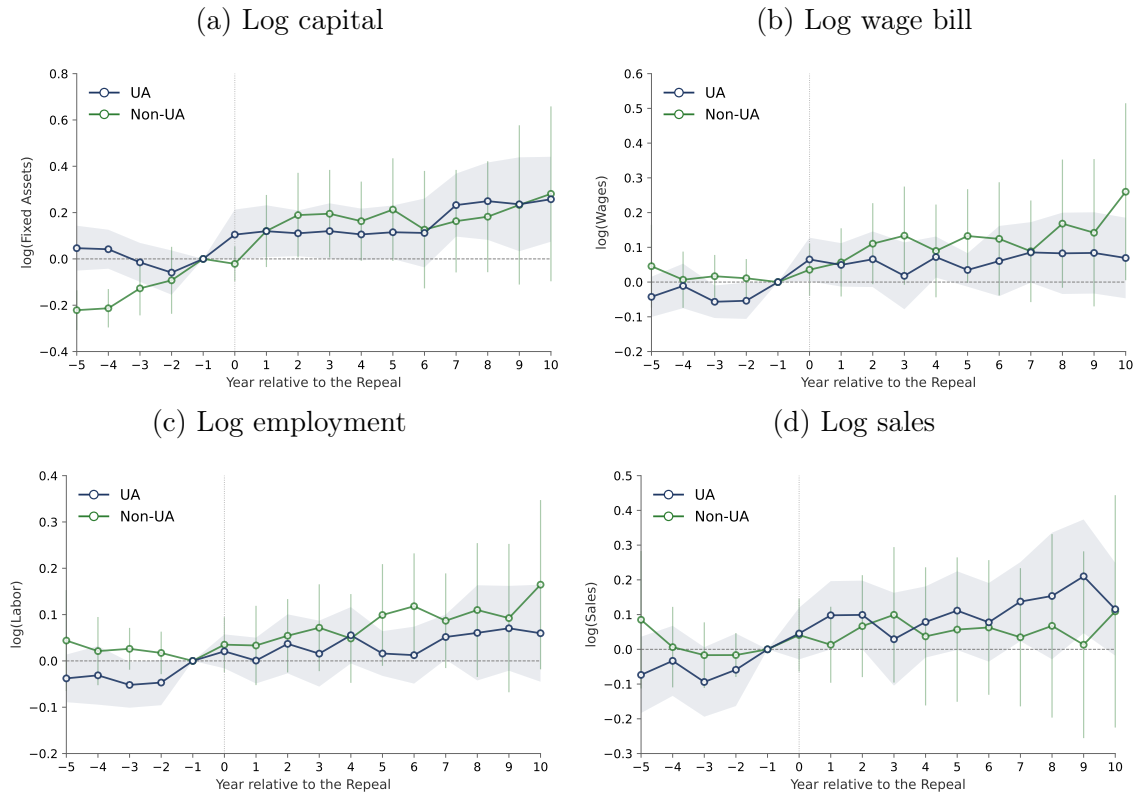


Figure B.9: Event study graphs for capital, wage bill, employment, and sales

Note: All dependent variables are in logarithms. Coefficients are normalized a year before the reform. The confidence interval is at the 95% level.

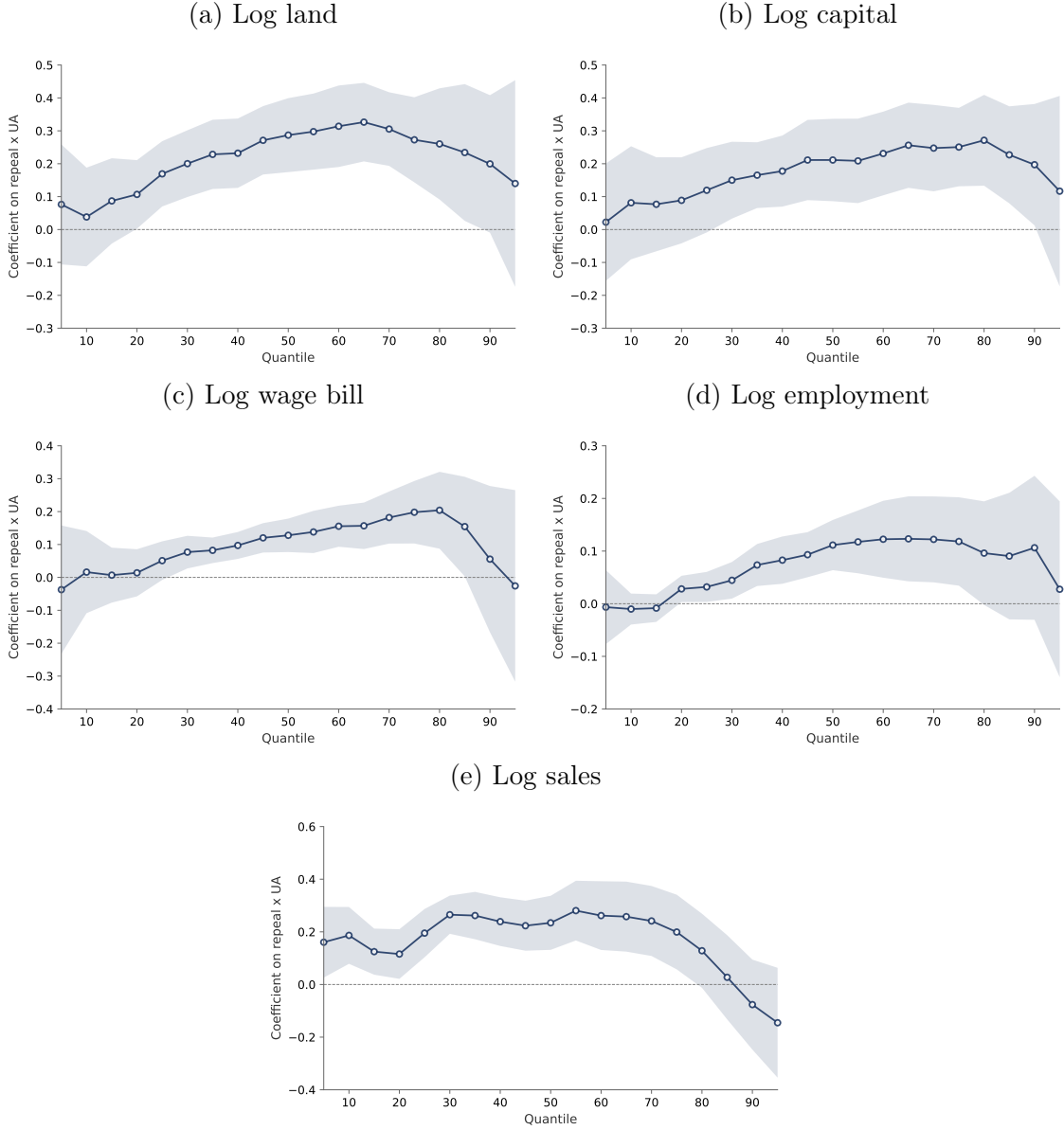


Figure B.10: Quantile treatment effects of the Repeal on firms in UA districts

Note: This figure reports the quantile regression estimates for the treatment effect of the Repeal Act on firm-level outcomes, from 5th to 95th quintiles, using the re-centered influence function approach of [Firpo et al. \(2009\)](#). All dependent variables are in logarithms. The confidence interval is at the 95% level, calculated using standard errors clustered at the state level.

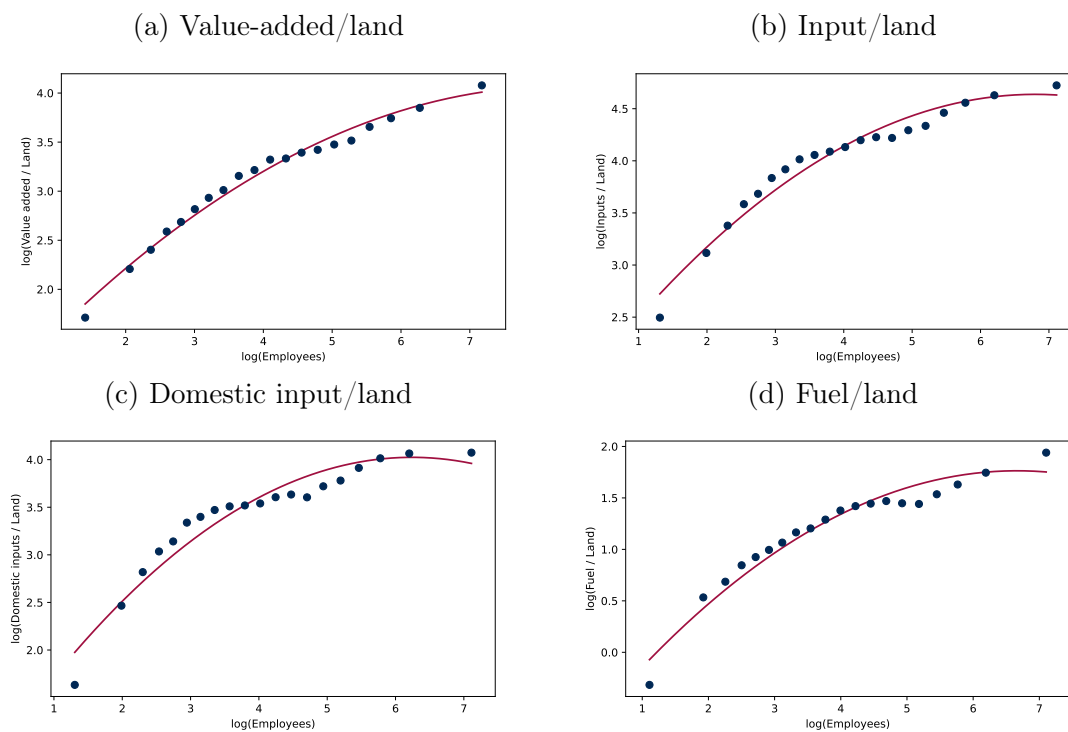


Figure B.11: Average value-added per land and firm size

Note: This figure reports the binned scatter plot and quadratic fitted line of log value-added and inputs per land against log employment: value-added (panel a), all intermediate inputs (panel b), domestic inputs (panel c), and fuels (panel d). All regressions control for 2-digit industry fixed effects and year fixed effects.

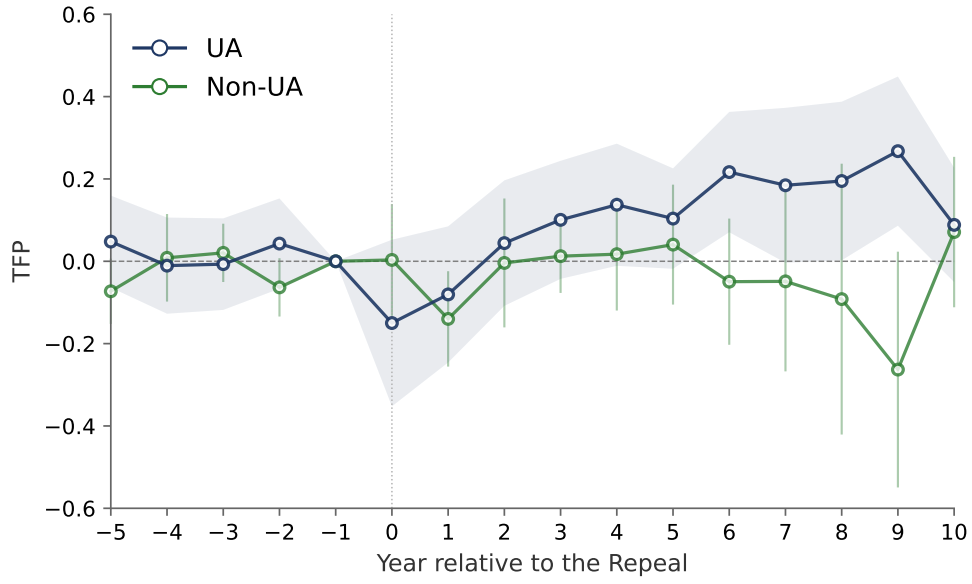


Figure B.12: Event study graphs for TFPQ

Note: The dependent variable (log TFPQ) is a production-function residual, estimated separately for each 2-digit industry by adapting the procedure of [De Ridder et al. \(2022\)](#) (see Section 4). Coefficients are normalized a year before the reform. The confidence interval is at the 95% level.

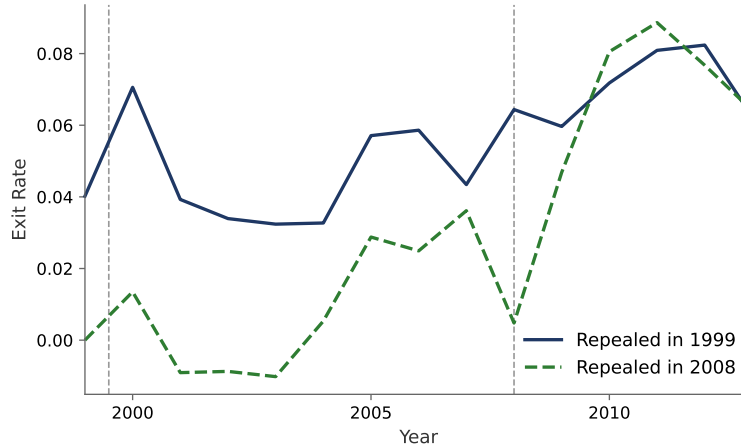


Figure B.13: Exit rates by repeal-timing cohort (descriptive)

Note: Weighted mean plant exit rates in the seventeen states and union territories that repealed the ULCRA in 1999 versus Andhra Pradesh and Maharashtra, the last states to repeal. Both series are shifted by a common constant equal to the late-repealing group's 1999 exit rate, so the vertical axis measures exit rates *relative to the late-repealing group in 1999* rather than exit rates in levels; the early cohort's positive value in 1999 reflects a pre-existing level difference. This exhibit is descriptive and is not the paper's identifying design: it compares cohorts of states rather than UA and non-UA plants within a state, it pools UA and non-UA plants, the later-repealing group contains only two states, and it uses formal (calendar) repeal dates rather than the ASI year containing the repeal, which is what the regressions use. Because the ASI status codes that identify exit begin only in 1998 and exit is identically zero in the first panel year, no pre-repeal observation exists for the 1999 cohort, so the figure displays no pre-treatment period and supports no parallel-trends test. Sampling-multiplier weights are used.

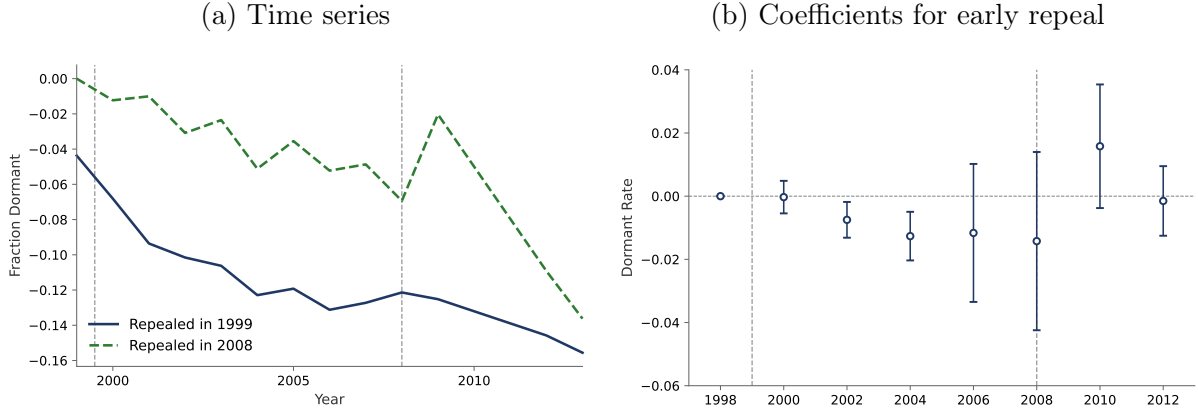


Figure B.14: Dormancy rates in the 1999 repeal cohort versus Andhra Pradesh and Maharashtra (descriptive)

Note: Panel (a) displays the time series of dormancy rates for the seventeen states and union territories that adopted the Repeal in 1999 (navy, solid) versus Andhra Pradesh and Maharashtra, the last states to repeal (green, dashed; labelled “Repealed in 2008” in the legend, their formal repeal dates being 2008 and 2007). Both series are shifted by a common constant equal to the late-repealing group’s 1999 dormancy rate, so the vertical axis measures dormancy *relative to the late-repealing group in 1999* rather than dormancy rates in levels. ‘Dormancy’ is defined by codes 2 and 3 in the variable ‘status of units,’ which include units found during the survey that were present at the location but did not initiate production and/or employ any staff during the reference period. The ASI did not record the dormancy codes in the 2010 and 2011 rounds, so neither series has an observation in those two years; panel (a) simply connects 2009 to 2012 across the gap, and the 2010–11 bin in panel (b) is estimated over rounds in which the indicator is identically zero and should not be read as an effect. Panel (b) plots the regression coefficients of our firm dormancy measure on an early repeal dummy variable, interacted with biennial dummies, controlling for industry-by-year, firm age, and state fixed effects; the pooled 1998–99 bin is the omitted reference. Like Figure B.13, this exhibit is descriptive and is not the paper’s identifying design: it compares cohorts of states rather than UA and non-UA plants within a state, it pools UA and non-UA plants, the later-repealing group contains only two states, and it uses formal (calendar) repeal dates rather than the ASI year containing the repeal, which is what the regressions use. Panel (a) uses sampling-multiplier weights; panel (b) is unweighted. The confidence intervals are set at the 95% level.

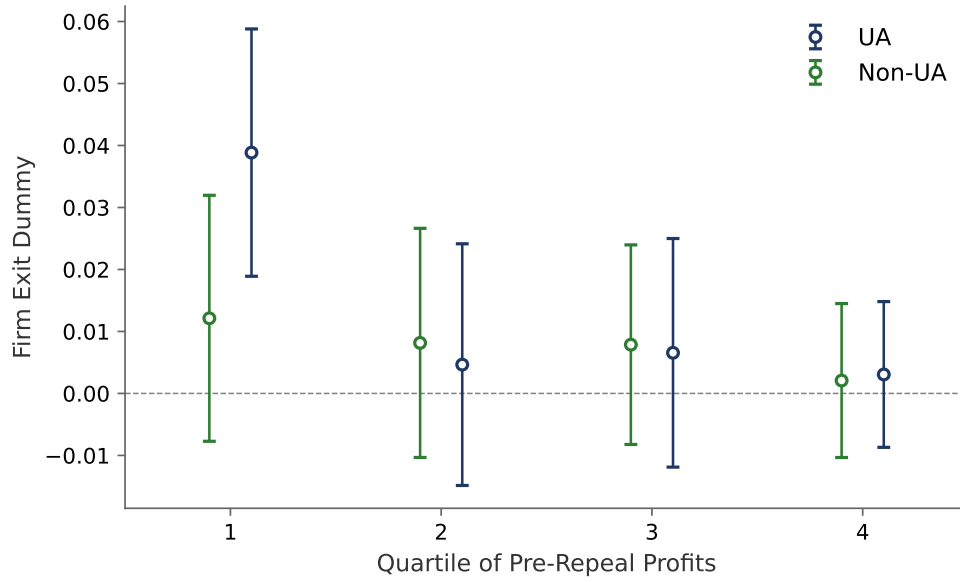


Figure B.15: Treatment effect of the Repeal on plant exit by pre-Repeal profits

Note: The dependent variable is a plant exit dummy, coded as 4 in the ‘status of units’ (as defined in Section 4.3). Quartile 1 represents the least profitable plants prior to the Repeal. Profit is measured as variable profits, calculated as sales revenue minus material and labor costs. Controls include a rural dummy, 2-digit industry fixed effects, firm age fixed effects, and land size quartile-by-year fixed effects. Standard errors are clustered at the state level. The confidence intervals are at the 95% level.

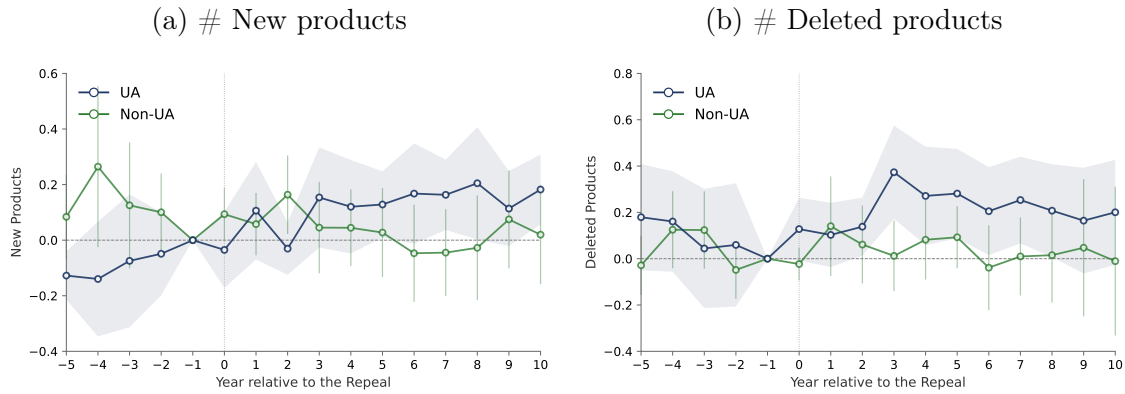


Figure B.16: Event study graphs for creative destruction

Note: The effects on adding/deleting products are estimated using the Poisson Pseudo-Maximum Likelihood regression. Coefficients are normalized a year before the reform. # New products is the number of new 5-digit products added to a firm’s product portfolio in a given year, and # Deleted products is the number of products deleted from the portfolio. The confidence interval is at the 95% level.

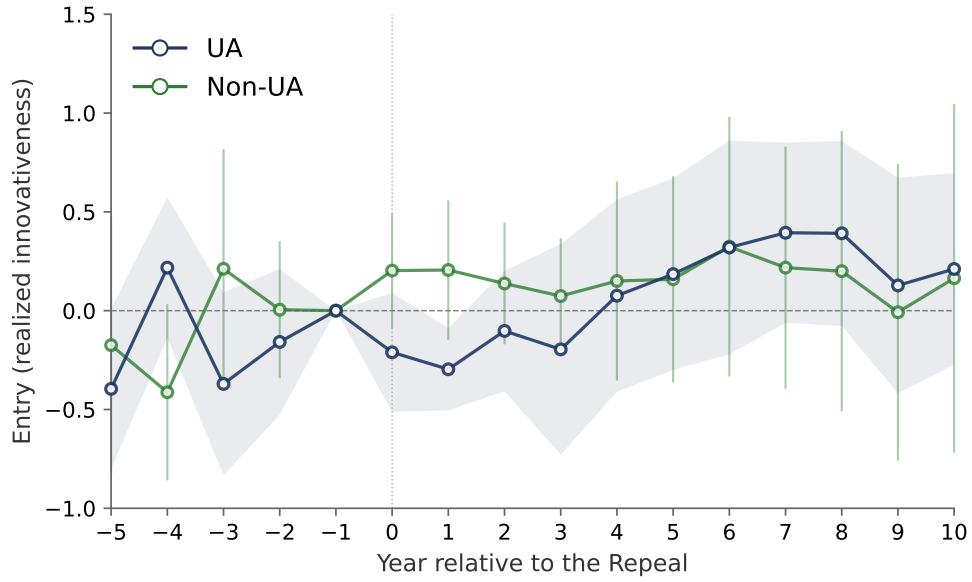


Figure B.17: Event study graphs for the entry of dynamic plants

Note: Dynamic plants are plants that have introduced new products during their life cycle. The data is aggregated to the district-year level using sampling weights. District-years with fewer than 30 plants are excluded. The analysis employs the Poisson Pseudo-Maximum Likelihood method, controlling for district fixed effects, year fixed effects, and district control-by-year fixed effects. These district controls-by-year are quartiles of two pre-Repeal metrics—the share of privately owned plants and the number of plants—each interacted with year fixed effects. A third candidate control, the pre-Repeal urban share, is deliberately excluded: among non-UA districts it carries no differential trend in innovative entry, so it is not a confound, but the entry response is concentrated in high-urban UA districts, so including it absorbs treatment-effect heterogeneity rather than omitted variation. Confidence intervals are set at the 95% level.

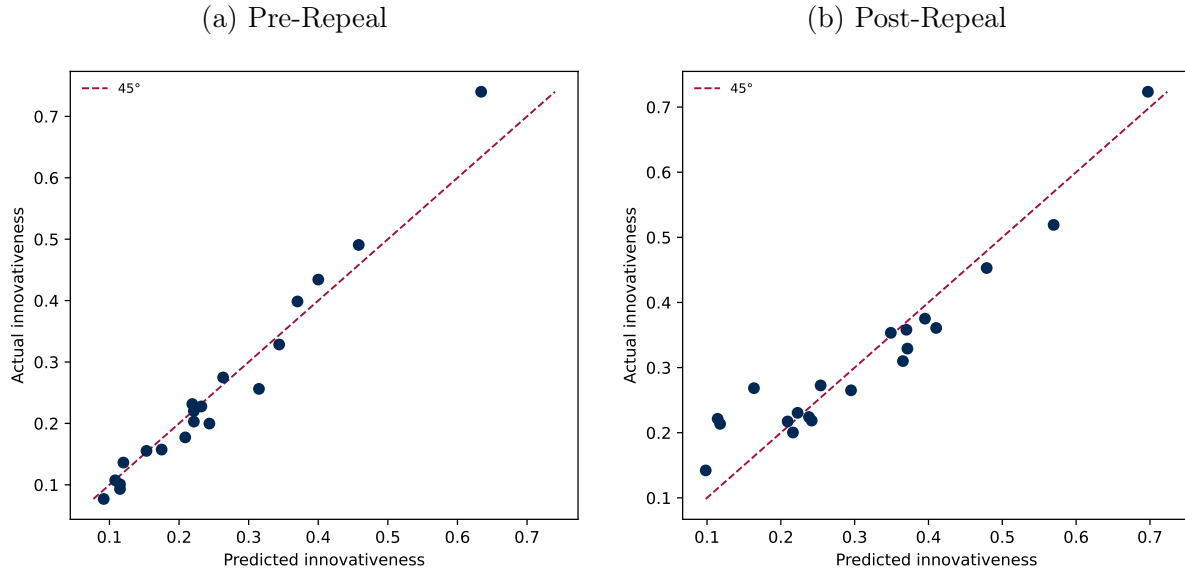
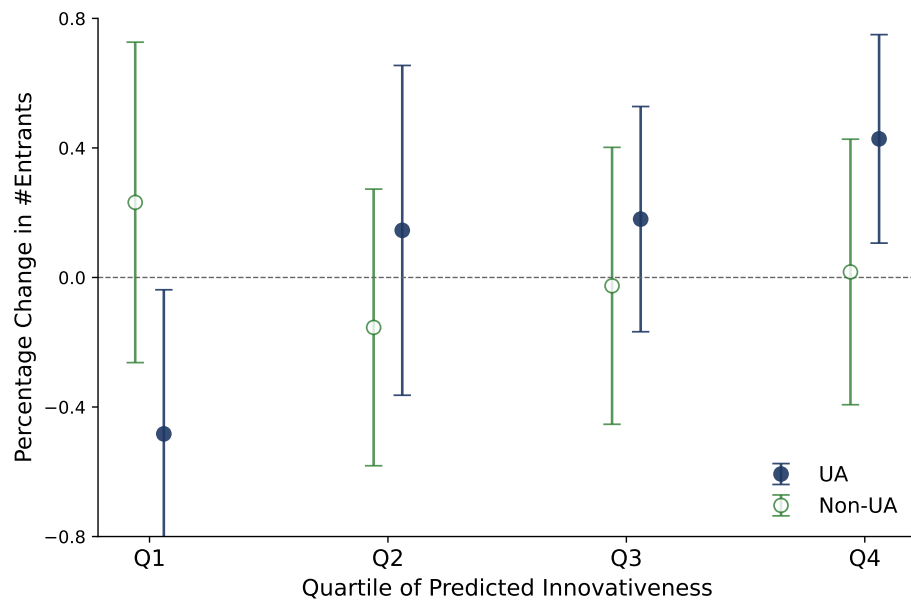


Figure B.18: Relationship between actual and LASSO-predicted dynamism

Note: This figure presents binned scatter plots comparing plants' average number of new products throughout their life cycle with our LASSO-predicted dynamism measure. Panels (a) and (b) show these relationships using pre-Repeal and post-Repeal data, respectively. Since the LASSO model is trained with pre-Repeal data, the left panel provides in-sample predictions, while the right panel offers out-of-sample predictions. Both panels include the correlation coefficients. The LASSO model is trained by regressing the number of new products introduced by plants throughout their life cycle on various plant-level characteristics recorded at entry, using pre-Repeal data. These characteristics encompass a range of dummies for private ownership, urban location, sole proprietorship, census scheme participation, importing foreign intermediates, and using computers, in addition to the total number of products. The model also includes the quadratic and interaction terms of these variables.



Note: This figure illustrates the treatment effects of the Repeal (β_2) on the number of entrants within each quartile of dynamism. Quartile 1 represents the least dynamic plants, while Quartile 4 represents the most dynamic, as determined by the LASSO-predicted dynamism measure.

Figure B.19: Treatment effects of the Repeal on entry across dynamism quartiles

Event study	Figure	Joint Wald statistic	p -value
<i>Main text figures</i>			
Land use	Figure 2	$\chi^2(4) = 7.14$	0.129
<i>Appendix figures</i>			
Land use, high-TFPR plants	Figure B.20	$\chi^2(4) = 21.86$	<0.001
Land use, low-TFPR plants	Figure B.20	$\chi^2(4) = 6.02$	0.197
Land use, high-TFPQ plants	Figure B.20	$\chi^2(4) = 15.97$	0.003
Land use, low-TFPQ plants	Figure B.20	$\chi^2(4) = 18.50$	<0.001
Capital	Figure B.9	$\chi^2(4) = 5.54$	0.236
Wage bill	Figure B.9	$\chi^2(4) = 8.66$	0.070
Employment	Figure B.9	$\chi^2(4) = 7.22$	0.125
Sales	Figure B.9	$\chi^2(4) = 6.09$	0.193
TFPQ	Figure B.12	$\chi^2(4) = 5.11$	0.276
New products	Figure B.16	$\chi^2(4) = 16.53$	0.002
Deleted products	Figure B.16	$\chi^2(4) = 5.83$	0.212
Dynamic entry	Figure B.17	$\chi^2(4) = 9.99$	0.041

Note: Each row reports the cluster-robust Wald test that all pre-Repeal differential coefficients (β_2 leads, together with the binned endpoint absorbing all earlier years) are jointly zero, computed from the same regression that underlies the listed figure. The statistic is invariant to the choice of reference period. The staggered-DiD estimator comparison (Figure B.8) re-estimates the land series with alternative estimators and is represented by the land row. Standard errors are clustered at the state level.

Table B.12: Joint pre-trend tests for the displayed event studies

B.2 Pre-trends: joint tests and sensitivity analysis

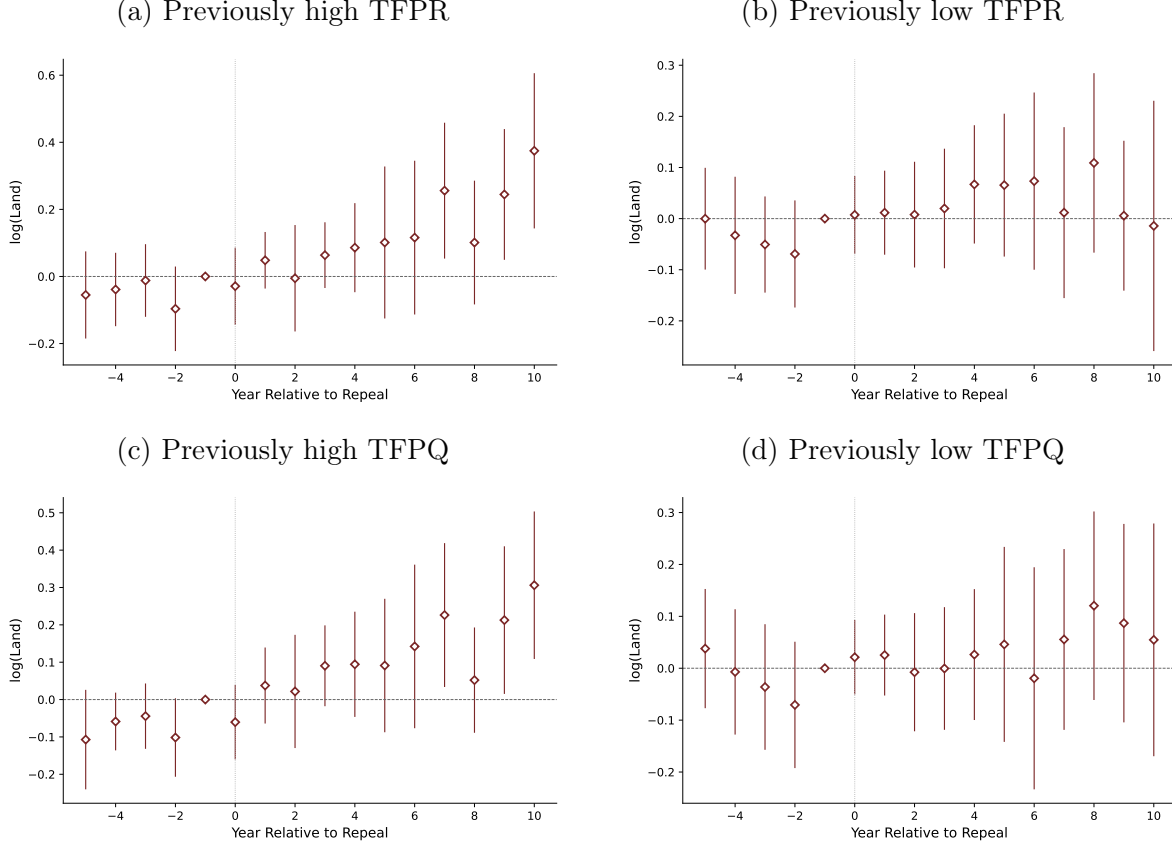
The main text reports a joint pre-trend test for the land event study and assesses the remaining figures through their plotted leads. This subsection collects the formal evidence in one place. For every event study displayed in the paper, Table B.12 reports a cluster-robust Wald test that the pre-Repeal differential coefficients—the β_2 leads, together with the binned endpoint that absorbs all earlier years—are jointly zero. Each statistic is computed from the same regression that underlies the corresponding figure, and it is invariant to the choice of reference period.

The five principal firm outcomes fail to reject flat pre-Repeal differential paths (p between 0.07 and 0.24), as do the TFPQ, deleted-products, and low-TFPR series. The tests do reject for three of the four productivity-split land panels (Figure B.20), for new products, and—marginally—for dynamic entry. Two considerations govern the weight these rejections can carry (Roth, 2022): a failure to reject is not evidence of parallel trends where power is limited, and a rejection need not indicate an economically meaningful violation, particularly

in large samples where a joint test aggregates several individually negligible leads. Magnitude and direction are therefore the relevant metrics. In the rejecting land splits and the new-products series, the estimated leads in the years approaching the reform are individually indistinguishable from zero at the five percent level and uniformly *negative*—between -0.01 and -0.14 —drifting away from the positive post-Repeal responses. The one individually significant pre-Repeal coefficient in these series is the new-products binned endpoint (-0.127 , absorbing all years at least four years before the reform), and it carries the same negative sign. A continuation of the pre-Repeal path would accordingly, if anything, understate the estimated effects. A declining relative pre-Repeal path for constrained high-productivity plants is, moreover, consistent with the paper’s mechanism: the land wedge binds progressively harder as productive plants expand, so UA plants fall increasingly behind their unconstrained counterparts until the constraint is lifted; the sensitivity analysis below makes this concrete for the high-TFPR panel. The dynamic-entry rejection reflects volatile leads of alternating sign in a sparse collapsed count panel; the entry-composition conclusion rests on Table 6 rather than on the event-study figure.

To quantify how much of a parallel-trends violation the conclusions could tolerate, Table B.13 reports the robust confidence sets of [Rambachan and Roth \(2023\)](#) for the land event study and the four productivity-split panels, targeting the first post-Repeal-year differential effect. We report two restriction classes. $\Delta^{SD}(M)$ bounds the period-to-period change in the slope of the differential trend by M , so it allows the estimated pre-Repeal path to continue approximately linearly into the post period; it is the natural class when the concern is trend-like drift. $\Delta^{RM}(\bar{M})$ allows post-Repeal violations up to $\bar{M}$ times the largest consecutive pre-Repeal deviation; because that benchmark is taken over *estimated* leads, it is mechanically conservative when the leads are noisy—for the land series the maximal consecutive pre-Repeal deviation is 0.065 log points, itself statistically indistinguishable from zero given lead standard errors of about 0.05.

Three results stand out. First, the first-year land effect (0.124, original confidence interval $[0.024, 0.224]$) remains positive and significant when the differential trend’s slope is allowed to change by up to $M = 0.01$ log points per year, and under relative-magnitude violations up to $\bar{M} = 0.3$ of the maximal—and statistically insignificant—observed pre-Repeal deviation. Second, the sensitivity analysis formalizes the direction argument: extrapolating the high-TFPR panel’s downward pre-Repeal drift (Δ^{SD} at $M = 0.01$) moves its first-year effect from an insignificant -0.03 to a significantly positive interval $[0.008, 0.196]$ —the estimated violations bias the split *against* finding an expansion by high-productivity plants. Third, the split panels’ event-study point estimates carry breakdown values of zero because their original



Note: The dependent variable is log-landholdings. TFPR and TFPQ are production-function residuals estimated for each 2-digit industry by adapting the procedure of [De Ridder et al. \(2022\)](#), as described in the text. The regression sample is segmented by whether firms' pre-Repeal average TFPR(Q) was above or below the district median. All regressions include plant fixed effects, year fixed effects, and controls incorporate both district pre-Repeal land endowment size bins and firm-specific land size bins, interacted with year fixed effects. Standard errors are clustered at the state level. The confidence interval is at the 95% level.

Figure B.20: The effect of the Repeal on land use by pre-Repeal TFP

first-year confidence intervals already include zero at conventional levels. The heterogeneity claim is established by the pooled interaction test in the main text (High–Low = 0.105, $p = 0.004$ for TFPR; 0.157, $p = 0.001$ for TFPQ); the event studies display its dynamics. The same bounds computed for the four production-outcome event studies (capital, employment, wage bill, and sales) are included in the replication package: three of the four first post-Repeal-year coefficients are not individually significant, and the wage-bill interval clears zero only marginally ($[0.002, 0.128]$), so the corresponding robust sets add little beyond the original intervals; the production results are established by the pooled estimates in Table 1.

	Estimate (first year)	Original 95% CI	$\Delta^{SD}(0.01)$ robust CI	$\Delta^{RM}(0.5)$ robust CI	Breakdown $M / \bar{M}$
Land use	0.124	[0.024, 0.224]	[0.003, 0.254]	[-0.017, 0.260]	0.010 / 0.3
Land use, high-TFPR plants	-0.029	[-0.144, 0.086]	[0.008, 0.196]	[-0.220, 0.109]	0.010 / 0.0
Land use, low-TFPR plants	0.007	[-0.069, 0.083]	[-0.078, 0.112]	[-0.116, 0.113]	none / 0.0
Land use, high-TFPQ plants	-0.060	[-0.160, 0.039]	[-0.086, 0.112]	[-0.227, 0.066]	none / 0.0
Land use, low-TFPQ plants	0.021	[-0.050, 0.093]	[-0.023, 0.151]	[-0.111, 0.116]	0.000 / 0.0

Note: Robust confidence sets of [Rambachan and Roth \(2023\)](#) for the first post-Repeal-year differential effect (β_2), computed from the three estimated leads and six estimated lags of each displayed event study. $\Delta^{SD}(M)$ bounds the period-to-period *change* in the slope of the differential trend by M (log points), so it allows the estimated pre-Repeal path to continue approximately linearly; $\Delta^{RM}(\bar{M})$ allows post-Repeal violations up to $\bar{M}$ times the largest observed consecutive pre-Repeal deviation. The breakdown value is the largest M (or $\bar{M}$) at which the robust confidence set excludes zero; it equals zero whenever the original confidence interval already includes zero, as it does for all four productivity-split panels, whose evidential weight is carried by the pooled difference-in-differences gradient test rather than by these event-study point estimates. Standard errors are clustered at the state level.

Table B.13: Sensitivity of the first post-Repeal-year effect to parallel-trend violations

Never partici- pated	Jammu and Kashmir; Kerala
1999	Gujarat; Haryana; Himachal Pradesh; Karnataka; Manipur; Meghalaya; Punjab; Rajasthan; Tripura; Uttar Pradesh; Uttarakhand; all Union Territories
2000	Chhattisgarh; Madhya Pradesh
2002	Odisha
2003	Assam
2006	Bihar
2007	Maharashtra
2008	Andhra Pradesh; Telangana
2011	Jharkhand
Still in effect	Tamil Nadu (state-specific land ceiling law remains in effect); West Bengal

Note: States are listed by their formal (calendar) repeal year. “All Union Territories” in the 1999 row comprises Chandigarh, Delhi, Dadra and Nagar Haveli, Daman and Diu, Pondicherry, the Andaman and Nicobar Islands, and Lakshadweep. Chhattisgarh and Telangana did not repeal independently: each inherits its parent state’s date (Madhya Pradesh and Andhra Pradesh, respectively), Chhattisgarh having been carved out in 2000 and Telangana in 2014, after our sample ends. The regression baseline instead uses fiscal-onset coding (the ASI April–March year containing the repeal) and dates Andhra Pradesh and Maharashtra to the 2006 JNNURM statewide commitment; see Section 2.

Table B.14: Timeline of the Repeal

B.3 Additional tables

Variable	N	Mean	p10	p50	p90
UA share (%)	597,356	50.0	0.0	0.0	100.0
Land stock (mn Rs.)	344,211	10.3	0.0	0.4	9.4
Fixed Assets (Million Rupee)	537,151	123.9	0.1	2.6	109.6
Wages (mn Rs.)	537,918	14.6	0.2	1.1	26.3
Sales (mn Rs.)	468,778	413.2	1.4	21.0	561.2
Land Adjustment Flows - Addition (Thousand Rupee)	339,093	359.5	0.0	0.0	19.2
$\mathbf{1}\{Add > 0\}$	339,848	11.1	0.0	0.0	100.0
Land Adjustment Flows - Deduction (Thousand Rupee)	338,808	67.2	0.0	0.0	0.0
$\mathbf{1}\{Deduct > 0\}$	339,016	3.0	0.0	0.0	0.0

Note: Reported statistics are weighted plant-level moments from the ASI analysis sample used in the GitHub pipeline.

Table B.15: Summary statistics of the ASI data

	(1)	(2)	(3)	(4)	(5)
	Land	Capital	Labor	Sales	MRPLand
UA	−0.030 (0.087)	−0.269** (0.111)	−0.131** (0.056)	−0.141** (0.058)	0.057 (0.070)
Linear trend	0.055*** (0.011)	0.182*** (0.014)	0.056*** (0.010)	0.108*** (0.011)	0.014 (0.010)
UA × trend	−0.014 (0.013)	−0.010 (0.012)	−0.002 (0.011)	−0.004 (0.013)	−0.001 (0.015)
Observations	92,557	160,436	163,847	138,821	78,769
R^2	0.098	0.169	0.120	0.170	0.097
Industry FE	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes

Note: Each column regresses a pre-Repeal outcome on the UA indicator, a linear time trend, and their interaction. These regressions are unweighted: the exercise tests whether UA and non-UA districts were on parallel paths before the reform, which is a statement about the estimation sample rather than about the population of registered plants. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table B.16: Pre-Repeal differences in time trends between UAs and non-UAs

	(1)	(2)
	Total land	Plant count
Repeal	0.079 (0.132)	-0.026 (0.089)
Repeal $\times$ UA	-0.012 (0.096)	-0.101* (0.056)
Observations	6,118	6,503
R^2	0.832	0.911
Year FE	Yes	Yes
District FE	Yes	Yes
Size $\times$ Year FE	Yes	Yes

Note: District-level regressions. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table B.17: Treatment effect of the Repeal on district-level total land and number of plants

	(1)	(2)	(3)
	MRPLand	MRPLand	MRPLand
Repeal	0.017 (0.044)	0.040 (0.040)	
Repeal $\times$ UA	-0.102** (0.048)	-0.137*** (0.045)	-0.185*** (0.040)
Observations	298,863	298,816	298,816
R^2	0.053	0.079	0.084
Year FE	Yes	Subsumed	Subsumed
District FE	Yes	Yes	Yes
Size $\times$ Year FE	Yes	Yes	Yes
3-digit Ind. $\times$ Year FE	No	Yes	Yes
State $\times$ Year FE	No	No	Yes

Note: The dependent variable is the *within-industry* marginal revenue product of land: $\log$ MRPLand (log sales over land) residualized on 2-digit-industry-by-year means. Level differences in land intensity across industries are already absorbed by the industry fixed effects; the industry-by-year adjustment additionally nets out industry-specific shocks over time, so the comparison is within industry and year. Column (1) is the baseline specification; columns (2)–(3) add 3-digit-industry-by-year and state-by-year fixed effects, respectively. ‘Land Size $\times$ Year’ are district average pre-treatment land-size bins interacted with year. The Repeal main effect is absorbed by the state-by-year fixed effects in column (3). MRPLand is measured as the ratio of sales revenue to the deflated book value of land. Because book value records acquisition cost rather than a current market valuation, a contemporaneous movement in land prices does not mechanically enter the denominator. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table B.18: Treatment effect of the Repeal on marginal revenue product of land

	(1)	(2)	(3)	(4)
	Land stock	Buy	Sell	Revalue
Repeal $\times$ UA	0.270*** (0.048)	0.244** (0.109)	0.221 (0.158)	-0.397 (0.297)
Observations	338,685	332,819	326,956	250,500
R^2	0.188			
Year FE	Yes	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes
Size $\times$ Year FE	Yes	Yes	Yes	Yes
State $\times$ Year FE	Yes	Yes	Yes	Yes

Note: Columns (2)–(4) report PPML estimates for land purchases, sales, and revaluation after adding state $\times$ year fixed effects. For the land-flow outcomes (purchases, sales and revaluations), positive values outside the 1st–99th percentile of the positive-value distribution are dropped, following the trimming applied in the source code; zeros are retained. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table B.19: Treatment effect of the Repeal on land adjustment: state $\times$ year FE

	Mean MRPLand		MRPLand dispersion		Land dispersion	
	(1)	(2)	(3)	(4)	(5)	(6)
	No FE	+ FE	No FE	+ FE	No FE	+ FE
<i>Panel A. Mean MRPLand</i>						
Lag 1						
Lag 2	0.008	0.003*				
	(0.005)	(0.001)				
Lag 3	−0.006	−0.001				
	(0.006)	(0.005)				
<i>Panel B. MRPLand dispersion</i>						
Lag 1						
Lag 2			−0.005	0.001		
			(0.011)	(0.005)		
Lag 3			0.008	−0.006		
			(0.009)	(0.004)		
<i>Panel C. Land dispersion</i>						
Lag 1						
Lag 2					−0.006	−0.011
					(0.014)	(0.007)
Lag 3					0.013*	0.002
					(0.006)	(0.003)
Observations	2,094	2,094	1,828	1,828	1,929	1,928
R^2	0.001	0.797	0.000	0.815	0.001	0.812
State FE	No	Yes	No	Yes	No	Yes
Year FE	No	Yes	No	Yes	No	Yes
Industry FE	No	Yes	No	Yes	No	Yes

Note: The dependent variable is repeal adoption. Column pairs report lagged MRPLand means, lagged MRPLand dispersion, and lagged land dispersion, with even columns adding state, year, and industry fixed effects. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table B.20: The impact of land misallocation and distribution on repeal timing

	(1)	(2)	(3)	(4)
	Land stock	Buy	Sell	Revalue
Repeal	0.025 (0.048)	0.075 (0.097)	0.129 (0.097)	0.122 (0.259)
Repeal $\times$ UA	0.164*** (0.046)	0.212* (0.112)	0.187 (0.123)	-0.314 (0.344)
Observations	338,681	333,227	324,832	208,705
R^2	0.190			
Year FE	Yes	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes
Size $\times$ Year FE	Yes	Yes	Yes	Yes
Ind. $\times$ Year FE	Yes	Yes	Yes	Yes

Note: Columns (2)–(4) report PPML estimates for land purchases, sales, and revaluation after adding industry $\times$ year fixed effects. For the land-flow outcomes (purchases, sales and revaluations), positive values outside the 1st–99th percentile of the positive-value distribution are dropped, following the trimming applied in the source code; zeros are retained. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table B.21: Treatment effect of the Repeal on land adjustment: sector $\times$ year FE

	OLS		LP		ACF	
	(1)	(2)	(3)	(4)	(5)	(6)
	High	Low	High	Low	High	Low
Repeal	0.010 (0.044)	0.051 (0.056)	0.023 (0.050)	-0.015 (0.048)	-0.032 (0.065)	0.011 (0.048)
Repeal $\times$ UA	0.143** (0.058)	0.097 (0.071)	0.166* (0.088)	0.061 (0.060)	0.100** (0.037)	0.043 (0.041)
Observations	46,621	45,073	43,901	44,576	45,957	47,242
R^2	0.880	0.872	0.869	0.882	0.889	0.890
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Size $\times$ Year FE	Yes	Yes	Yes	Yes	Yes	Yes
3-digit Ind. $\times$ Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Note: Alternative TFP estimators for the pre-Repeal split. All specifications include plant fixed effects. These regressions are unweighted. Because the sample is restricted to plants with a measurable pre-Repeal productivity level and observed on both sides of the reform, it is a selected subsample rather than a representative cross-section of registered plants, so the ASI sampling multipliers used elsewhere in the paper do not recover a population quantity here. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table B.22: Within-plant land growth by high/low pre-Repeal productivity plants: alternative measures of TFP

	(1)	(2)	(3)	(4)
	Capital	Labor	Wages	Sales
Repeal $\times$ UA	0.175*** (0.054)	0.068** (0.028)	0.084*** (0.028)	0.152*** (0.046)
Observations	530,008	533,802	531,119	464,755
R^2	0.257	0.151	0.217	0.246
Year FE	Yes	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes
Size $\times$ Year FE	Yes	Yes	Yes	Yes
State $\times$ Year FE	Yes	Yes	Yes	Yes

Note: Columns (1)–(4) are in logarithms. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table B.23: Treatment effect of the Repeal on firm production: state $\times$ year FE

	(1)	(2)	(3)	(4)
	Capital	Labor	Wages	Sales
Repeal	0.091 (0.086)	−0.010 (0.033)	0.009 (0.044)	0.024 (0.049)
Repeal $\times$ UA	0.117** (0.045)	0.033* (0.019)	0.051** (0.019)	0.088*** (0.023)
Observations	530,005	533,799	531,116	464,752
R^2	0.259	0.155	0.220	0.252
Year FE	Yes	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes
Size $\times$ Year FE	Yes	Yes	Yes	Yes
Ind. $\times$ Year FE	Yes	Yes	Yes	Yes

Note: Columns (1)–(4) are in logarithms. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table B.24: Treatment effect of the Repeal on firm production: sector $\times$ year FE

	(1)	(2)	(3)
	OLS	LP	ACF
Repeal	0.001 (0.074)	-0.022 (0.080)	-0.002 (0.073)
Repeal $\times$ UA	0.139** (0.066)	0.173*** (0.059)	0.119* (0.066)
Observations	359,848	358,500	359,432
R^2	0.092	0.081	0.080
District FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Size $\times$ Year FE	Yes	Yes	Yes

Note: Columns report alternative TFPQ estimators with district and year fixed effects. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table B.25: Treatment effect of the Repeal on TFPQ: alternative measures

	(1)	(2)	(3)	(4)	(5)
	TFPR OLS	TFPR LP	TFPR ACF	Price	Markup
Repeal	-0.000 (0.010)	-0.014 (0.022)	0.001 (0.012)	-0.001 (0.072)	0.016 (0.021)
Repeal $\times$ UA	0.017** (0.008)	0.047** (0.021)	-0.006 (0.011)	-0.110* (0.062)	-0.016 (0.019)
Observations	430,301	430,300	430,298	389,963	368,953
R^2	0.034	0.156	0.101	0.090	0.111
District FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Size $\times$ Year FE	Yes	Yes	Yes	Yes	Yes

Note: All specifications include district and year fixed effects. Firm-level prices are a quantity-weighted index of product unit values (ex-factory value divided by physical quantity), standardized within comparable products and aggregated to the firm; markups are estimated on the same quantity data. TFPR is revenue-based and TFPQ (Table 3) is quantity-based, so the two differ by this same price index. The TFPQ gain and the price decline are therefore two views of one movement rather than independent pieces of evidence: the estimates here and in Table 3 satisfy $TFPQ \approx TFPR - \text{price}$ (0.118 versus $0.017 + 0.110$) up to differences in estimation sample. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table B.26: Treatment effect of the Repeal on TFPR, firm-level prices, and markup

	(1)	(2)	(3)	(4)	(5)
	Land	Capital	Wages	Labor	Sales
Repeal	0.052 (0.080)	-0.122 (0.105)	-0.057 (0.059)	-0.022 (0.040)	-0.120 (0.088)
Repeal $\times$ UA	0.235** (0.098)	0.158 (0.104)	0.055 (0.065)	0.060 (0.059)	0.106 (0.087)
Observations	335,008	523,898	528,178	527,278	463,705
Year FE	Yes	Yes	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes	Yes

Note: Imputation estimates following [Borusyak et al. \(2023\)](#): untreated potential outcomes are imputed from a fixed-effects model fit on not-yet-treated observations, and the resulting treatment effects are projected onto the UA indicator. The ‘Repeal’ row is therefore the average effect for non-UA plants and the ‘Repeal $\times$ UA’ row the differential effect for UA plants. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table B.27: Accounting for potential bias in staggered difference-in-differences: imputation approach based on [Borusyak et al. \(2023\)](#)

	(1)	(2)	(3)	(4)	(5)
	Land	Capital	Wages	Labor	Sales
Repeal	0.069 (0.070)	0.236*** (0.084)	0.049 (0.066)	0.012 (0.047)	0.021 (0.064)
Repeal $\times$ UA	0.177*** (0.055)	0.144*** (0.049)	0.053* (0.026)	0.050* (0.025)	0.115** (0.047)
Observations	930,233	1,358,440	1,362,792	1,359,669	1,213,837
R^2	0.208	0.319	0.243	0.181	0.293
Stack $\times$ District FE	Yes	Yes	Yes	Yes	Yes
Stack $\times$ Year FE	Yes	Yes	Yes	Yes	Yes
Stack $\times$ Industry FE	Yes	Yes	Yes	Yes	Yes
Stack $\times$ Rural FE	Yes	Yes	Yes	Yes	Yes
Stack*Land Size*Year FE	Yes	Yes	Yes	Yes	Yes

Note: Stacked difference-in-differences following [Cengiz et al. \(2019\)](#): each repeal cohort is paired with the never-treated units in its own sub-experiment, and the sub-experiments are stacked. Fixed effects are the baseline set interacted with the sub-experiment, and the regression is weighted by the ASI sampling multiplier, so the estimates are comparable in level with Table 1. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table B.28: Accounting for potential bias in staggered difference-in-differences: stacked DiD based on [Cengiz et al. \(2019\)](#)

	(1)	(2)	(3)	(4)	(5)	(6)
	Land	Capital	Employment	Wage Bills	Sales	TFPQ
Repeal	0.105 (0.083)	0.023 (0.032)	−0.028 (0.039)	−0.020 (0.026)	0.015 (0.035)	−0.035 (0.132)
Repeal × UA	−0.028 (0.060)	0.064** (0.028)	0.022 (0.016)	0.032** (0.014)	0.026 (0.025)	0.193 (0.118)
Observations	185,147	185,147	185,147	185,147	185,147	185,147
R^2	0.915	0.956	0.941	0.954	0.945	0.788
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Size × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Plant FE	Yes	Yes	Yes	Yes	Yes	Yes

Note: Within-plant counterpart of Table 1: every column adds plant fixed effects to that table’s specification and all columns are estimated on one consistent sample—plant-years reporting all six outcomes simultaneously, which requires membership in the TFPQ estimation sample. The coefficients therefore identify within-plant adjustment by incumbents observed under both regimes, holding the set of operating plants fixed, and are directly comparable across columns. They are not comparable in level to Table 1, whose columns use the full, column-specific cross-sections of operating plants without plant fixed effects and so combine incumbent expansion with changes in operating-plant composition; neither table measures aggregate manufacturing outcomes, which additionally reflect entry and exit. The UA main effect is absorbed by the plant fixed effects. TFPQ is the quantity-based production-function residual of Table 3. Estimates are weighted by the ASI sampling multipliers, as in Table 1. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table B.29: Within-plant treatment effect of the Repeal on a consistent sample

Table B.29 re-estimates the production outcomes of Table 1 with plant fixed effects, on one consistent sample of plant-years reporting every outcome, so the coefficients isolate within-plant adjustment by incumbents observed under both regimes. Incumbents raise capital and wage bills significantly; employment, sales, and physical productivity move in the same direction but imprecisely; and the average incumbent’s landholding is flat. Read together with Table 2, which shows that within-plant land expansion is concentrated in high-productivity incumbents, the pattern implies that the aggregate land response in Table 1 operates through reallocation—expansion by more productive incumbents and changes in the set of operating plants—rather than uniform land accumulation by the average incumbent, which is the mechanism the model formalizes.

	(1)	(2)	(3)	(4)	(5)	(6)
	Land	MRPLand	Capital	Employment	Wage Bills	Sales
Repeal	0.007 (0.052)	0.013 (0.041)	0.037 (0.098)	-0.025 (0.036)	-0.010 (0.048)	-0.028 (0.048)
Repeal $\times$ UA	0.190*** (0.056)	-0.099** (0.048)	0.145** (0.059)	0.063** (0.026)	0.080*** (0.025)	0.148*** (0.045)
Observations	338,686	298,863	530,008	533,802	531,119	464,755
R^2	0.183	0.053	0.253	0.148	0.214	0.243
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Size $\times$ Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Note: Robustness of the principal outcomes to the treatment-timing convention for Andhra Pradesh and Maharashtra. The baseline dates their treatment to the 2006 statewide JNNURM repeal commitment (footnote 15); this table recodes the two states to their formal repeal years and re-estimates the specifications of Table 1 unchanged. Column (2) is the within-industry MRPLand outcome of that table.

Table B.30: Treatment effect of the Repeal under the formal AP/MH repeal dates

Table B.30 re-estimates the six principal outcomes with Andhra Pradesh and Maharashtra recoded from the baseline 2006 JNNURM-commitment dating to their formal repeal years. No coefficient moves by more than 0.012 and every outcome retains its significance level, so the treatment-timing convention for the two late-repealing states carries none of the results.

	(1)	(2)	(3)
	Base	+ Size-Year	+ Ind-Year
Repeal	−0.007 (0.008)	−0.006 (0.008)	−0.003 (0.007)
Repeal $\times$ UA	0.011** (0.004)	0.011** (0.004)	0.008** (0.004)
Observations	249,183	249,172	249,166
R^2	0.055	0.059	0.067
Year FE	Yes	Yes	Yes
District FE	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes
Size $\times$ Year FE	No	Yes	Yes
Ind. $\times$ Year FE	No	No	Yes

Note: All specifications include firm controls. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table B.31: Treatment effect of the Repeal on exit among non-dormant plants

	4-digit		3-digit	
	(1)	(2)	(3)	(4)
	New	Deleted	New	Deleted
Repeal	0.004 (0.053)	−0.042 (0.075)	−0.044 (0.072)	−0.071 (0.077)
Repeal $\times$ UA	0.113** (0.057)	0.114* (0.061)	0.148** (0.067)	0.132* (0.073)
Observations	264,424	272,645	264,238	272,253
Year FE	Yes	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes
Size $\times$ Year FE	Yes	Yes	Yes	Yes

Note: Columns (1)–(2) use 4-digit product definitions; columns (3)–(4) show the corresponding 3-digit robustness checks. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table B.32: Treatment effect of the Repeal on creative destruction: 4-digit product

Table B.33 separates within-plant portfolio churn from the composition of observed plants. Adding plant fixed effects to the turnover PPML restricts identification to continuing plants that ever innovate; among them the Repeal raises product additions and *lowers* deletions, so continuing innovators expanded their portfolios on net, and the baseline deletion response reflects in part the product retirements that precede exit—itsself a treatment effect documented in Table 4.

Two forces, both set in motion by the Repeal itself, reconcile the sign change on deletions. The first is a selection asymmetry in what the panel records. A plant’s pre-Repeal deletion history enters the sample in full, but a plant drawing a large post-Repeal deletion burst is disproportionately one that exits, and its worst years leave the recorded sample—through the endpoint convention, survey attrition, and gaps—so the within-plant comparison sets a truncated post-period against an untruncated pre-period. Because the Repeal raised exit precisely in treated markets, this truncation is stronger there and pushes the estimated differential downward. The second force is economic rather than mechanical: deletions in these data include a plant’s own decision to discontinue a product, and by lowering the effective cost of the land supporting marginal lines, the Repeal made previously borderline products worth keeping. Continuing plants accordingly pruned less even as they added more—the same portfolio widening that appears as increased product scope in Table B.34. Both readings imply that the aggregate rise in deletions is carried by the exit margin rather than by continuing plants, which is how the main text presents it.

	(1)	(2)	(3)	(4)
	#New Products	#Del. Products	#New Products	#Del. Products
Repeal	0.004 (0.049)	-0.035 (0.085)	0.011 (0.067)	0.062 (0.094)
Repeal $\times$ UA	0.132*** (0.051)	0.140** (0.058)	0.173*** (0.062)	-0.142** (0.058)
Observations	264,437	272,666	162,918	157,221
Year FE	Yes	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes
Size $\times$ Year FE	Yes	Yes	Yes	Yes
Plant FE			Yes	Yes

Note: Columns (1)–(2) repeat the baseline PPML product-turnover estimates of Table 5. Columns (3)–(4) add plant fixed effects, so identification comes only from within-plant variation; a plant whose outcome is zero in every observed year is perfectly fit by its own fixed effect and is dropped by the Poisson likelihood, so these columns estimate the within-plant response among plants that ever add (or ever delete) a product. The UA main effect is absorbed by the plant fixed effects. Within continuing innovators the Repeal raises additions and *lowers* deletions, so the baseline deletion response loads on the product retirements that precede exit rather than on portfolio churn among continuing plants.

Table B.33: Product turnover within plants

	(1)	(2)	(3)
	Base	+ State-Year	+ State-Year + Ind-Year
Repeal	0.002 (0.016)		
Repeal $\times$ UA	0.006 (0.011)	0.016 (0.012)	0.008 (0.013)
Observations	422,453	422,453	422,445
R^2	0.265	0.270	0.281
Year FE	Yes	Yes	Yes
District FE	Yes	Yes	Yes
Size $\times$ Year FE	Yes	Yes	Yes
State $\times$ Year FE	No	Yes	Yes
Ind. $\times$ Year FE	No	No	Yes

Note: Log-number-of-products specifications under progressively richer fixed effects. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table B.34: Treatment effect of the Repeal on product scope

	(1)	(2)	(3)
	All	Bin 1	Bin 2
Repeal	0.091 (0.119)	0.119 (0.152)	-0.117 (0.205)
Repeal $\times$ UA	-0.044 (0.151)	-0.177 (0.168)	0.441*** (0.149)
Observations	5,396	5,354	5,082
Year FE	Yes	Yes	Yes
District FE	Yes	Yes	Yes
Dist. size $\times$ Year FE	Yes	Yes	Yes
Private share $\times$ Year FE	Yes	Yes	Yes

Note: Columns split entrants by LASSO-predicted innovativeness: a LASSO trained on pre-Repeal entrants predicts realized product introduction over the plant's life cycle from characteristics observed at entry. Bin 2 contains entrants whose predicted innovativeness exceeds the 0.29 cutoff (the higher predicted innovativeness bin, roughly 27% of entrants); Bin 1 contains entrants at or below the cutoff. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table B.35: Treatment effect of the Repeal on firm entry: LASSO predicted innovativeness

C Additional empirical results

C.1 Data construction and measurement

ASI panel and district harmonization. The ASI reporting period is the Indian fiscal year, which begins on April 1 and ends on March 31; throughout the paper, a survey year is the calendar year in which that fiscal year begins. Although the official ASI data provide plant identifiers only after 1998, the separately purchased files contain identifiers beginning in 1993. The publicly available ASI files do not contain the district information required for our analysis. Following [Martin et al. \(2017\)](#), we match the panel files to an older cross-sectional version that reports districts through 2009, using time-invariant plant characteristics and opening and closing variables. After 2009, we use the panel structure to infer districts for the approximately two-thirds of plants that also appear before 2010; the results are robust to ending the sample in 2009. Because Indian district boundaries changed frequently, we harmonize all locations to time-invariant 1991 districts.

Land measurement. Our measure is the average book value of land at the beginning and end of the year. Book value records acquisition cost and changes with market conditions only through revaluation, which is rare—fewer than 1% of firms report a positive revaluation—while land does not depreciate. Land rented under arrangements that are not capitalized in fixed assets does not enter the measure. Such payments appear instead in a separate category of other expenditures that combines rent for land with royalties on mines and quarries; fewer than 10% of firms ever report a positive value in that category. Land is also reported less consistently than other asset categories: it is nonmissing for about 58% of plant-years and is absent from the 1997 survey round, which accounts for the smaller samples used for land-based outcomes.

Treatment timing. The repeal dates in our policy data are formal calendar dates, whereas ASI years run from April to March. We assign each repeal to the ASI year containing its formal date and code the subsequent ASI years as post-Repeal. The distinction matters for repeals enacted from January through March: the central Repeal Act of 11 January 1999 is assigned to ASI year 1998, Madhya Pradesh’s February 2000 repeal to 1999, and Andhra Pradesh’s 27 March 2008 repeal to 2007. For Maharashtra and Andhra Pradesh, the baseline instead dates treatment to their public statewide 2006 JNNURM commitment to repeal, whose institutional basis is documented in [Appendix E.3](#). Recoding these two states

to their formal repeal years moves no principal coefficient by more than 0.012 (Appendix Table B.30). Excluding them preserves the signs and magnitudes: the land effect is 0.170 with $p = 0.051$, with the wider standard error reflecting the loss of 2 of 27 clusters and roughly one-quarter of the sample; capital strengthens, while MRPLand attenuates to -0.071 and becomes insignificant, a change well within one standard error.

Productivity measurement. We construct TFPR and TFPQ in two stages. First, we correct observed output—deflated sales for TFPR and quantity sold for TFPQ—for measurement error by regressing it on time fixed effects and a third-order polynomial in input use. For TFPQ, we also control for prices and market shares within five-digit industries to account for imperfect competition, following De Ridder et al. (2022). Second, we use corrected output as the dependent variable in an IV-GMM procedure that estimates the revenue elasticity for TFPR and the output elasticity for TFPQ; the two productivity measures are the corresponding residuals. Physical output is deflated sales net of firm-level prices. For multi-product firms, we standardize each product’s unit price by the sales-weighted mean price of comparable products and then construct the firm price as the sales-weighted average of these standardized product prices, again following De Ridder et al. (2022).

C.2 Outcome construction for selection and innovation

Alternative MRPLand classification. One possible heterogeneity design would classify plants by their pre-Repeal MRPLand, adapting to land the pre-reform MRPK design of Bau and Matray (2023). Land adjusts slowly, however, so plants with high MRPLand may already be expanding their landholdings and would then exhibit faster subsequent land growth even without the Repeal. The corresponding event studies display the predicted mean-reverting pre-period drift: joint tests of the pre-Repeal coefficients reject at $p = 0.04$ for the high-MRPLand group and $p = 0.006$ for the low-MRPLand group. We therefore do not use this split; the underlying estimates remain available in the replication package.

Exit and dormancy. Before 1998, the ASI *status of units* variable distinguished only open and closed plants. Beginning in 1998, it reports more detailed reasons for inactivity and disappearance. We classify a plant as dormant under code 2, which denotes a unit present at the specified location that employed staff during the reference period but did not commence production or produced nothing, or code 3, which denotes a unit present at the location that neither employed staff nor produced during the reference period. We classify

a plant as exiting under code 4 (“deleted: found non-existent within 3 years”), code 6 (“non-response due to non-existence, owner not traceable”), code 11 (“deleted: found non-existent after more than 3 years”), or code 12 (“deleted due to deregistration”). We exclude the non-response codes for plants that remain in existence but submit no return (codes 5 and 7–10). Dormancy is not itself an exit: a dormant plant counts as exiting only if the ASI subsequently assigns it one of the deletion codes.

Because the detailed status codes begin only with the 1998 redesign and the earliest repeals take effect in fiscal 1998, no plant can be classified as ever dormant strictly before treatment in the earliest cohort. As a check, we restrict the sample to states repealing in fiscal year 2002 or later and classify plants using dormancy observed strictly before their state’s repeal. In the Column (1) specification of Table 4, $\text{Repeal} \times \text{UA}$ is 0.048 (standard error 0.019) for predetermined ever-dormant plants and 0.019 (standard error 0.009) for never-dormant plants, preserving the full-history ordering. Because only five treated states remain, cluster-robust inference in this exercise is indicative rather than exact.

Product turnover. Product additions and deletions are constructed from the ASI product schedules using five-digit ASICC codes over 1995–2009; codes are not comparable across the 1994 and 2010 classification revisions. A product counts as added in its first year of appearance unless that is the plant’s first year in the product data, and as deleted in its last year of appearance unless that is the plant’s last year. Excluding plant endpoints prevents a plant’s appearance or disappearance from mechanically registering its entire portfolio as additions or deletions. The rule does not remove product shedding in the years before a plant exits, which is a real part of the exit process and is recorded like any other deletion.

Appendix Table B.33 adds plant fixed effects to the PPML and therefore identifies within-plant portfolio changes among continuing plants that ever innovate. The addition response remains positive, at 0.173 ($p = 0.006$), while the deletion response becomes negative, at -0.142 ($p = 0.015$). The addition result is thus robust within plants, while the baseline deletion response combines continuing-plant churn with product retirements preceding exit. The discussion accompanying that table explains how endpoint-related recording asymmetry and reduced voluntary pruning among surviving plants reconcile the change in sign.

C.3 Land adjustments

Land market transactions. The main text shows that the Repeal increased land use and land *purchases* without a commensurate increase in land *revaluation*, consistent with lower

frictions in transacting land rather than a pure appreciation of existing holdings. Here we decompose the transaction response into the likelihood of transacting (extensive margin) and a floored-log measure of transaction scale, separately for purchases and sales, and find that the Repeal raised activity on both dimensions for both types of transaction, though three of the four estimates are significant only at the 10 percent level.

Table C.36 reports this decomposition. The extensive-margin columns (1) and (3) employ a Probit model; columns (2) and (4) report estimates for the floored-log outcome $\log(\max\{Y, y_{\min}\})$, which sets all zero values to the smallest positive land adjustment before taking logarithms. As Chen and Roth (2023) show, log-like transformations of outcomes with zeros do not isolate the intensive margin: the floored-log coefficient values movements between 0 and $y_{\min}$ at zero, but it still combines transaction incidence above $y_{\min}$ with transaction size, and its magnitude depends on the choice of floor. We therefore read columns (2) and (4) as treatment effects on the floored unconditional distribution of land adjustments rather than as volume effects conditional on transacting. Treated UA plants raised the probability of *buying* land by 0.9 percentage points (average marginal effect, relative to a sample mean of 10.9%), and the floored-log estimate implies an 8.1% increase in purchase activity. The Repeal also increased land *sales* on both dimensions: the likelihood of selling rose by 0.5 percentage points (relative to a 3.0% mean), with a 3.1% increase in the floored-log measure.⁵¹ That the reform raised turnover on *both* sides of the market—acquisitions and disposals alike—is the signature of lower transaction frictions rather than a one-sided asset revaluation.

⁵¹Columns (1) and (3) of Table C.36 report Probit index coefficients; the percentage-point effects quoted here are the corresponding average marginal effects. The floored-log regressions are run on the full sample with zeros recoded to the smallest positive adjustment; they are not conditional-on-transacting estimates.

	Add		Sell	
	(1)	(2)	(3)	(4)
	Extensive	Intensive	Extensive	Intensive
Repeal	0.017 (0.045)	0.008 (0.042)	−0.035 (0.035)	−0.016 (0.016)
Repeal × UA	0.052* (0.029)	0.078* (0.039)	0.070** (0.035)	0.031* (0.016)
Observations	334,588	334,583	334,307	334,302
R^2	0.039	0.030	0.033	0.010
Year FE	Yes	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes
Size × Year FE	Yes	Yes	Yes	Yes

Note: Columns (1)–(2) report land purchases (“Add”); columns (3)–(4) report land sales. Within each pair, the first column is the extensive margin (a Probit for the likelihood of transacting) and the second is a floored-log specification (log adjustment with zeros set to the smallest positive value); following Chen and Roth (2023), the latter is a treatment effect on the floored unconditional distribution, not an intensive-margin effect conditional on transacting. For the land-flow outcomes (purchases, sales and revaluations), positive values outside the 1st–99th percentile of the positive-value distribution are dropped, following the trimming applied in the source code; zeros are retained. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table C.36: Treatment effect of the Repeal on land adjustment: extensive vs. intensive

Table C.37 corroborates this pattern with PPML estimates on the full sample of plant–years, the vast majority of which record no transaction. Land *purchases* increase significantly (Column 2), while land *revaluations* show no detectable change (Column 4). Land *sales* carry a positive point estimate (Column 3) but are imprecisely estimated in levels.⁵² The extensive-margin and floored-log estimates above, which pool information across the many zero and small transactions, recover the sell-side response that the level specification is underpowered to detect. On balance, the reform is associated with greater activity in land markets on both sides—most sharply in acquisitions—rather than a pure appreciation of existing holdings, reinforcing the interpretation that the Repeal eased the *frictions* of transacting land rather than merely raising its price. Tables B.19 and B.21 re-estimate the same four PPML specifications with state-by-year and sector-by-year fixed effects, respectively; the land-stock and purchase responses survive both, while sales and revaluation remain imprecise.

⁵²Land transactions are sparse—about 3% of plant–years record a positive land sale and fewer than 0.3% a revaluation—so the level-PPML sales and revaluation columns are estimated with limited power.

	(1)	(2)	(3)	(4)
	Land	Buy	Sell	Rev.
Repeal	0.005 (0.057)	0.092 (0.089)	0.148 (0.104)	−0.176 (0.279)
Repeal × UA	0.202*** (0.056)	0.224** (0.104)	0.183 (0.133)	−0.034 (0.317)
Observations	338,686	333,805	330,681	297,070
R^2	0.183			
Year FE	Yes	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes
Size × Year FE	Yes	Yes	Yes	Yes

Note: Column (1) reports log-landholdings. Columns (2)–(4) report PPML estimates for land purchases, sales, and revaluation. For the land-flow outcomes (purchases, sales and revaluations), positive values outside the 1st–99th percentile of the positive-value distribution are dropped, following the trimming applied in the source code; zeros are retained. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table C.37: Treatment effect of the Repeal on landholdings and adjustments

Flexible land inputs. We further demonstrate that the Repeal reduced land adjustment costs, making it easier for firms to divest land in response to profitability shocks. Following Guiso et al. (2005), Fagereng et al. (2018), and Chatterjee et al. (2023), we measure plant-level profitability shocks by unanticipated changes in gross value added that cannot be explained by plant-specific characteristics and aggregate fluctuations. Specifically, we regress the logarithm of value-added (VA_{it}) on plant (δ_i), industry-year (δ_{jt}), and state-year (δ_{st}) fixed effects, as shown below:

$$\ln(VA_{it}) = \delta_i + \delta_{jt} + \delta_{st} + \epsilon_{ijst}.$$

The regression residuals, ϵ_{ijst} , thus capture shocks to a plant’s “profitability”, including variation in both technical efficiency and product appeal.⁵³ We define a dummy variable $Shock_{i,t-1} = 1$ if the residual is negative in the previous year.⁵⁴ We then interact this measure of negative shocks with our baseline DiD regressors, while controlling for plant,

⁵³An alternative method for measuring productivity shocks involves estimating the residual of the revenue function, or Total Factor Productivity (TFP), by deducting the cost of inputs from sales revenue. However, as noted by Decker et al. (2020), this TFP measurement is susceptible to errors due to omitted variables. In contrast, the plant fixed effects in our approach effectively absorb any plant-specific factors that do not change over time.

⁵⁴Note that all the estimates would be symmetric if we defined a positive shock as having positive residuals in the previous year.

industry-year, and state-year fixed effects, with standard errors clustered at the state level. Our identification leverages within-plant variation in responsiveness to negative shocks before and after the Repeal, compared to control firms in the same industry-year and state-year cells.

Table C.38 reports the regressions for landholdings (Column 1), number of employees (Column 2), and fixed assets (excluding land) in Column 3. Two patterns stand out. First, land adjustments appear considerably less flexible than adjustments in labor and other fixed assets such as machines and plants: facing negative shocks, plants divest their landholdings by 9.1%, compared with a 14.6% and 16.4% decline in labor and capital, respectively, as indicated by the first row. Each of these own-input responses is precise, but the three outcomes come from separate regressions estimated on different samples, so the individual tests against zero do not test the equality of the land, labor, and capital responses; we read the cross-input ranking as a comparison of point estimates. Second, plants in UA districts were, if anything, even less responsive to negative shocks before the Repeal (second row of Column 1, +3.7%), though this differential is not statistically significant.

The triple interaction in the last row of Column (1) is the coefficient of interest: it measures whether treated UA plants became more willing to shed land in response to a negative shock after the Repeal. Its sign is consistent with that interpretation, but the estimate is imprecise (-0.056 , standard error 0.040) and not significant at conventional levels, so we read it as suggestive rather than conclusive. Taken at face value, the implied post-Repeal responses of treated UA plants—the sum of the four shock-interacted coefficients in each column—would narrow the gap between land and the other inputs without closing it: -0.102 for land, against -0.158 for labor and -0.163 for capital. That remaining gap is statistically meaningful: on the common sample of plant-years observed for all three inputs ($N = 78,954$, where the estimates barely move), stacking the three outcomes with every regressor and fixed effect interacted with the outcome indicator—so that each column’s regression is nested exactly—and clustering by state, the post-Repeal shock response of treated UA plants remains significantly weaker for land than for labor ($p = 0.019$) or capital ($p < 0.001$); equality of the triple interactions themselves is rejected against capital ($p = 0.033$) but only marginally against labor ($p = 0.057$). We therefore treat the flexible-land-inputs evidence as corroborative of, rather than independent proof of, the transaction-cost interpretation developed above.

The corresponding triple interactions for labor and capital (last row of Columns 2 and 3) are close to zero and likewise insignificant, so the reform shows no sign of having changed the flexibility of those inputs. The contrast is therefore one of sign and magnitude rather than

	(1)	(2)	(3)
	Land	Labor	Capital
Negative shock _{<i>t</i>-1}	-0.095*** (0.028)	-0.158*** (0.024)	-0.179*** (0.016)
Negative shock _{<i>t</i>-1} × UA	0.036 (0.030)	-0.039 (0.025)	0.005 (0.025)
Negative shock _{<i>t</i>-1} × Repeal	0.013 (0.027)	0.009 (0.024)	0.005 (0.016)
Repeal × UA	0.088** (0.034)	-0.038 (0.026)	0.043 (0.046)
Negative shock _{<i>t</i>-1} × Repeal × UA	-0.056 (0.040)	0.030 (0.026)	0.006 (0.026)
Observations	79,470	106,935	107,046
<i>R</i> ²	0.924	0.949	0.971
Firm FE	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes
Ind. × Year FE	Yes	Yes	Yes

Note: All regressions include firm, state × year, and industry × year fixed effects. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table C.38: Input adjustments following negative shocks

of statistical significance: only land carries a point estimate pointing toward greater post-Repeal flexibility, which is consistent with the reform having relaxed land-market constraints specifically rather than frictions in labor and capital markets. More firmly, the Repeal × UA coefficient in Column (1) implies a statistically significant 9.2% within-plant increase in landholdings among plants that had not experienced a recent negative profitability shock. Because the shock is an innovation to value added net of plant, industry-year, and state-year averages, and the plant fixed effects absorb persistent profitability differences, this coefficient identifies expansion by plants in good recent conditions, not reallocation toward persistently more profitable plants; read alongside the (imprecise) post-Repeal shedding among shocked plants, it is consistent with landholdings tracking current conditions more closely after the Repeal.

C.4 Heterogeneity across the plant size distribution

We next examine the distributional consequences of the Repeal using the (re-centered) influence function approach proposed by [Firpo et al. \(2009\)](#). The quantile regression estimates in Figure [B.10](#) indicate that, for land, capital, employment, and the wage bill, the Repeal had small and insignificant, albeit positive, effects on plants at the lower end of the size distribution. These effects gradually increase, peaking between the 60th and 80th percentiles, and then fade at the very top of the distribution; sales are the exception, with effects that are already positive and significant at the bottom. This pattern is consistent with plants in the upper-middle of the size distribution benefiting most from the easing of land-market frictions, since they had greater need to buy, sell, and adjust landholdings.

To further probe this hypothesis, we conduct an analysis in the spirit of [Hsieh and Olken \(2014\)](#), examining the relationship between firm size and the average revenue product of land. Panel (a) of Figure [B.11](#) shows that, within each 2-digit industry and year, larger plants—as measured by employment—exhibit a systematically higher average revenue product of land. Under a Cobb–Douglas production technology, a higher *average* revenue product directly implies a higher *marginal* revenue product of land. This pattern is therefore consistent with larger plants facing higher effective land costs, reflected in their elevated returns to land.⁵⁵

⁵⁵A potential alternative explanation is that larger plants charge higher markups rather than face higher land costs. Panel (b) of Figure [B.11](#) addresses this concern by showing that larger plants also have higher input expenditures per unit of land. Another concern is that larger plants may rely more heavily on higher-quality or imported inputs, as emphasized by [Kugler and Verhoogen \(2012\)](#). To mitigate this possibility, panel (c) shows that the positive relationship between input value per unit of land and plant size persists when restricting attention to domestic inputs. Finally, panel (d) documents a similar pattern for fuel inputs, which are plausibly homogeneous across plants.

D Socioeconomic impacts of the Repeal

In the main text, we demonstrated that the Repeal of the Urban Land Ceiling and Regulation Act (ULCRA) unleashed manufacturing growth and improved the allocative efficiency of land. In this section, drawing on village-level data from three rounds of the Population Census and the 2011 Socio-Economic and Caste Census, we systematically investigate the socioeconomic impacts of the Repeal. Using detailed employment information from the Population Census, we explore how the Repeal Act facilitated the reallocation of agricultural laborers to non-agricultural sectors. The rich data on income, wealth, and housing in the Socio-Economic and Caste Census—available only as a single post-Repeal cross-section—allows us to examine whether the municipalities most exposed to the Repeal show signs that its industrial development came at the expense of the poor. Utilizing household expenditure data from the National Sample Survey, we also demonstrate that the Repeal had a negligible impact on residential rents in Urban Agglomerations. We conclude this section by noting that poverty rates decline with proximity to UAs, a pattern consistent with the Repeal’s development effects, although the cross-sectional SECC data cannot establish causality.

D.1 Data

Our labor market data are sourced from the Socioeconomic High-resolution Rural-Urban Geographic Dataset (SHRUG),⁵⁶ which integrates data from multiple rounds of the population and economic census. Specifically, we utilize the 1991, 2001, and 2011 village-level Population Census data from the Primary Census Abstract and Village Directory tables. This dataset offers comprehensive information on village infrastructure, demographics, employment, occupation, and population, which we use to construct variables related to labor supply in the agricultural, manufacturing, and service sectors, as well as self-employment. Additionally, we gather village-level income, wealth, and housing data from the Socioeconomic and Caste Census (SECC), a universal enumeration of household assets conducted in 2011. The SECC details household assets that are readily assessable, including roof and wall materials, number of rooms, and assets like agricultural equipment, vehicles, and mobile phones. By mapping data from the population census and the SECC to time-invariant municipalities—including both villages ($n=590,648$) and towns ($n=7,528$)—the SHRUG enables us to trace the evolution of these municipalities both before and after the Repeal. We discuss

⁵⁶Asher and Novosad (2020) provides details on the data construction, accessible at <https://www.devdatalab.org/shrug>.

variable construction along with the results.

We supplement this municipality-level data with a manually collected ULCRA dataset, which includes the repeal time for each state and the geographic coverage of the 1976 Act. The Act specifies numerous Urban Agglomerations (UAs) along with their associated towns and villages. We systematically gathered coordinates for over six hundred locales listed in these delineations through extensive online research. To ensure the accuracy of our data, we cross-referenced each locale with the Indian Village Directory to verify their existence and correct spellings, particularly for those not readily found online.

We overlay the map of UA locales with the map of municipalities to delineate groups of municipalities that were more or less exposed to the Repeal, determined by their proximity to UAs. Since the exact geographic boundaries of UAs are not known, we assume they are defined by the encompassing set of UA locales. We calculate the distance between each municipality and these UA locales. Figure D.21 illustrates this mapping for the Hyderabad UA. Red dots represent the UA locales under Hyderabad, while the colored polygons depict municipalities with available Population Census data. Darker colors indicate greater distances from the municipalities to these locales—up to 50 km. Larger polygons at the center of the circle identify towns around the Hyderabad metro—areas likely within the boundaries of the Hyderabad UA.

D.2 Identification strategy

To estimate the causal effects of the Repeal and to capture potential spatial spillover to nearby municipalities, we implement a Triple-Difference design following [Gechter and Tsivanidis \(2023\)](#). This design compares changes in outcome variables before and after the Repeal, between municipalities that are closer to UAs and those that are further away. Specifically, we group municipalities into 5km-distance rings around their closest UA, up to a distance of 50km. This non-parametric approach allows us to study the spatial effects of the policy. Municipalities outside the 50km radius around a UA are excluded from the analysis.

The main analysis relies on a spatial Triple-Difference model as described below:

$$y_{ijt} = \sum_{d=1}^{d=9} \beta_d (D[d_i = d] \times \text{POST}_{jt}) + X_i \times \text{Year}_t + \sigma_t + \sigma_i + \sigma_{jt} + \epsilon_{ijt} \quad (\text{D.71})$$

where y_{ijt} represents outcomes such as the share of agricultural laborers or the log of total population in municipality i , around UA j in year t . $D[d_i = d]$ indicates whether a munici-

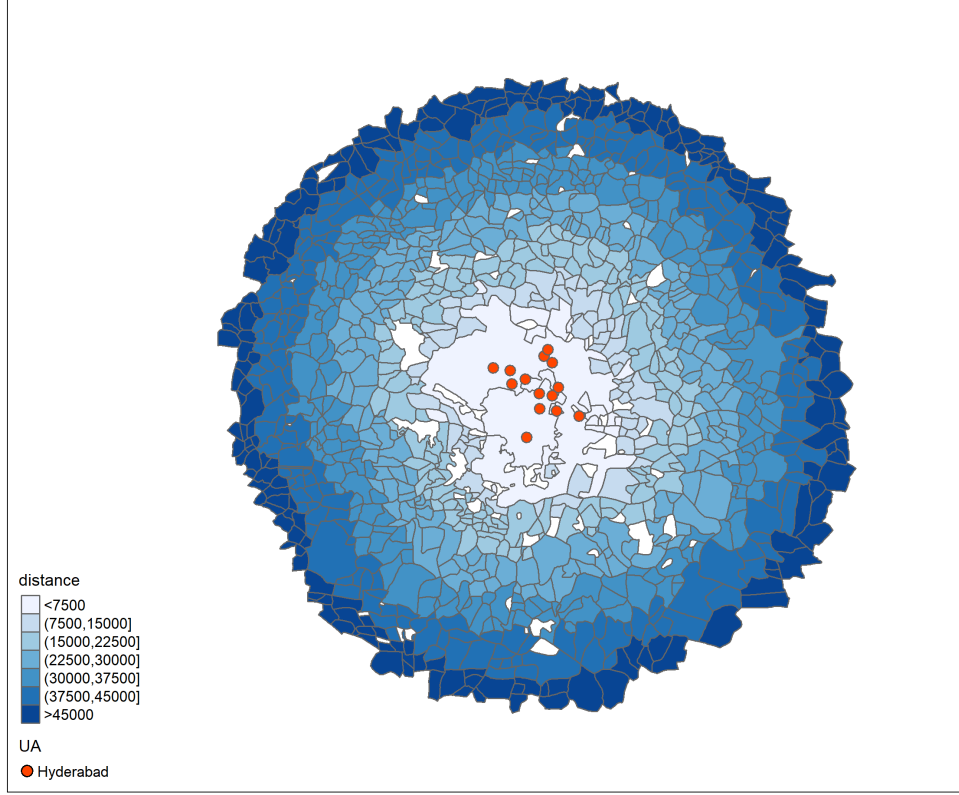


Figure D.21: UA locales under Hyderabad UA and nearby municipalities

pality i is in distance bin d to the nearest UA, with $d = 1$ representing municipalities within a 5km distance to the UA—likely municipalities within the confines of the UA—and extending up to 50km. Distance bin $d = 10$ (45-50km) is omitted and serves as the reference category. Our final sample has approximately 120,000 municipalities in a given year. We interact the distance dummies with a post-Repeal dummy, $POST_{jt}$, which is set to 1 if UA_j is located in a state that had repealed the ULCRA by year t . The model includes municipality fixed-effects σ_i and year fixed effects σ_t . We also include UA-by-year fixed effects σ_{jt} to ensure that we are comparing relative growth among municipalities—those closer and further away—from the same Urban Agglomeration over time. Additional controls $X_i \times Year_t$ include various municipality-level baseline covariates interacted with year fixed effects. ϵ_{ijt} is the error term; standard errors are clustered at the UA-year level, accounting for correlation across municipalities around the same UA within a census year.⁵⁷ The β_d are the parameters of interest, capturing the triple-difference estimates. They capture the differential changes in outcome

⁵⁷The spatial analyses in this appendix were estimated with code that predates the paper's replication pipeline and is not part of the replication package. We report the clustering level as recorded in the original figure notes; the exact covariance construction could not be independently re-verified.

variables across municipalities in distance bin d that experienced the Repeal, relative to those in the reference category, and contrast these changes with municipalities in states that have not yet implemented the Repeal.

Our identification assumption is that the relative growth rates of municipalities near UAs compared to those farther away do not differ significantly between states that adopted the Repeal early and those that adopted it later. This assumption could be violated if, for example, municipalities closer to urban centers in early-adopter states grow differentially faster than those in late-adopter states, even in the absence of the Repeal. While this assumption is untestable, we address potential concerns in two ways. First, we control for various baseline covariates—such as population size, distance to the nearest town, and proximity to roads—interacted with year fixed effects. These controls are intended to mitigate any pre-existing differential growth trends between closer and farther municipalities across early and late adopter states. Secondly, we perform a placebo test by substituting the distance-to-UA dummies with distance-to-town dummies. If the differential growth patterns were primarily driven by proximity to urban centers rather than the Repeal, we would expect to see similar patterns in the placebo test.

D.3 Results from the Population Census

Effects on structural transformation. We begin our analysis by examining the effects of the Repeal on structural transformation, which could be facilitated through both labor supply and demand channels. On one hand, the expansion of manufacturing firms post-Repeal is likely to create more job opportunities in the industrial sector. On the other hand, by removing arbitrary building and construction regulations, the Repeal could decrease urban rent, thereby increasing the labor supply in non-agricultural sectors.

Using data from three rounds of the Population Census, we construct municipality-level shares of cultivators, agricultural hired laborers, other workers (include both manufacturing and service workers), and workers in household enterprises, among main workers. Main workers are those who work for more than six months in their jobs. Figure D.22 plots the coefficients β_d of our empirical equation (D.71) with each dot indicating the coefficient for a specific distance bin. Panels (b) and (c) show that municipalities closer to UAs experience an increase in the share of other workers and a decrease in the share of agricultural hired laborers. For more distant municipalities ($> 25\text{km}$), the estimates for β_d turn out to be small and statistically insignificant from the reference group (45-50km), suggesting that employment changes between municipalities in further distance do not differ systematically.

The spatial spillovers on structural change are substantial in magnitude. For example, municipalities that are 10-15 km away from UAs witness an increase in the share of other workers by about 10 percentage points, which corresponds to approximately 30% of the sample mean (31.5%). Similarly, these municipalities observe a decrease in agricultural hired laborers by 4 percentage points, amounting to about 17% of the sample mean (25%). By contrast, Panels (a) and (d) suggest that changes in the shares of cultivators and workers in household enterprises are smaller in magnitude and often insignificant. Moreover, Panels (e) and (f) demonstrate that the Repeal not only changes the employment shares but also accelerates the growth of the total population and the number of main workers in municipalities close to UAs.

Effects on amenities. One might wonder how amenities—such as roads, schools, and electricity—respond endogenously to structural changes. These responses are particularly interesting as they could serve as potential mechanisms for the industrial growth results reported in our main text. For example, if the Repeal removed land constraints, it could allow developers in UAs to construct more roads, which in turn could boost manufacturing growth. To investigate this, we utilize municipality-level infrastructure data to construct various indicators of amenities, including the number of primary and high schools, the presence of paved and unpaved roads, and the availability of electricity for agricultural and general purposes. We then apply our spatial difference-in-differences (DiD) specification to assess the impact of the Repeal on these amenities. Note that, unlike our baseline specification, which controls for baseline amenities-by-year fixed effects, this analysis includes only baseline quartiles of population and distance to towns, interacted with year fixed effects. Interestingly, in Figure D.23 we do not observe accelerated growth of amenities in municipalities closer to UAs. In fact, municipalities near UAs experience somewhat slower growth in paved roads, although this effect is relatively modest in magnitude.

D.4 Results from the 2011 Socio-Economic and Caste Census

We now turn to household income, housing conditions, wealth, and poverty, using data from the 2011 Socio-Economic and Caste Census (SECC). Note that this is the only round of the SECC that contains household income and wealth information. Consequently, we are unable to estimate our baseline spatial DiD specification, as all UAs were treated by 2011. As an alternative, we examine the relative differences in the *levels* of outcomes between places closer to UAs and those further away. These distance gradients are associations rather than

treatment effects: they cannot separate the effects of the Repeal from pre-existing differences between municipalities closer to and farther from UAs. We therefore read them as a check on whether the areas most exposed to the Repeal show signs of deterioration, not as causal estimates.

Housing conditions. We begin with housing conditions. As noted by [Gechter and Tsvanidis \(2023\)](#), urban redevelopment could have adverse effects on the local population, particularly if local dwellers are forcibly relocated. In our context, if residential land near UAs is converted to industrial land following the Repeal, this might lead to a deterioration of housing conditions in nearby municipalities. To assess this, we consider several indicators of housing conditions, including the share of households that own or rent a house, the share of homeless individuals, the average number of rooms per house, and the proportion of houses with solid walls and roofs.

Figure [D.24](#) demonstrates that municipalities closer to UAs tend to have a smaller share of households that own a house, but a higher share renting. The differences in the share of homeless individuals and the average number of rooms are negligible and statistically insignificant. Additionally, municipalities nearer to UAs are more likely to have houses with solid walls and roofs. Taken together, these gradients show no sign of worse housing conditions near UAs. They cannot isolate the effect of the Repeal from pre-existing differences by proximity, but they are consistent with no deterioration in housing conditions following the Repeal.

Income and poverty. We then examine municipality-level income, poverty rate, and estimated consumption. Although consumption was not directly surveyed by the SECC, the SHRUG provides imputed consumption information, for each municipality, based on another household survey (IHDS-II, 2011-12), which includes similar asset, income, and land data as the SECC. SHRUG reports mean predicted consumption and the share of the population below the poverty line (27 Rupees/day) for each municipality. Figure [D.25](#) shows that municipalities closer to UAs tend to have higher income, more likely to have salaried wage, consume more, and exhibit a lower poverty rate. The poverty-rate gradient extends well beyond UA boundaries—municipalities 25 km away from UAs have a poverty rate approximately 2 percentage points lower than the reference group, which constitutes about 6.5% of the sample mean (31%). By contrast, the gradients in income and in salaried wage from the government tend to be more localized and diminish more rapidly with distance. In fact, the differences in earning more than 10,000 Rupees per month and in having a government salaried wage are

only significant within the 0-5 km distance bin, suggesting that these patterns are primarily driven by municipalities *inside* UAs. By contrast, the gradient in private sector salaried wages extends farther and is substantial in magnitude. For instance, municipalities that are 20-25 km away tend to have 1 percentage point more households earning a salaried wage from the private sector, which constitutes approximately 25% of the sample mean (3.9%).

We then examine how the main sources of household income vary with proximity to UAs. Consistent with our findings on structural transformation, Figure D.26 demonstrates that municipalities closer to UAs are less likely to report cultivation as the main source of income, and more likely to derive income from manual labor. Additionally, the share of households who report begging or foraging as the main source of income does not vary with proximity to UAs. This cross-section cannot rule out displacement that occurred before 2011 or pushed households beyond the 50km radius, but it shows no concentration of destitute households near the treated UAs.

D.5 Placebo test

As previously noted, our identification assumption is that the relative growth rates of municipalities near Urban Agglomerations (UAs) compared to those farther away do not differ significantly between states that adopted the Repeal early and those that adopted it later. This assumption could be violated if, for instance, municipalities closer to urban centers in early-adopter states exhibit differentially faster growth than those in late-adopter states, even in the absence of the Repeal. In our baseline regression, we always control for various baseline covariates—such as population size, distance to the nearest town, and proximity to roads—interacted with year fixed effects, to mitigate any pre-existing differential growth trends.

As an additional strategy, we perform a placebo test by substituting the distance-to-UA dummies with distance-to-town dummies. Should our previous results be driven primarily by proximity to urban centers rather than the Repeal itself, we would expect to observe similar patterns in the placebo test. Figures D.27 and D.28 replicate the longitudinal Population Census analyses using the distance to the nearest town. These placebo estimates are significantly noisier, exhibit no systematic patterns, and tend to be considerably smaller in magnitude, supporting the causal interpretation of the spatial DiD results. Figures D.29, D.30, and D.31 repeat the cross-sectional SECC analyses with distance to the nearest town. Because the SECC results are themselves associational, these figures serve as specificity checks rather than falsification tests: the housing, income, and income-source gradients are specific

to proximity to UAs and are not reproduced by proximity to towns in general.

D.6 Rent and housing prices

Another question left unaddressed in the main text concerns whether the increase in the value of firms' landholdings is driven by the increase in the amount of land or in land prices. Unfortunately, there is no systematic data on property prices across districts in India. To address this, we construct several measures of residential rent expenditure using household-level consumption data. The rationale behind this approach is that if urban land prices increased dramatically following the Repeal, we would expect a corresponding rise in rent expenditures. This examination is important in its own right. Suppose the Repeal facilitated the conversion of residential land to industrial land, thereby pushing up rents; this could potentially have negative effects on urban renters.

The consumption data are sourced from the Indian National Sample Survey (NSS), which provides pooled cross-sectional household consumer expenditure data. The full-scale quinquennial ('thick') rounds survey approximately 100,000 households every five years.⁵⁸ For our analysis, we utilize the data from the 50th (1993-94), 55th (1999-2000), and 61st (2004-05) rounds, corresponding to the fifth, sixth, and seventh quinquennial rounds.

The NSS dataset contains information on household consumption expenditure and quantity consumed across approximately 300 product categories. Rents are reported in four categories:

- House and garage actual rent, which includes rent for residential buildings and garages for private vehicles.
- Hotel lodging charges, paid by any household member for staying in a hotel.
- Other consumer rent, which includes charges for consumer goods (like furniture) and decorations on ceremonial occasions.
- Imputed house and garage rent for urban households living in owned or otherwise occupied houses without paying rent, based on the prevailing rate of rent for similar accommodations in surrounding areas.

To capture the overall rental expenditure, we use the sum of the first three categories—the actual rent paid by the household over the last 30 days. Note that this composite is broader

⁵⁸In addition to the quinquennial rounds, there are also annual NSS rounds that survey fewer households, ranging from 20,000 to 50,000 households.

than residential rent: alongside house and garage rent, it includes hotel lodging charges and rentals of consumer goods. Urban households that own their house typically pay no actual rent but report an imputed rent. Our combined-rent measure therefore takes the imputed house and garage rent as its primary component and falls back on the actual-rent composite only for households whose imputed rent is missing or zero; for households reporting a positive imputed rent, the combined measure equals the imputed rent even when they also report actual rental spending such as hotel lodging.

Both rent expenditure and imputed rent are prorated to reflect monthly rent amounts. One limitation of our analysis is the lack of data on floor area, which is available only from the 62nd round of the NSS onward. Consequently, we are unable to calculate rent per square meter as in [Harari \(2020\)](#). This limitation may restrict our ability to standardize rent expenditures across household sizes and wealth. Therefore, we derive two outcome variables for each measure: the share of rent over the household’s total monthly expenditure over the last 30 days, and the rent expenditure per household member.

We construct a crosswalk to map districts in each NSS round to the time-invariant 1991 population census district boundaries, along with our policy and Repeal data. Subsequently, we aggregate household-level rent information to the district level by calculating the average rent expenditure using the sampling weights. Using the district-round level panel, we examine the treatment effects of the Repeal on our rent measures using the DiD design. This involves comparing rents in districts that had already adopted the Repeal to those that had not, and considering districts that contain any Urban Agglomerations versus those that do not. Our regressions include district fixed effects, year fixed effects, and baseline district population quintile bin-by-year fixed effects.

Table [D.39](#) reports the estimates of the treatment effects on four measures of rents. Columns (1) and (3) present the share of actual and combined rents over the household’s total monthly expenditure over the last 30 days, respectively. Columns (2) and (4) present the actual and combined rents per household member, respectively. Rent *levels* show no response to the Repeal: the per-member actual and combined rent estimates (Columns 2 and 4) are statistically insignificant for both the non-UA effect and the UA interaction, as is the actual-rent share (Column 1). The one exception is the combined-rent share (Column 3), whose interaction term is positive and marginally significant (+0.024, or 2.4 percentage points, $p = 0.04$). We read this cautiously: its own level counterpart in Column (4) shows no significant increase, so the share movement is not corroborated where a genuine rent-price increase would also have to appear, and an expenditure share can move through the total-expenditure denominator as easily as through rents. Taken together, we conclude that the

	(1)	(2)	(3)	(4)
	Rent share	Log rent p.c.	Combined share	Combined log p.c.
Repeal	−0.008 (0.012)	−0.048 (0.245)	0.000 (0.017)	0.066 (0.176)
Repeal × UA	−0.001 (0.011)	−0.022 (0.222)	0.024** (0.011)	0.044 (0.165)
Observations	2,034	2,034	2,065	2,065
R^2	0.459	0.587	0.580	0.659
Year FE	Yes	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes
Population*Year FE	Yes	Yes	Yes	Yes

Note: District-level rent regressions on the NSS district-round panel (Consumer Expenditure Rounds 50, 55, and 61). Robust standard errors clustered at the state level are in parentheses. Significance stars on the Repeal and Repeal × UA coefficients reflect wild cluster bootstrap p -values (Webb six-point distribution, 9,999 replications) where available. Significance levels: * 10%, ** 5%, *** 1%.

Table D.39: Treatment effect of the Repeal on rents

Repeal had minimal impacts on residential rent levels in UA districts, with at most weak evidence of a higher housing-services expenditure share. This provides suggestive evidence that our findings regarding firms' landholdings are unlikely to be driven solely by changes in land prices.

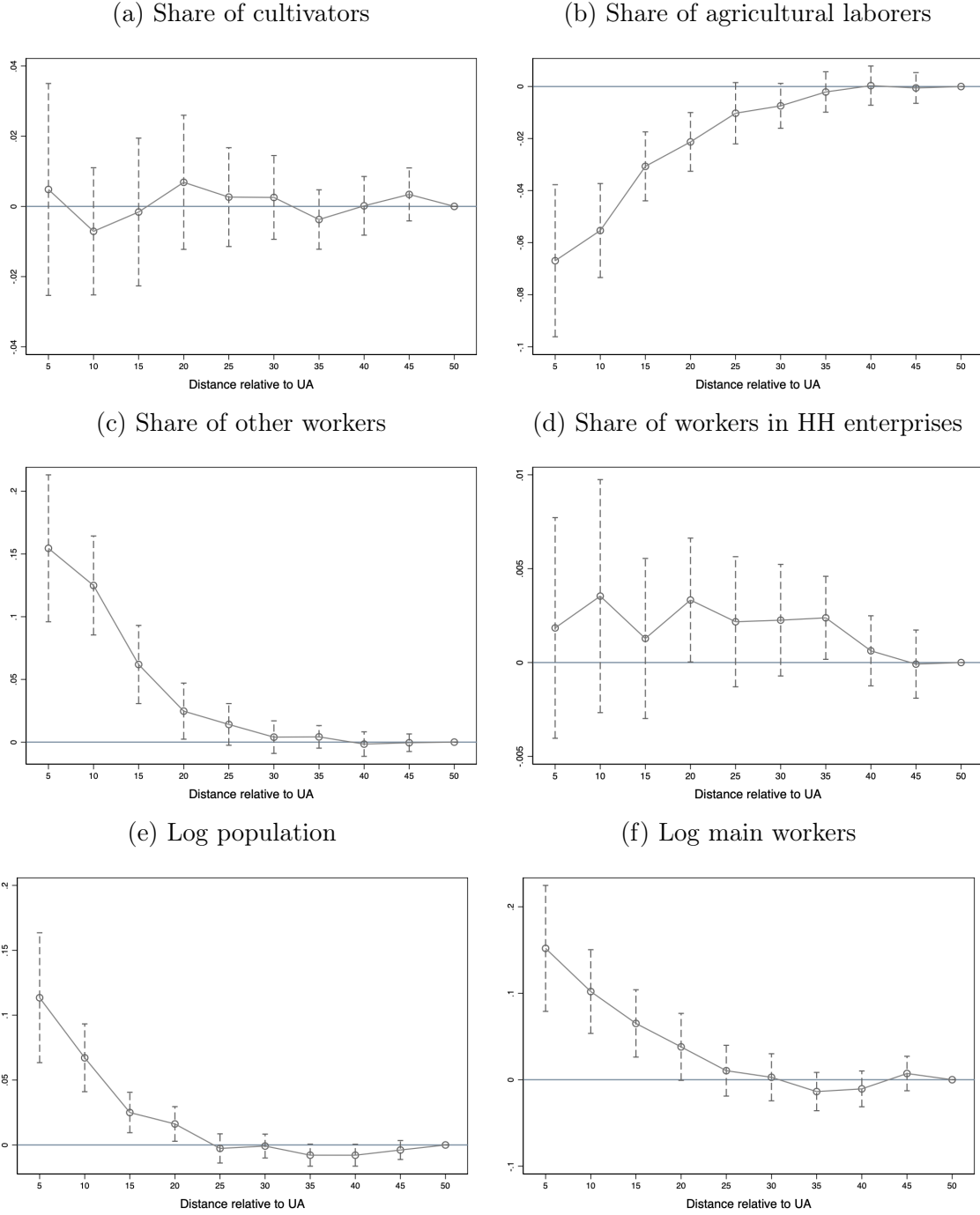


Figure D.22: Spatial DiD effects on employment shares

Note: This figure displays the spatial difference-in-differences (DiD) effects of the Repeal on the shares of cultivators, agricultural hired laborers, other workers, and workers in household enterprises among main workers (panels a-d), and logged population and main workers (panels e-f). The dots represent the estimated parameters β_d from our empirical equation (D.71), where each subscript d corresponds to a distance on the horizontal axis; for example, the coefficient at $d = 1$ indicates the municipality is within 5 km of a UA. The model accounts for municipality fixed effects, year fixed effects, and UA-by-year fixed effects. Controls include quartiles of baseline population, road presence, electrification, and distance to towns, each interacted with year fixed effects. Standard errors are clustered at the UA-year level. Dashed lines represent 95% confidence intervals.

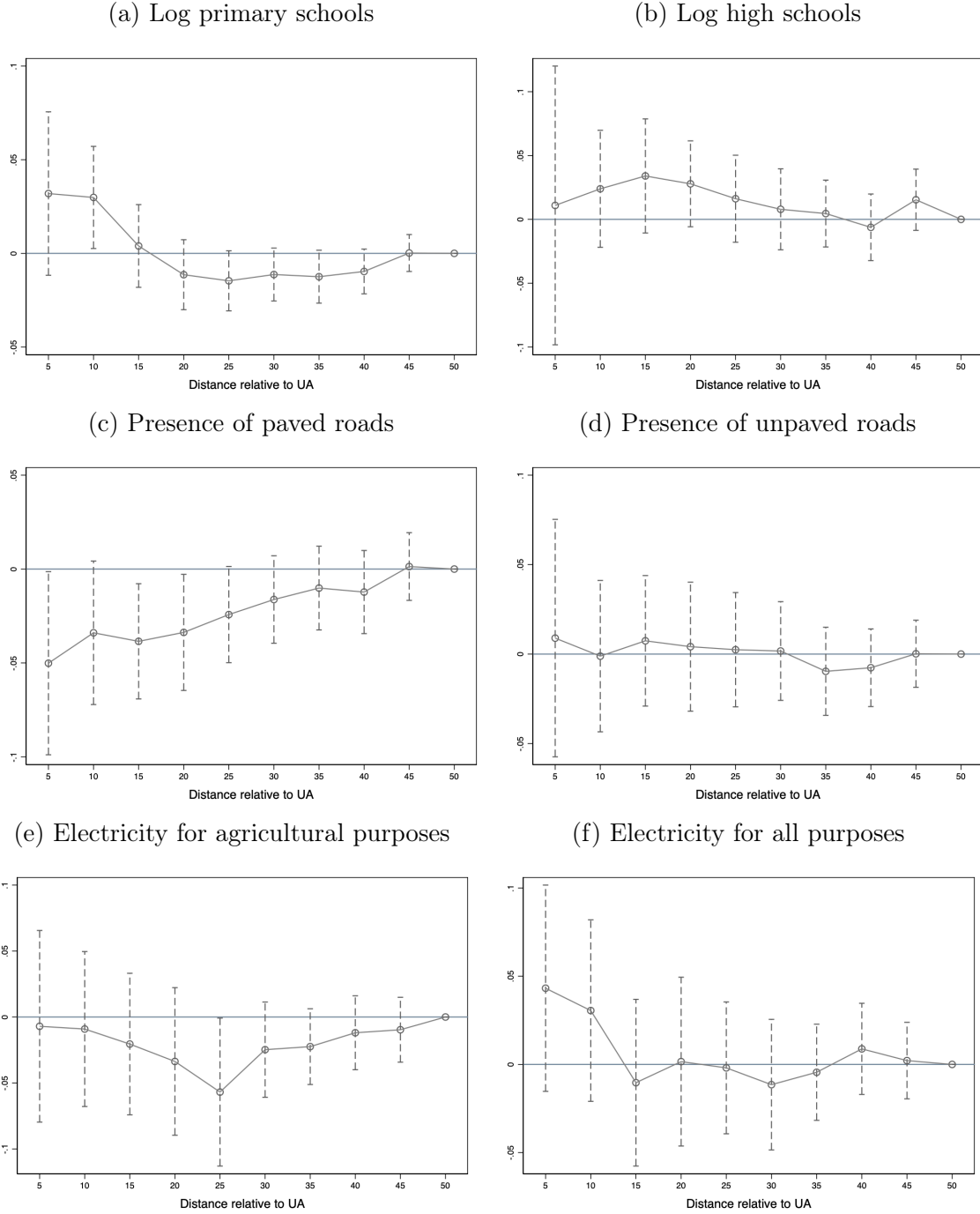


Figure D.23: Spatial DiD effects on amenities

Note: This figure displays the spatial difference-in-differences (DiD) effects of the Repeal on logged number of primary and high schools, the presence of paved and unpaved roads, and the availability of electricity for agricultural and general purposes. The dots represent the estimated parameters β_d from our empirical equation (D.71), where each subscript d corresponds to a distance on the horizontal axis; for example, the coefficient at $d = 1$ indicates the municipality is within 5 km of a UA. The model accounts for municipality fixed effects, year fixed effects, and UA-by-year fixed effects. Controls include quartiles of baseline population and distance to towns, each interacted with year fixed effects. Standard errors are clustered at the UA-year level. Dashed lines represent 95% confidence intervals.

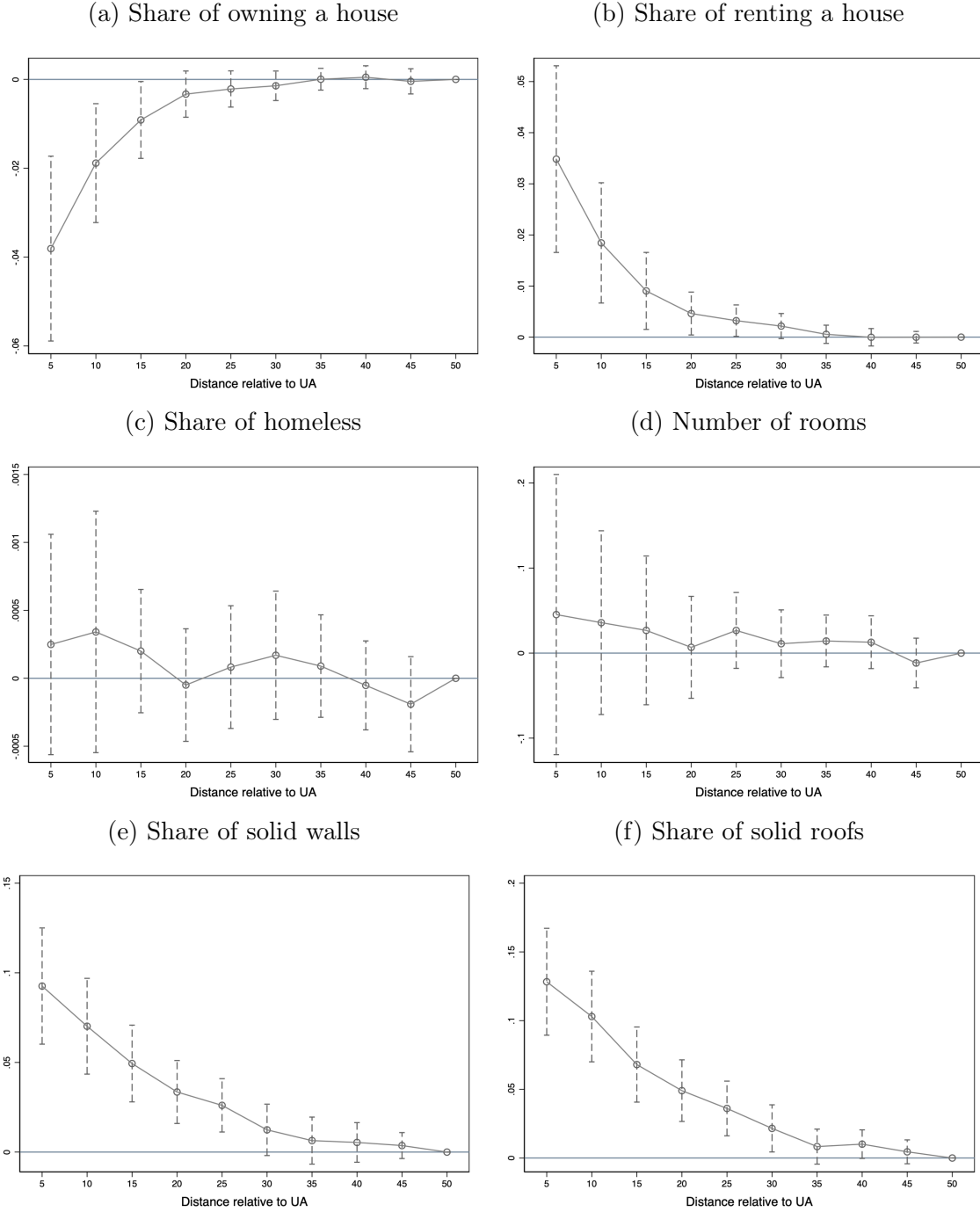


Figure D.24: Proximity to UAs and housing conditions

Note: This figure displays how several housing indicators vary with proximity to UAs: the share of households that own or rent a house, the share of homeless individuals, the average number of rooms per household, and the proportion of houses with solid walls and roofs. Each dot represents the estimated parameter β_d , where each subscript d corresponds to a specific distance along the horizontal axis. For instance, a coefficient at $d = 1$ suggests that the municipality is within 5 km of a UA. The model incorporates UA fixed effects, and controls include quartiles of baseline population, road presence, electrification, and distance to towns. Standard errors are clustered at the UA level. Dashed lines indicate 95% confidence intervals.

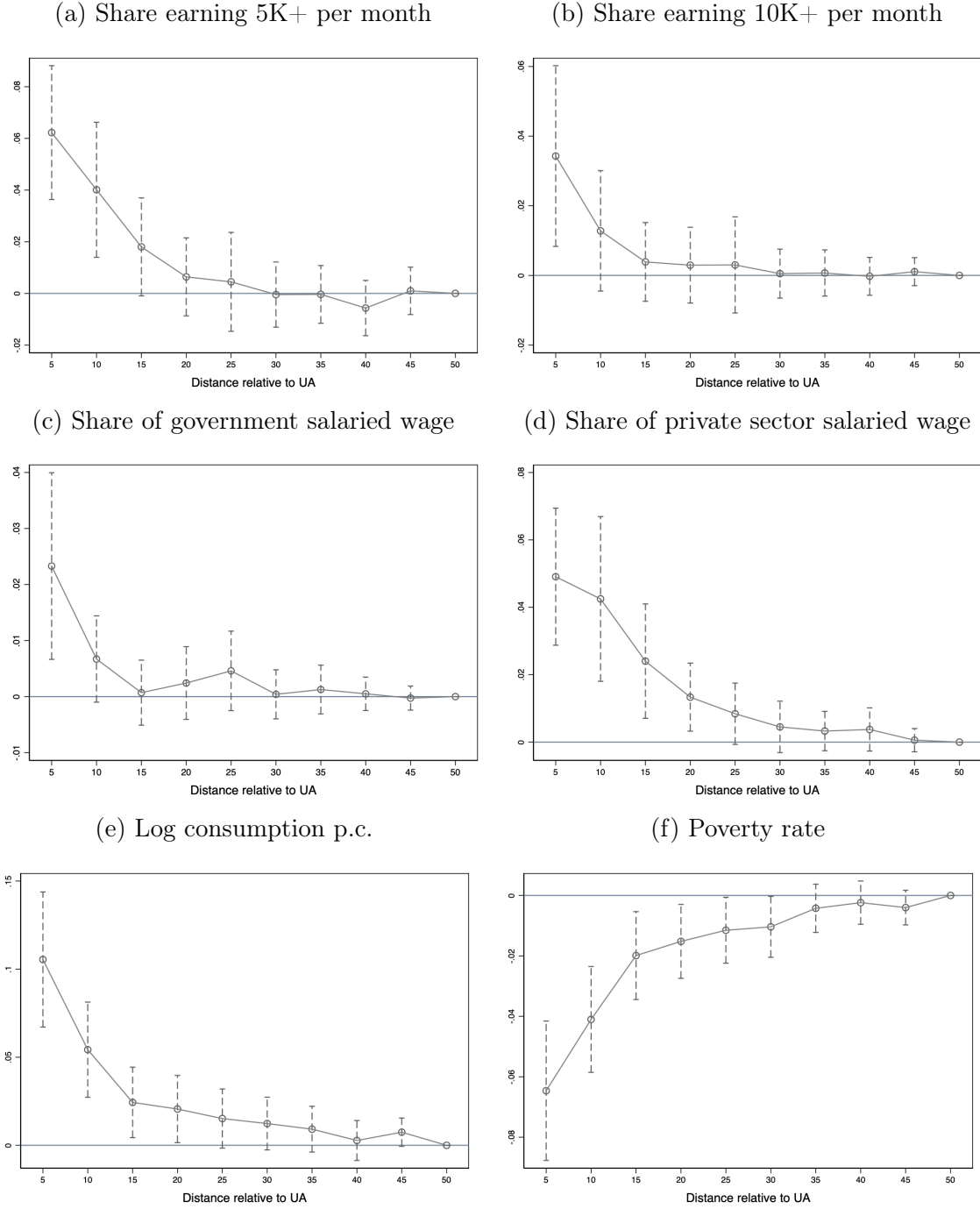


Figure D.25: Proximity to UAs, income, and poverty

Note: This figure displays how municipality-level income, poverty rate, and estimated consumption vary with proximity to UAs. Consumption information is imputed based on another household survey (IHDS-II, 2011-12). SHRUG reports mean predicted consumption and the share of the population below the poverty line (27 Rupees/day) for each municipality. Each dot represents the estimated parameter β_d , where each subscript d corresponds to a specific distance along the horizontal axis. For instance, a coefficient at $d = 1$ suggests that the municipality is within 5 km of a UA. The model incorporates UA fixed effects, and controls include quartiles of baseline population, road presence, electrification, and distance to towns. Standard errors are clustered at the UA level. Dashed lines indicate 95% confidence intervals.

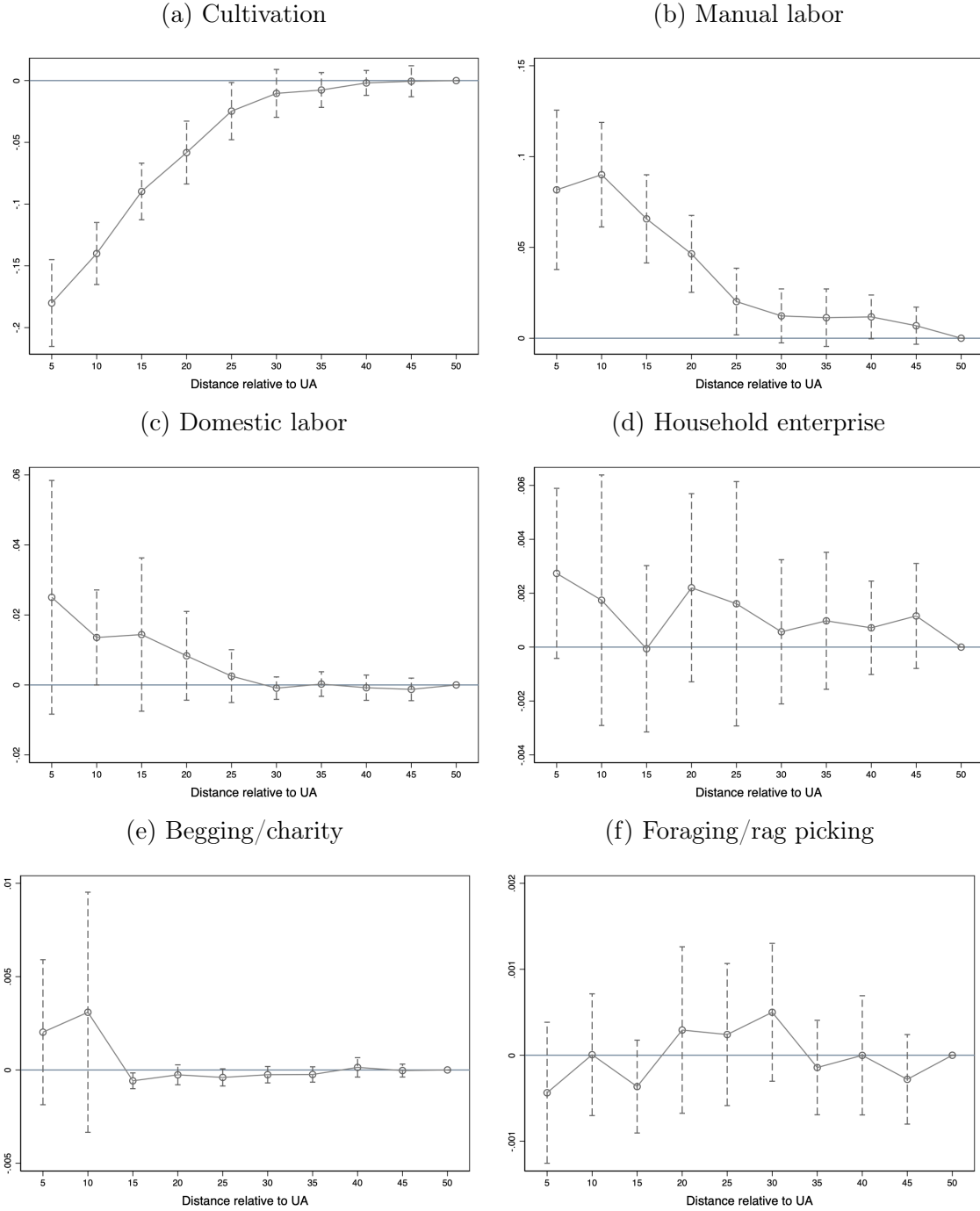


Figure D.26: Proximity to UAs and income sources

Note: This figure displays how the main source of household income varies with proximity to UAs. Each dot represents the estimated parameter β_d , where each subscript d corresponds to a specific distance along the horizontal axis. For instance, a coefficient at $d = 1$ suggests that the municipality is within 5 km of a UA. The model incorporates UA fixed effects, and controls include quartiles of baseline population, road presence, electrification, and distance to towns. Standard errors are clustered at the UA level. Dashed lines indicate 95% confidence intervals.

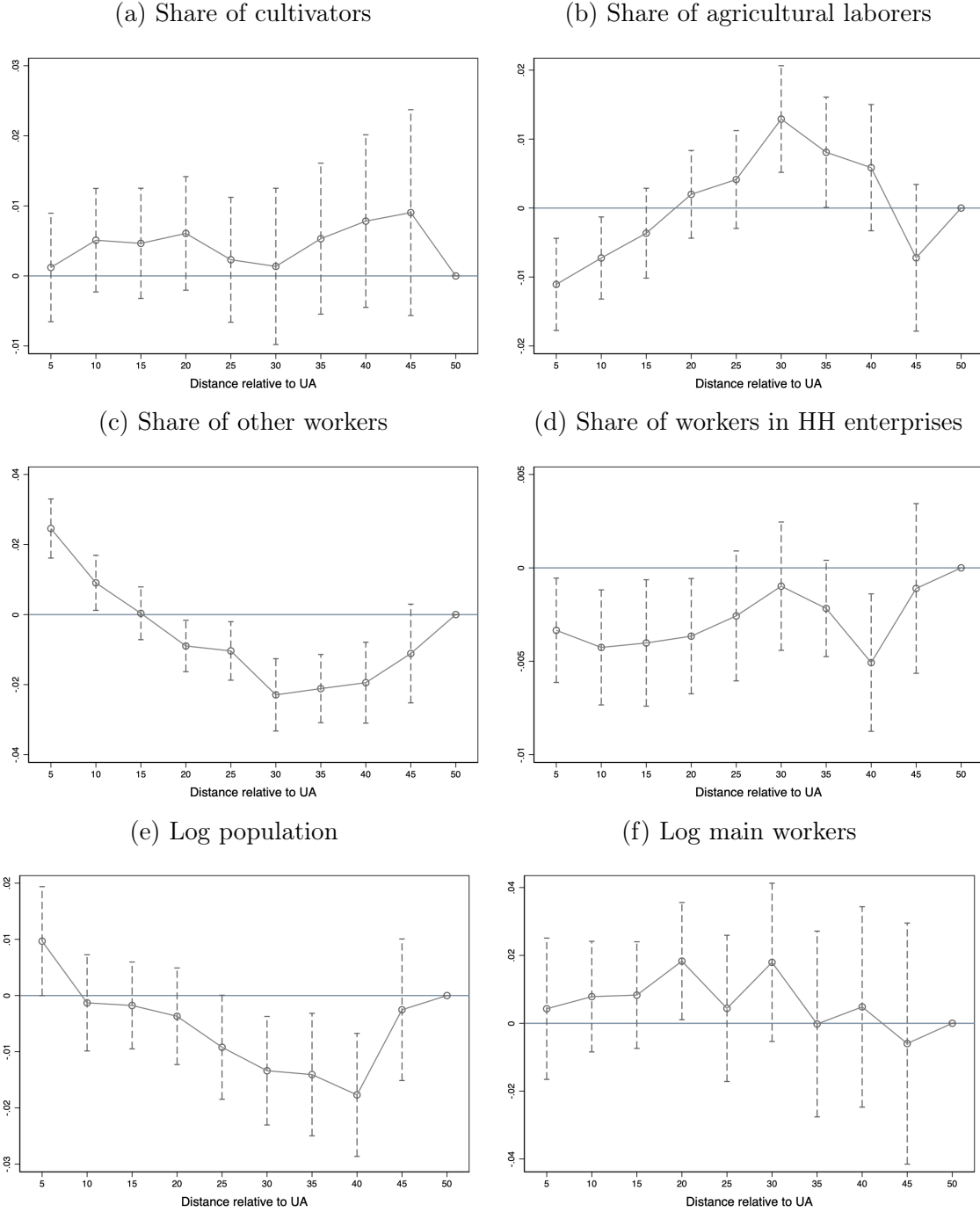


Figure D.27: Placebo: spatial DiD effects on employment shares

Note: This figure displays the spatial difference-in-differences (DiD) effects of the Repeal on the shares of cultivators, agricultural hired laborers, other workers, and workers in household enterprises among main workers (panels a-d), and logged population and main workers (panels e-f). The dots represent the estimated parameters β_d from our empirical equation (D.71), where each subscript d corresponds to a distance on the horizontal axis; for example, the coefficient at $d = 1$ indicates the municipality is within 5 km of a town. The model accounts for municipality fixed effects, year fixed effects, and UA-by-year fixed effects. Controls include quartiles of baseline population, road presence, electrification, and distance to towns, each interacted with year fixed effects. Standard errors are clustered at the UA-year level. Dashed lines represent 95% confidence intervals.

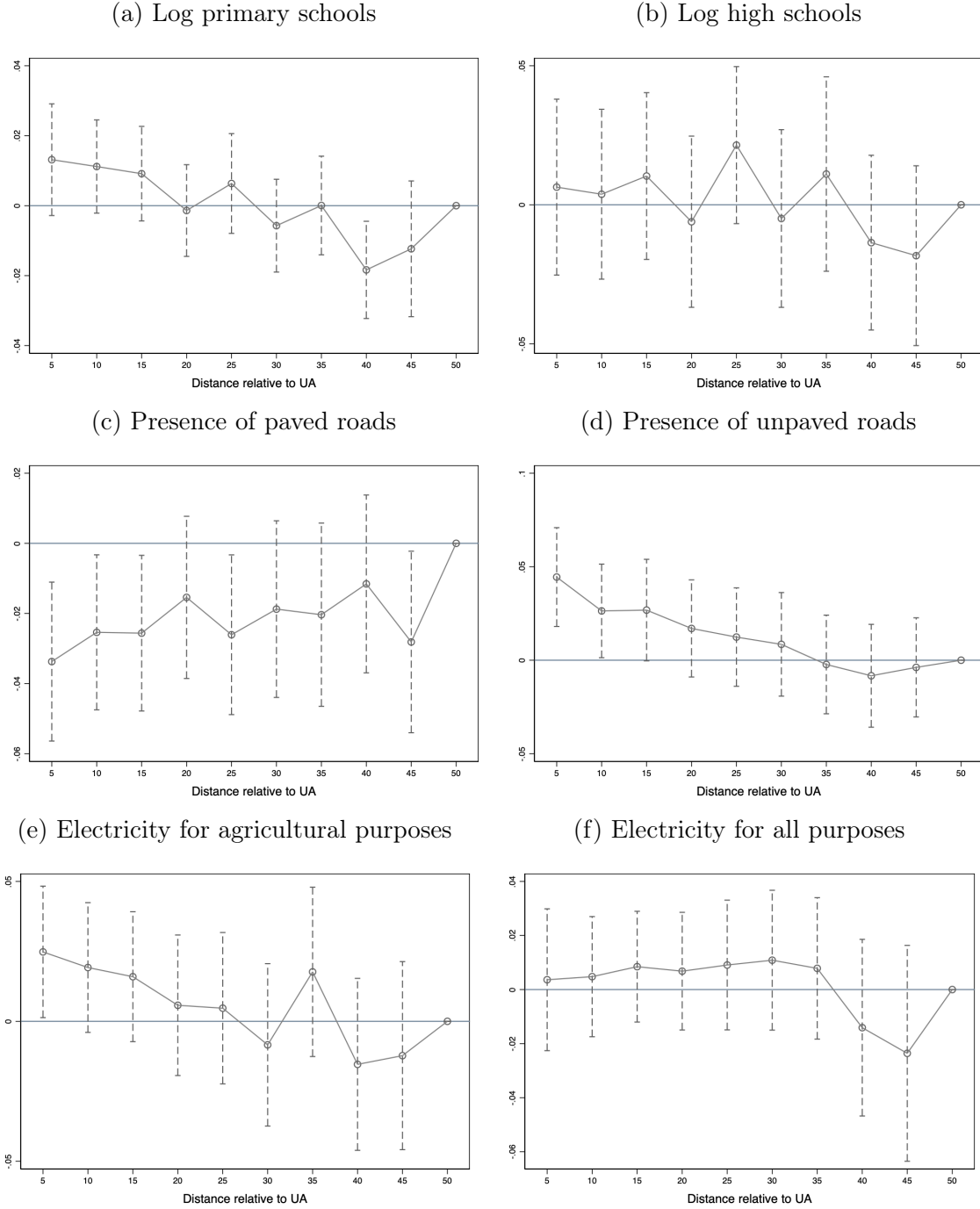


Figure D.28: Placebo: spatial DiD effects on amenities

Note: This figure displays the spatial difference-in-differences (DiD) effects of the Repeal on logged number of primary and high schools, the presence of paved and unpaved roads, and the availability of electricity for agricultural and general purposes. The dots represent the estimated parameters β_d from our empirical equation (D.71), where each subscript d corresponds to a distance on the horizontal axis; for example, the coefficient at $d = 1$ indicates the municipality is within 5 km of a town. The model accounts for municipality fixed effects, year fixed effects, and UA-by-year fixed effects. Controls include quartiles of baseline population and distance to towns, each interacted with year fixed effects. Standard errors are clustered at the UA-year level. Dashed lines represent 95% confidence intervals.

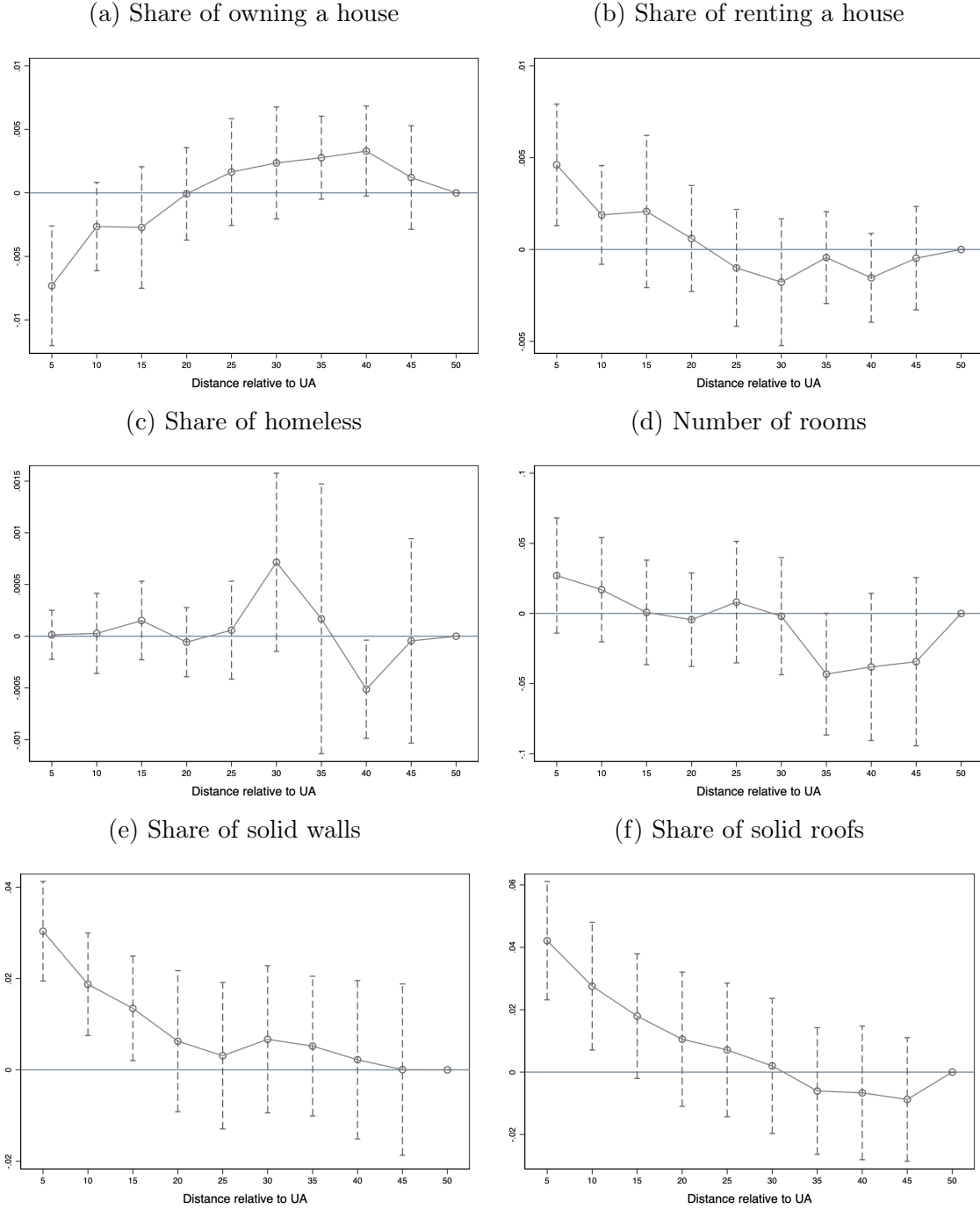


Figure D.29: Placebo: proximity to towns and housing conditions

Note: This figure displays how several housing indicators vary with proximity to towns: the share of households that own or rent a house, the share of homeless individuals, the average number of rooms per household, and the proportion of houses with solid walls and roofs. Each dot represents the estimated parameter β_d , where each subscript d corresponds to a specific distance along the horizontal axis. For instance, a coefficient at $d = 1$ suggests that the municipality is within 5 km of a town. The model incorporates UA fixed effects, and controls include quartiles of baseline population, road presence, electrification, and distance to towns. Standard errors are clustered at the UA level. Dashed lines indicate 95% confidence intervals.

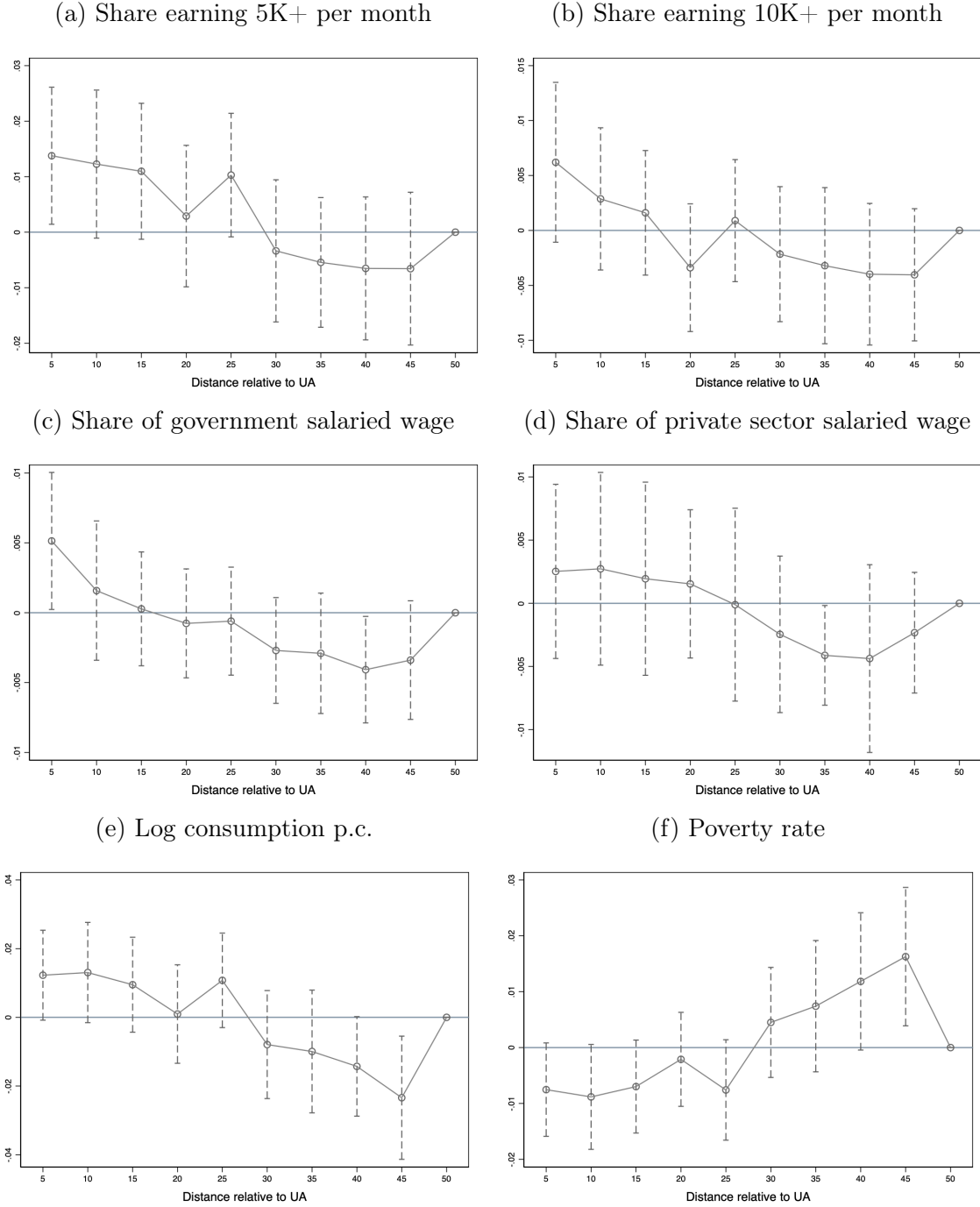


Figure D.30: Placebo: proximity to towns, income, and poverty

Note: This figure displays how municipality-level income, poverty rate, and estimated consumption vary with proximity to towns. Consumption information is imputed based on another household survey (IHDS-II, 2011-12). SHRUG reports mean predicted consumption and the share of the population below the poverty line (27 Rupees/day) for each municipality. Each dot represents the estimated parameter β_d , where each subscript d corresponds to a specific distance along the horizontal axis. For instance, a coefficient at $d = 1$ suggests that the municipality is within 5 km of a town. The model incorporates UA fixed effects, and controls include quartiles of baseline population, road presence, electrification, and distance to towns. Standard errors are clustered at the UA level. Dashed lines indicate 95% confidence intervals.

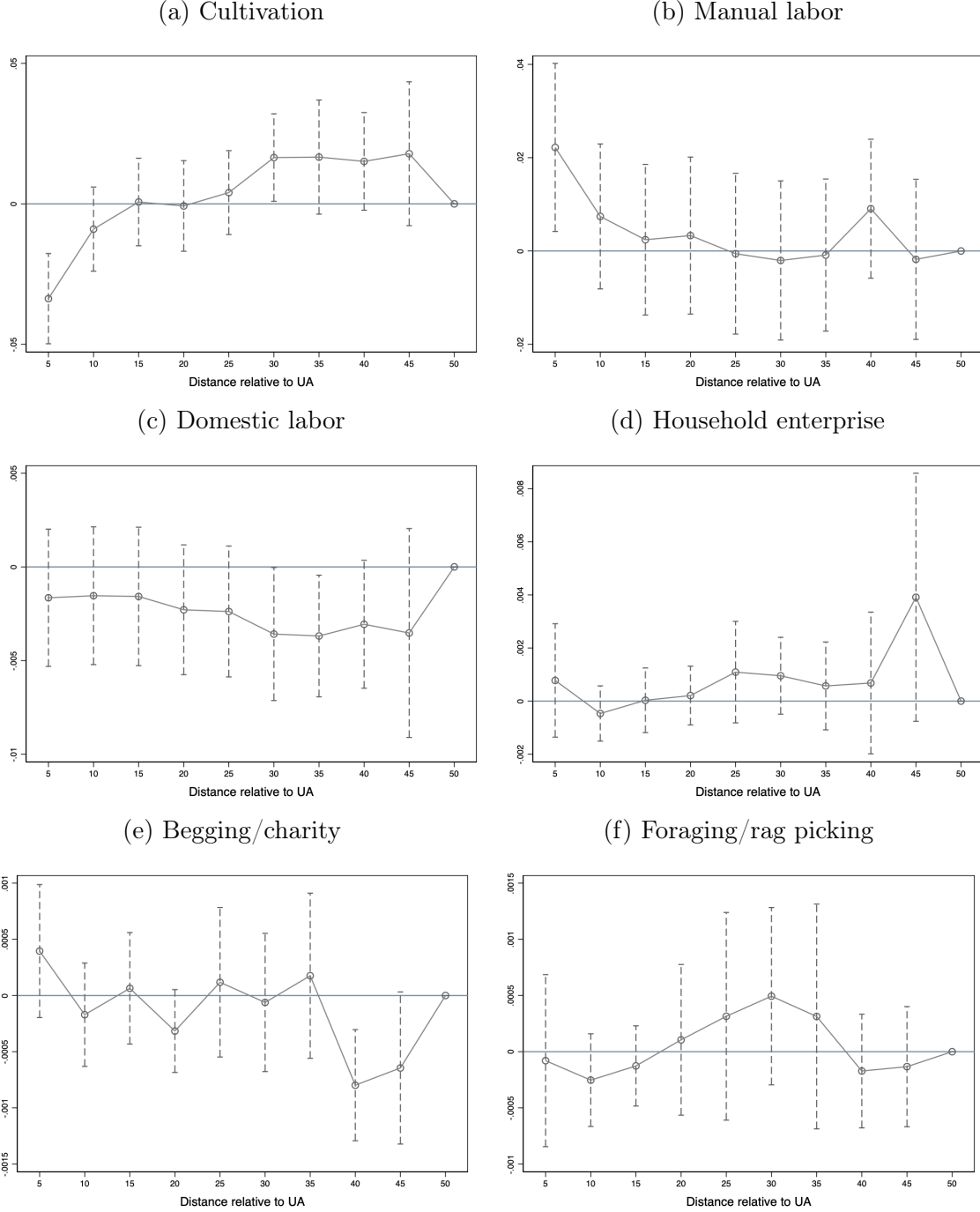


Figure D.31: Placebo: proximity to towns and income sources

Note: This figure displays how the main source of household income varies with proximity to towns. Each dot represents the estimated parameter β_d , where each subscript d corresponds to a specific distance along the horizontal axis. For instance, a coefficient at $d = 1$ suggests that the municipality is within 5 km of a town. The model incorporates UA fixed effects, and controls include quartiles of baseline population, road presence, electrification, and distance to towns. Standard errors are clustered at the UA level. Dashed lines indicate 95% confidence intervals.

E The ULCRA in law and practice

This appendix provides the statutory and institutional detail underlying the summary in Section 2. It first documents the Act’s geographic coverage, ceiling schedule, compensation rules, transfer restrictions, and exemptions; it then explains the constitutional mechanism and chronology of repeal. The final subsection presents the industrial cases that show how these provisions operated in practice.

E.1 Coverage, ceilings, and compensation

The ULCRA applied within designated urban agglomerations, defined around core cities and their contiguous suburban areas using the 1971 Census. Seventy-three UAs were notified to guide implementation, of which 64 were covered in practice. Seventeen states adopted the Act and it applied to all union territories. Several states and union territories without an urban agglomeration above the 200,000-person threshold nevertheless adopted it, including Haryana, Himachal Pradesh, Manipur, Meghalaya, and Tripura. Conversely, Jammu & Kashmir, Kerala, and Tamil Nadu did not bring the central ULCRA into force in their territory despite having qualifying UAs, and instead legislated their own landholding rules.

Within covered UAs, the Act imposed the following ceiling schedule. The ceilings applied broadly to individuals, families, firms, companies, and associations and encompassed land held through ownership, lease, or mortgage.

Category	1971 Census criterion	Maximum vacant land per person
A	Delhi, Mumbai, Kolkata, and Chennai	500 square metres
B	Population above 1 million	1,000 square metres
C	Population from 300,000 to 1 million	1,500 square metres
D	Population from 200,000 to 300,000	2,000 square metres

Land above the applicable ceiling was subject to compulsory acquisition at statutory compensation. For income-generating land, compensation was eight and one-third times the average annual net income over the preceding five years. For other land, the Act imposed maximum rates of five to ten rupees per square metre depending on the agglomeration category. Total compensation was capped at two lakh rupees, a nominal ceiling fixed in 1976 and worth approximately \$4,400 by the time of the Repeal at the then-prevailing exchange rate of roughly 45 rupees per U.S. dollar.

E.2 Transfers, buildings, and exemptions

Section 26 governed transfers of vacant land within the ceiling. A holder wishing to sell, mortgage, gift, or lease the land had to notify the competent authority. The state's sixty-day purchase option arose only for a sale: mortgages, gifts, and leases required notice but conferred no purchase right. If the state exercised its option, the price was compensation under the Land Acquisition Act of 1894, and Section 26(3) fixed the valuation date at the date of notice, so the holder forfeited any appreciation between notice and completion. Section 28 enforced the requirement at the registry by barring registration until notice had been given. Section 26 carried no sunset and remained in force until repeal.

The restrictions were stronger for land above the ceiling. Sections 5(3) and 10(4) deemed any transfer before the excess had been declared and notified to be null and void. Section 15 required a statement within three months of any later acquisition that pushed a holder above the ceiling and subjected the excess to the same machinery. The Act therefore constrained the accumulation of land as well as its disposal. Private industrial firms had no categorical exemption: Section 19 covered only governments, government companies, military institutions, banks, and charitable trusts, while relief for private firms ran through the discretionary and revocable exemptions of Section 20.

The distinction between vacant land and built property was correspondingly important. Section 2(q) excluded land occupied by a building together with appurtenant land, but Section 2(g) generally capped that allowance at five hundred square metres. Factory yards, open storage, and land held for expansion could therefore remain exposed even when the building footprint was excluded. The Act also capped built-up area where a structure contained a dwelling unit, at 300 square metres in Category A and B cities and 500 square metres in Category C and D cities.

A separate provision, Section 27, required prior written permission to transfer land *with a building*, but it was not operative during our sample. In *Maharao Sahib Shri Bhim Singhji v. Union of India*, [1981] 1 SCC 166, a Constitution Bench of the Supreme Court struck down Section 27(1) insofar as it restricted transfers of land with a building within the ceiling area—which, because above-ceiling land vested in the state, covered substantially all land remaining in private hands. Independently, Section 27 applied by its own terms only for ten years from the commencement of the Act or the date of construction, whichever was later, and exempted leases of ten years or less throughout.

E.3 Adoption and repeal under Article 252

The ULCRA was enacted by Parliament under Article 252 after resolutions by eleven states. Article 252(2) reserved amendment or repeal of the resulting parliamentary law to Parliament. The Supreme Court applied that rule to the ULCRA itself in *Union of India v. Valluri Basavaiah Chaudhary*, (1979) 3 SCC 324 (Constitution Bench), and again in *Thumati Venkaiah v. State of Andhra Pradesh* (1980), holding that the state legislature “stood denuded of its power to legislate on that subject.” Adopting states could not unilaterally amend the central Act. Its statutory restrictions were therefore common across adopting states, although implementation through notifications, exemptions, and enforcement could differ. Tamil Nadu provides the contrast: it passed no Article 252(1) resolution, enacted its own Urban Land (Ceiling and Regulation) Act in 1978, amended it independently, and repealed it through its own legislation.

The same constitutional mechanism governed repeal. Parliament’s 1999 Repeal Act applied initially in the union territories and in Haryana and Punjab, the states at whose request Parliament passed it. In other adopting states, the central Act remained in force until the state legislature adopted the Repeal Act by resolution. Article 252 thus fixed the legal mode of repeal without fixing its timing, which remained a state political choice. This is why the paper treats repeal timing as potentially endogenous, tests whether it is related to pre-Repeal land-market conditions (Table B.20), and verifies the results using only the first, centrally triggered 1999 repeal episode.

Maharashtra and Andhra Pradesh illustrate the distinction between formal enactment and public commitment. Maharashtra’s two Houses adopted repeal resolutions on 29 November 2007 and the notification was dated and published on 1 December 2007. Andhra Pradesh’s repeal took effect on 27 March 2008 under G.O.Ms. No. 603. The dates are recorded in the subsequent litigation over the repeals, and the Maharashtra date is corroborated by [Pethe and Nallathiga \(2011\)](#). Both states had already committed statewide in 2006 to repeal the ULCRA as a stated condition for central infrastructure funding under JNNURM. The baseline therefore dates treatment in these states to 2006; the principal estimates are essentially unchanged under their formal repeal dates (Table B.30). Appendix F separately assesses JNNURM investment as a potential confounder. Telangana, which remained part of Andhra Pradesh throughout the sample, is recorded as Andhra Pradesh.

West Bengal adopted the central ULCRA but did not adopt the Repeal Act, while Tamil Nadu continued to operate its separate state ceiling law. Urban land ceilings therefore remained in force in both states. West Bengal was governed by the Left Front from 1977 to

2011 and is notable for its comparatively extensive land expropriation and redistribution (Raychaudhuri, 2004). The analysis excludes West Bengal and Tamil Nadu as institutional outliers.

E.4 Industrial land in practice

The ULCRA's reach over industrial land is easier to see in individual proceedings than in the text of the Act. The evidence below combines two adjudicated cases showing how the ceiling applied to working plants with one aggregate figure showing that industrial holdings were not a marginal part of what the Act caught. Together they support the reading in Section 2: the restrictions reached land inside working industrial sites and land that plants wanted to acquire, and relief was available only at the discretion of the competent authority.

A working factory, most of its site declared surplus. In *Agra Concrete Pipe Co. v. Competent Authority, Agra*, AIR 1987 All 232, the holder was an operating manufacturer of concrete pipes. Of premises totalling 10,881.2 square metres, the competent authority declared 6,771.8 square metres—about 62 percent—to be surplus vacant land. The dispute was not over whether the firm was a genuine manufacturer but over which of its production assets earned an appurtenant-land allowance: the firm argued that its curing tanks were structures for this purpose, and the court accepted that they were, while declining to treat two adjoining sheds as separate structures. The point of interest is not the outcome but the terms of the argument. The question before the court was how much of an operating industrial site the Act reached, and the answer was most of it.

The arithmetic of an exemption. A Competent Authority order of 25 July 1985 under Section 8(4), reproduced in *Acme Housing India Pvt. Ltd. v. State of Maharashtra*, 2026:BHC-AS:8418-DB (Bom. HC), sets out the computation for a factory site at Thane held by Shah Malleable Castings. Of a total holding of 54,500 square metres, the authority deducted the built-up area of 6,756, appurtenant land of 13,512, internal roads of 4,643.30, statutory open space of 6,965.50, and a retainable 2,000, leaving 20,622.80 square metres—38 percent of the site—as surplus vacant land.

This example is useful precisely because the allowances were generous. The appurtenant figure of 13,512 square metres is twice the built-up area, far beyond the five hundred square metres the statute prescribes, and the order further set aside land for roads and open space. Even on that basis, more than a third of a working foundry's site was surplus. A plant that

received only the statutory allowance would have fared considerably worse.

Industrial holdings in the aggregate. The two cases above are individual proceedings, and no aggregate figure can establish that they are representative of how the Act was applied site by site. What the aggregate evidence can establish is that the Act's reach into industrial holdings was not marginal. Reviewing the Act's implementation, [Acharya \(1989\)](#) reports that in Madras, of roughly 1,700 hectares of potentially available excess land, only about 700 hectares were held by individuals while the remaining 1,000 hectares were held by industry.⁵⁹ His own argument is that the Act's excess-land estimates rested on the unrealistic premise that large urban parcels were held by individuals. The figure speaks to ownership, not use: it does not record whether the industrial land identified as excess lay inside working sites, as in the cases above, or in parcels held apart from production. What it does show is that industrial holdings accounted for the majority of the excess vacant land identified in a major agglomeration—wherever within a holding the Act's bite fell, it fell mostly on industry.

What the cases do and do not show. These proceedings establish that the ceiling reached land inside working industrial sites, and that firms had no exemption as of right: relief ran through the discretionary and revocable route of Section 20, which Maharashtra administered as a standing docket of industrial exemption schemes. They do not establish that the Act operated as a binding cap on how much land a firm could hold. As [Section 5](#) notes, national acquisitions under the Act were small relative to the land notified as excess, and evasion was widespread. The picture consistent with both facts is the one the model adopts: a legal constraint that bound on paper, was negotiated and evaded in practice, and therefore raised the cost of holding and transacting industrial land rather than fixing its quantity.

⁵⁹[Acharya \(1989\)](#), p. 48, citing Planning Commission, *Report of the Task Force on Planning of Urban Development* (New Delhi, 1983), p. 140. We cite the figure as Acharya reports it.

F JNNURM

In this section, we assess the sensitivity of our estimates of the Repeal’s effects to the Jawaharlal Nehru National Urban Renewal Mission (JNNURM), a large urban-infrastructure program launched in 2005–06 that overlapped with the later Repeal episodes and under which Andhra Pradesh and Maharashtra made the statewide repeal commitment that dates their treatment year in our baseline coding. Our data on the program’s implementation come from the audit report by [Grant Thornton \(2011\)](#), which served as the appraisal agency for the JNNURM. The report is based on information collected from a pre-approved sample of 41 of the 65 mission cities, representing roughly two-thirds of all JNNURM cities. The data includes city-level information on the number of projects approved and completed, as well as both approved and actual expenditures for the Urban Infrastructure and Governance (UIG) and Basic Services to the Urban Poor (BSUP) schemes, as of 2011.

To measure the intensity of JNNURM implementation, we construct four measures: logged per capita expenditure and the number of sanctioned projects in each scheme. Because the ASI locates plants by district, we aggregate the city-level audit data to 1991-census districts—summing projects and expenditures across audited cities and dividing district expenditure by the audited cities’ population—and assign each plant the intensity of the district in which it is located. Districts with no audited city are coded as zero exposure. This zero-fill is innocuous outside the mission, but it introduces measurement error within it: of the 62 districts containing a JNNURM mission city, 40 appear in the audit sample, so the remaining 22 mission districts carry an intensity of zero. Low completion rates compound this problem—only 17.8% of sanctioned UIG projects and 1.5% of sanctioned BSUP projects were completed by 2011 ([Comptroller and Auditor General of India, 2012](#))—so sanctioned counts are a noisy measure of realized implementation. Both features attenuate the intensity coefficients toward zero. We accordingly treat the interactions of these four measures with a post-2006 dummy variable as exposure-intensity diagnostics rather than as estimates of the JNNURM’s causal effects: absent a conditional parallel-trends argument for implementation intensity—and given that realized expenditure may itself respond to post-2006 local conditions—these interactions do not identify effects of the program. The object of interest is instead the stability of our estimated effects of the Repeal, as reported in the main text, when the interactions are included. As a complementary check that does not depend on the audit sample, we also re-estimate the baseline specification excluding UAs covered by the JNNURM, based on the full mission roster in our policy data (63 cities across 62 districts; the mission’s official list later grew to 65 cities).

Tables [F.40](#), [F.41](#), [F.42](#), [F.43](#), and [F.44](#) report our results on firms’ landholdings, fixed assets, number of employees, sales revenue, and TFPQ. Column (1) reports the baseline estimates as a benchmark. Column (2) reports the estimates of the Repeal’s effects when dropping firms in districts whose UA was covered by the JNNURM. In Columns (3)-(4), we use the full sample of data and add the project-count and expenditure intensity controls, respectively. Our estimates in Column (2) become smaller by roughly 15% on average (except for TFPQ) when dropping UAs covered by the JNNURM, but generally remain statistically significant—with the exception of employment, whose estimate becomes imprecise. Importantly, the estimated Repeal’s effects in Columns (3)-(4) are very close to the baseline estimates reported in Column (1). The intensity interactions themselves are uniformly small and statistically insignificant, although given the audit-coverage gaps and low completion rates noted above, we read them as noisy exposure diagnostics rather than as estimates of the program’s effects. Collectively, the evidence indicates that the JNNURM is unlikely to bias our estimates of the Repeal’s effects.

	(1)	(2)	(3)	(4)
	Baseline	Drop JNNURM UAs	Projects	Expenditure
Repeal	0.005 (0.057)	0.026 (0.057)	0.016 (0.055)	0.012 (0.056)
Repeal $\times$ UA	0.202*** (0.056)	0.185*** (0.051)	0.185*** (0.052)	0.194*** (0.050)
BSUP projects			0.018 (0.021)	
UIG projects			-0.002 (0.011)	
BSUP expenditure p.c.				-0.046 (0.053)
UIG expenditure p.c.				0.043 (0.043)
Observations	338,686	235,346	338,686	338,686
R^2	0.183	0.212	0.183	0.183
District FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Size $\times$ Year FE	Yes	Yes	Yes	Yes

Note: Column (1) is the baseline specification. Column (2) drops JNNURM-covered UAs. Columns (3)–(4) add JNNURM project and expenditure controls, respectively. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table F.40: Effects of the Repeal on landholdings: account for JNNURM

	(1)	(2)	(3)	(4)
	Baseline	Drop JNNURM UAs	Projects	Expenditure
Repeal	0.081 (0.104)	0.155 (0.104)	0.090 (0.104)	0.093 (0.104)
Repeal $\times$ UA	0.145** (0.058)	0.124* (0.064)	0.135** (0.059)	0.128** (0.057)
BSUP projects			−0.001 (0.009)	
UIG projects			0.003 (0.003)	
BSUP expenditure p.c.				−0.005 (0.025)
UIG expenditure p.c.				0.015 (0.021)
Observations	530,008	340,990	530,008	530,008
R^2	0.254	0.310	0.254	0.254
District FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Size $\times$ Year FE	Yes	Yes	Yes	Yes

Note: Column (1) is the baseline specification. Column (2) drops JNNURM-covered UAs. Columns (3)–(4) add JNNURM project and expenditure controls, respectively. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table F.41: Effects of the Repeal on fixed assets: account for JNNURM

	(1)	(2)	(3)	(4)
	Baseline	Drop JNNURM UAs	Projects	Expenditure
Repeal	−0.029 (0.038)	0.000 (0.038)	−0.030 (0.040)	−0.033 (0.039)
Repeal × UA	0.067** (0.026)	0.055 (0.042)	0.069** (0.027)	0.073** (0.027)
BSUP projects			−0.003 (0.006)	
UIG projects			0.001 (0.003)	
BSUP expenditure p.c.				0.005 (0.025)
UIG expenditure p.c.				−0.007 (0.019)
Observations	533,802	343,519	533,802	533,802
R^2	0.148	0.172	0.148	0.148
District FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Size × Year FE	Yes	Yes	Yes	Yes

Note: Column (1) is the baseline specification. Column (2) drops JNNURM-covered UAs. Columns (3)–(4) add JNNURM project and expenditure controls, respectively. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table F.42: Effects of the Repeal on employees: account for JNNURM

	(1)	(2)	(3)	(4)
	Baseline	Drop JNNURM UAs	Projects	Expenditure
Repeal	-0.021 (0.049)	0.022 (0.054)	-0.014 (0.050)	-0.018 (0.051)
Repeal $\times$ UA	0.160*** (0.046)	0.119* (0.058)	0.152*** (0.046)	0.156*** (0.045)
BSUP projects			-0.003 (0.006)	
UIG projects			0.004 (0.003)	
BSUP expenditure p.c.				-0.015 (0.027)
UIG expenditure p.c.				0.015 (0.020)
Observations	464,755	301,786	464,755	464,755
R^2	0.244	0.287	0.244	0.244
District FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Size $\times$ Year FE	Yes	Yes	Yes	Yes

Note: Column (1) is the baseline specification. Column (2) drops JNNURM-covered UAs. Columns (3)–(4) add JNNURM project and expenditure controls, respectively. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table F.43: Effects of the Repeal on sales: account for JNNURM

	(1)	(2)	(3)	(4)
	Baseline	Drop JNNURM UAs	Projects	Expenditure
Repeal	0.004 (0.073)	0.002 (0.092)	0.003 (0.073)	0.000 (0.073)
Repeal $\times$ UA	0.120** (0.056)	0.182** (0.065)	0.122** (0.056)	0.124** (0.052)
BSUP projects			−0.002 (0.009)	
UIG projects			0.000 (0.003)	
BSUP expenditure p.c.				0.017 (0.024)
UIG expenditure p.c.				−0.017 (0.017)
Observations	365,512	241,142	365,512	365,512
R^2	0.489	0.514	0.489	0.489
District FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Size $\times$ Year FE	Yes	Yes	Yes	Yes

Note: Column (1) is the baseline specification. Column (2) drops JNNURM-covered UAs. Columns (3)–(4) add JNNURM project and expenditure controls, respectively. Robust standard errors clustered at the state level are in parentheses. Significance levels: * 10%, ** 5%, *** 1%.

Table F.44: Effects of the Repeal on TFPQ: account for JNNURM